

GEOLOGICAL FACTS;
OR,
THE CRUST OF THE EARTH,
WHAT IT IS,
AND WHAT ARE ITS USES.

BY
THE REV. W. G. BARRETT,
ROYSTON.

"There is in this universe a stair, or manifest scale of creatures, rising not disorderly nor in confusion, but with a comely method and proportion."—SIR THOMAS BROWNE, *Religio Medici*, 1642.

LONDON :
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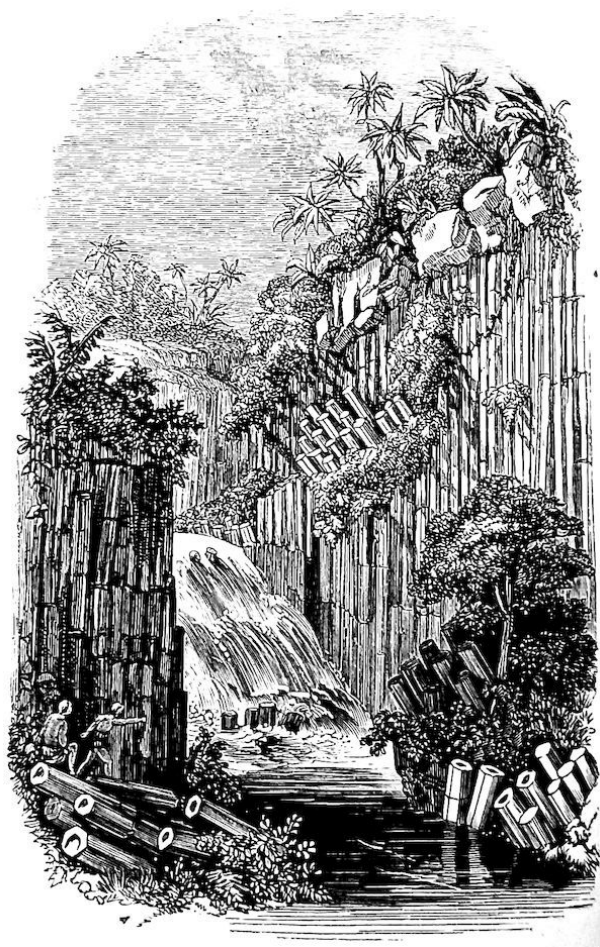
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BASALTIC COLUMNS AT REGIA, IN
MEXICO

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PREFACE.

Last written, and generally last read, a Preface has nevertheless become so integral a part of every book, that I may presume, I trust, upon the attention of my readers while I ask them to indulge me with a little prefatory egotism.

This book of mine goes forth into the world with many misgivings on the part of its author. How he came to write it was thus-wise. He was settled in a quiet country town in Cambridgeshire, Royston to wit, as a Dissenting Minister; around him he found a number of young persons, who did not believe they had “finished their education” because they had left school, and who were anxious to avoid the usual littleness and small talk of such towns, by earnest attention to actual study. And so it came to pass that a Geological class was formed, which, meeting every week, afforded real stimulus for private work, and led to the consultation of the best standard works, happily available through the well-conditioned library of the town.

The result of those classes is this little book. What is written here was mostly, if not all, said there; and, urged to publish, the author feels a pleasure in dedicating this book to the class, composed almost entirely of young ladies, who found in these studies one of their chief delights, and whose private collections have been greatly assisted by the hints thus obtained.

I do not pretend to *teach the Science of Geology*; I aspire simply to give a taste for this noble and elevating physical study; and, imperfect as this little manual, written in the few hours of capricious leisure snatched from an incessant strain of engagements, must be, I shall only be too happy, if one and another lay aside *my* book, and go up higher to Lyell, Sedgwick, Buckland, Murchison, Ansted, Miller, and others.

Possibly my stand-point as a minister of religion may have given unconsciously a too theological tone to some of the chapters, especially the last; if such is the case, I beg leave to apologise for such an error by the candid statement, that I have come into contact frequently with minds who have not hesitated to express the doubts I have endeavoured to resolve.

I am quite sure that if we, whose calling is with the greatest and the deepest truths that can touch the heart of the real world in which we live and move and have our being, encourage those whom we meet in the free intercourse of social life, to express their doubts, however painful the form of that expression may sometimes be, we shall be far better prepared to meet the wants of our age than if we shut ourselves up in our studies, and exclude ourselves by conventional devices from God's great world of thought and action that is vibrating so palpably around us, many of whose most painful throbs are occasioned by a supposed contradiction between Science and Scripture.

At the feet of the Master I desire to serve, I lay this little book, beseeching Him to regard it as a labour of love, and to use it as an aid to the faith of others in the inspired books of Nature and of Revelation.

Removed from the happy town where this book was written, it only remains to add, that for many of the fossils figured in the following pages I am indebted to the kindness of the Rev. Mr. Meeke, at that time the Unitarian Minister of Royston, whose cabinet was always open to my use, and whose courtesy and catholic kindness I thus desire to record; while to Miss Butler, one of my Geological class, I am indebted for all the drawings and devices which will doubtless make this book more attractive than it could otherwise have been.

In the words of Archdeacon Hare, I close this brief prelection: "So imperfectly do we yet understand the redemption wrought for us by Christ; and so obstinate are we in separating what God has united, as though it were impossible for the Tree of Knowledge to stand beside the Tree of Life. Yet in the redeemed world they do stand side by side, and their arms intermingle and intertwine, so that no one can walk under the shade of the one, but he will also be under the shade of the other."

W. G. B.

MANCHESTER,

July, 1855.

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GEOLOGY.

CHAPTER I.

INTRODUCTORY.

“In the beginning God created the heaven and the earth.”

MOSES.

Geology is the history of the *crust* of this planet. This history we compile, not from old records or moth-eaten state papers, not from antiquarian research or the study of ancient coins, but from actual and painstaking examination of the materials composing this crust. How suitable is this word *crust*, will be seen at once, when it is considered that its thickness in all probability does not exceed eighty miles, a mere fraction of the distance to the earth's centre. Of this eighty miles we know pretty accurately the character and arrangement of some seven or eight miles; not that we have ever penetrated so far beneath the surface in a straight line, the deepest mines not exceeding 1800 or 1900 feet; but, by putting together the thicknesses of the various strata, with which we are well acquainted, we reach this conclusion without much hesitation. Professor Whewell has well observed, that “an earth greater or smaller, denser or rarer, than the one on which we live, would require a change in the structure and strength of all the little flowers that hang their heads under our hedges. There is something curious in thus considering the whole mass from pole to pole and from circumference to centre, as employed in keeping a snowdrop in the position most suited to its vegetable health.”^[1] When we come to examine this crust, several appearances of a striking character reward our toil. At first, and before we proceed in our investigations more minutely, we find that there are only two varieties of rocks in all the vast arrangements spread out before us. Some rocks we find to be unstratified, and others we find to be stratified: from the absence of all fossils in the former of these, and from

their crystalline character, we conclude that these were formed by the action of *fire*, and therefore we call them *Igneous* or *Plutonic* rocks. From the sedimentary character and from the numerous fossils of the stratified rocks, we conclude these to have been formed by the action of *water*, and we therefore call these *Aqueous* or *Neptunian* rocks. Following out these investigations, we meet with other facts equally interesting: we find that the Plutonic or unstratified rocks lie generally at the base of all the others, and that where they come up to the surface and crop out from other rocks, or rise in towering mountain heights above them, this has been the result of igneous action from beneath, and that this elevation has disturbed the surrounding strata from the horizontal position in which we imagine them to have been first arranged. The extent of the Plutonic rocks is immense; in *Europe*, the Scandinavian mountains, the Alps, the Pyrenees, and the Carpathians; in *Asia*, the Himalayan, the Caucasian, and the Altai mountains; in *Africa*, the Atlas mountains and the Cape of Good Hope; in *America*, nearly the whole of the two continents, and in *Australia*, its southern part;—all these wide regions of the globe are composed of those igneous or Plutonic rocks to which we give the names of granite, basalt, porphyry, trap, &c. &c. Finding the surrounding strata disturbed by depression, or upheaval in consequence of the giant claims of these older rocks that appear to have risen out of the centre of the earth in a red-hot or semifluid condition, and then suddenly to have cooled down, we begin to examine this upper and sedimentary strata, and here the most delightful and romantic results are obtained. We find no two courses or formations in these sedimentary rocks alike. Rising up from the granite, we meet in the clay strata corals and trilobites, the first fossil forms of ancient life with which we are acquainted: we rise higher still, for our imaginary start is from the bottom of the earth's crust, and we meet in other rocks fossil fishes of an extraordinary shape, and once, doubtless, possessed of extraordinary functions; higher yet, and touching the coal measures, we come to vast forests of palms and ferns, that by chemical changes and mechanical pressure have been converted into our mineral coal, the vast fields of which constitute the real diamond mines of Great Britain; higher yet, exploring the Lias and Oolitic groups, the huge remains of ancient saurians (animals of the lizard species) fill us with awe and wonder, and make us rejoice that they had no successors in the next strata; higher yet, and we reach the last period of the earth's history, previous to the introduction of man, and

enormous “four-footed beasts,” the mastodon, the megatherium and others, astonish us by their gigantic proportions, and evidently herbivorous habits; and last of all we rise to the surface and breathe freely in company with our fellow man, made in the image of God, to inhabit this world as his palace, and to interpret its mysteries as its priest.

We may probably put these general results into a more popular form,—for we reserve the details to a seriatim examination of each formation,—by the following quotation from a modern and extensively useful writer: “We distinguish four ages of nature, corresponding to the great geological divisions, namely—

“1. The *primary* or *palæozoic*^[2] age, comprising the Lower Silurian, the Upper Silurian, and the Devonian. During this age there were no air-breathing animals. The fishes were the masters of creation. We may therefore call it the *Reign of Fishes*.

“2. The *secondary* age, comprising the carboniferous formation, the trias, the oolitic, and the cretaceous formations. This is the epoch in which air-breathing animals first appear. The reptiles predominate over the other classes, and we may therefore call it the *Reign of Reptiles*.

“3. The *tertiary* age, comprising the tertiary formations. During this age, terrestrial mammals of great size abound. This is the *Reign of Mammals*.

“4. The *modern* age, characterized by the appearance of the most perfect of created beings. This is the *Reign of Man*.”^[3]

From this brief but necessary outline of “the treasures of the deep” which lie before us we may proceed to make a few preliminary remarks on the moral and theological aspects of this science. Many persons have supposed that the statements of Scripture and the alleged facts of Geology are at variance, and, forgetful that some of the devoutest minds of this and other countries have been equal believers in both, have too summarily dismissed geology from their notice as a study likely to lead to infidelity. To such we would briefly remark, that it is utterly impossible there can be any contradiction between the written volume of Inspiration and the outspread volume of Creation. Both are books written by the same hand, both are works proceeding from the same ever blessed and beneficent Creator. We believe in the plenary inspiration of the Bible, and we believe equally in the plenary inspiration of Nature; both are full of God, for in them both He is all and in all; and he who is the deepest and the most reverent student of

both will not be long before he comes to the conclusion that not only is there no disharmony, no discrepancy and no contradiction between them, but that they are both harmonious utterances of the one infinite and ever blessed God.

“In reason’s ear they *both* rejoice,
And utter forth a glorious voice;
For ever singing as they shine,
‘The hand that made us is divine.’”

Let us remember that Geology has nothing to do with the history of man, nor with God’s government of man; to the Bible, and only there, do we go for information on these points. Geology gives us the history and the succession of *the things and beings* that were created and made, we believe, incalculable ages before man was placed on the face of the earth. Possibly at times some new discovery in geology may appear to contradict our long received interpretations of isolated passages in Scripture, in which case the modesty of science compels us to reexamine our data, while our reverence for the word of God teaches us to revise our interpretations. As Dr. Chalmers once remarked, “*the writings of Moses do not fix the antiquity of the globe; if they fix anything at all, it is only the antiquity of the species.*” We believe that the same God who, in anticipation of the spiritual wants of the human race, graciously promised from the beginning of man’s transgression, that “the seed of the woman should bruise the serpent’s head,” laid up for him “in the bowels of the earth those vast stores of granite, marble, coal, salt, and the various metals, the products of its several revolutions; and thus was an inexhaustible provision made for his necessities, and for the developments of his genius, ages in anticipation of his appearance.”^[4]

Truth is, and always must be, coincident. There can be no real contradiction between the truth of Scripture and the truth of Science. Whatever is true in one department of God’s agency, must be true when compared with his works in any other department. As an illustration we may notice one particular in which Geology and Scripture move towards the same point in proving the recent introduction of man. We take up a chart of the earth’s crust, and examine it so far as that crust is open to our investigation: eight miles depth or height we know pretty accurately, and in all these accumulations we find one concurrent testimony. If we take the Azoic period of the earth’s crust, and search through the granitic rocks of

Scotland, Wales, or Cornwall; or if we pass on to the Palæozoic period, and examine the Old Red Sandstone, the Carboniferous system, or other formations; or, extending our researches, investigate the secondary formations, the Lias, the Oolite, and the Chalk, and so on until we arrive at the Tertiary period of the earth's history; *all the testimony is one*; there is no contradiction; there are no fossil boats or sofas; no fossil beds or books; no fossil boys and girls; no fossil knives and forks; so far as the teachings of Geology go throughout all these vast periods it says, "there was not a man to till the earth;" they declare that man is not so old as the earth, and that all its fossil remains are pre-Adamite.

Now why should this truth be supposed to lie against the teaching of Scripture? The object of Moses in the first chapter of Genesis, is to teach us that all existing matter owes its origin to the God of the Bible, and not to any of the idols of the heathen. "In the beginning," says that oldest historical record with which we are acquainted, "God created the heaven and the earth;" that is, we apprehend, at some period of the earth's history, in all probability an undefined and incalculable distance from the present time, God created all matter out of nothing, the universe, these heavens and this earth, began to be at the word of God.

"*But afterwards*," says Dr. Pye Smith, in his translation of these words, "the earth was without form and void;" undergoing, we believe, those vast geological changes, those deposits of metal, and those slow accumulations of mineral wealth, by which it was fitted to become the temple, the palace, the workshop, and the home of man. "The first sentence in Genesis is a simple, independent, all-comprehending axiom to this effect, that matter elementary or combined, aggregated only or organised, and dependent, sentient, and intellectual beings, have not existed from eternity either in self-continuity or succession; but had a beginning; that their beginning took place by the all-powerful will of one Being, the Self-existent, Independent and Infinite in all perfection; and that the date of that beginning is not made known."^[5]

Dr. Redford says, "We ought to understand Moses as saying, that indefinitely far back and concealed from us in the mystery of eternal ages prior to the first moment of mundane time, God created the heavens and the earth;" and Dr. Harris in the same strain writes thus, "The first verse in Genesis was designed by the Divine Spirit to announce the absolute

origination of the material universe by the Almighty Creator; and it is so understood in the other parts of holy writ; passing by an indefinite interval, the second verse describes the state of our planet immediately prior to the Adamic creation, and the third verse begins the account of the six days' work."

On this subject we will quote but one brief sentence more—and we have preferred using these quotations to stating the question in our words, thoroughly accordant as they would have been. In Dr. Hitchcock's valuable work, entitled "The Religion of Geology," he says, "The time is not far distant when the high antiquity of the globe will be regarded as no more opposed to the Bible than the earth's revolution round the sun and on its axis. Soon shall the horizon where Geology and Revelation meet be cleared of every cloud, and present only an unbroken and magnificent circle of truth."

Let these thoughts be borne in mind while we pursue our examination of the solid crust of this globe. We do *not*

"drill and bore
The solid earth, and from its strata thence
Extract a register, by which we learn
That He who made it and revealed its date
To Moses was mistaken in its age."

Nowhere do we find the age of the globe revealed either to Moses or any other inspired writer; we believe that as science has nothing to fear from the Bible, so the Bible has nothing to fear from the widest intellectual range of study. We ponder in devout amazement over these unwritten records of the earth's bygone history: we find 'sermons in stones' as we light on some delicate fern, or elegant vertebrate animal, preserved in the deposits of past ages, and the hieroglyphics of nature and the distincter utterances of the Bible prompt the same exclamation,—“How marvellous are thy works, O God, in wisdom hast thou made them all!” “Waste, and disorder, and confusion we nowhere find in our study of the crust of the earth; instead of this we find endless examples of economy, order, and design; and the result of all our researches carried back through the unwritten records of past time, has been to fix more steadily our assurance of the existence of one Supreme Creator of all things; to exalt more highly our conviction of the immensity of His perfections, of His might and majesty, His wisdom and His goodness, and all-sustaining providence; and to penetrate our

understandings with a profound and sensible perception of the high veneration man's intellect owes to God. The earth from her deep foundations unites with the celestial orbs that roll through boundless space, to declare the glory and show forth the praise of their common Author and Preserver; and the voice of natural religion accords harmoniously with the testimonies of revelation, in ascribing the origin of the universe to the will of one Eternal and Dominant Intelligence, the Almighty Lord and supreme First Cause of all things that subsist; the same yesterday, to-day, and for ever; before the mountains were brought forth, or ever the earth and the world were made, God from everlasting and world without end."—*Buckland's Bridgewater Treatise*.

"Come, frankly read the rocks, and see
In them the Earth's biography;
Let mountain piled on mountain tell
Its antique age; and every shell
In fossil form its tale unfold,
Of life's bright day through time untold;
And gathering use from great and small,
See good in each, but God in all."

CHAPTER II.

PRELIMINARIES.

“Of old hast thou laid the foundation of the earth.”

DAVID.

As yet we have only been *talking* about the crust of the earth; we shall now return and enter upon its actual examination. It will not be necessary for us personally to descend into the abysmous caverns that lie beneath our feet, nor, with hammer in hand, to go forth and explore the district of country in which we may happen to dwell: we may do all this by and by, when we know both *how* and *what* to observe. Meanwhile, with such teachers as Buckland, Sedgwick, Murchison, Pye Smith, Hugh Miller, De la Beche, Lyell, Owen and others, we may for some while to come be only tarry-at-home travellers; for in a true sense, in this department of knowledge, “other men have laboured, and we enter into their labours.” Let us now look at the crust of the earth, as it may be represented in two imaginary sections. Suppose we could make a vertical section of the earth’s crust, and cut straight down some eighty miles till we reached the central mass of incandescence that we believe lies beneath this crust, or Erdrinde (*earth-rind*), as the Germans call it, and then bring out this section to daylight, it would present something very much like the following appearance.

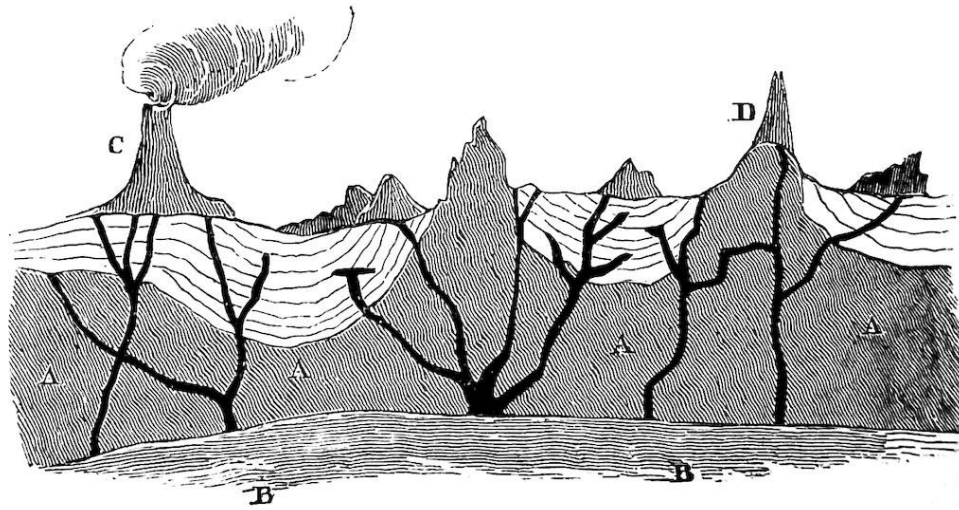


DIAGRAM I.

Here the granite A will be observed forming the supposed boundary between the superlying strata and the fire B below to which we have just referred, and thus will be seen the origin of all plutonic rocks. Here too will be seen how the granite is not confined to the lower levels, but rises, as mentioned in the first chapter, far above all the other strata, and forms some of the highest peaks on the face of the globe.^[6] Here, too, will be seen how the granite is frequently traversed by veins of trap-dykes, those black-looking branches, which rise often above the whole mass of metamorphic and stratified rocks, often occasioning great difficulties in mining operations. Here, too, the student will see how, supposing the theory of a central globe of heat to be founded in fact, the volcanoes that are now active, C, form, as the volcanoes that are extinct, D, once did, the safety valves of this mighty mass of incandescence, B; and in the same way may be seen how certain strata may be above the granite, or above any other formation, though they do not overlie them, and how the lowest strata, being formed first, is said to be older than any superlying strata, notwithstanding any accidental arrangement produced by upheaval or depression. For, in “consequence of the great commotions which the crust of the globe has undergone, many points of its surface have been elevated to great heights in the form of mountains; and hence it is that fossils are sometimes found at the summit of the highest mountains, though the rocks containing them were originally formed at the bottom of the sea. But, even when folded or partly broken, their relative age may still be determined by

an examination of the ends of their upturned strata, where they appear or crop out in succession, at the surface or on the slopes of mountains.”^[7]

But to make this view of the subject clearer, let us imagine that some Titanic power was granted us to push down these towering masses of granite to their original situation, below the metamorphic and stratified rocks, by which means we should at the same time restore these curved and broken strata to their originally horizontal position; and let us suppose that we were now again to descend to the foundations of the earth for the purpose of making another vertical section; then the crust of the earth would present to us an arrangement something like the leaves of a book, or the coats of an onion, arranged in successive and uninterrupted layers, or in concentric and unbroken circles. Such a diagram must of course be imaginary, and unless it is taken into connexion with the previous remarks, it is more likely to bewilder than to assist the beginner. Let it again be urged upon the reader, that such a chart as we are about to lay before him is only intended to give him an idea of the succession of these formations and systems, and that the details found in it are anticipatory of many future references to it on the student’s part. Let it not be supposed that this is a mere barren research into dry facts that have no connexion with our truest welfare; for if, as Lord Bacon somewhere finely observes, all study is to be valued “not so much as an exercise of the intellect, but as a true discipline of humanity,” then what study is calculated to be more useful than Geology, in enlarging and purifying the powers of the mind, by teaching us how harmonious, and orderly, and economic are the works of God; in removing all narrow notions of the extent and age of this solid globe, which from the beginning had its origin in the almighty will of God; in checking the presumptuous or the chilling inferences of a sceptical philosophy, by everywhere pointing out the design, skill, and adaptations of an ever-present and most beneficent Creator; and in chastening those overweening ideas of ourselves which both ignorance and knowledge may create and foster, by saying to us, in the language of God himself, as we stand amazed in the presence of huge pre-Adamite vestiges of creation, “Where wast thou,” vain man, “when I laid the foundations of the earth? Declare if thou hast understanding.” (Job xxxviii. 4.)

“Among these rocks and stones, methinks I see
More than the heedless impress that belongs
To lonely nature’s *casual* work; they bear

Here, in the words of another writer, we would add for the reader's guidance, that "the unstratified or igneous rocks occur in no regular succession, but appear amidst the stratified without order or arrangement; heaving them out of their original horizontal positions, breaking up through them in volcanic masses, and sometimes overrunning them after the manner of liquid lava. From these circumstances they are, in general, better known by their mineral composition than by their order of occurrence. Still it may be convenient to divide them into three great classes; *granite*, *trappean*, and *volcanic*—*granite* being the basis of all known rocks, and occurring along with the primary and transition strata; the *trappean*, of a darker and less crystalline structure than the granite, and occurring along with the secondary and tertiary rocks; and the *volcanic*, still less crystalline and compact, and of comparatively recent origin, or still in process of formation." This the student will observe by another reference to the previous diagram; but, in looking at the one now before him, we must also add for his further guidance,—for we are presuming that we address those who need initiation into the rudiments of this science, and the circumstance that we never met with a preliminary treatise that quite satisfied us, or helped such intelligent youth as were prying into the apparently cabalistic mysteries of the earth's structural divisions, is one strong inducement to the present undertaking;—we must add, that "it must not be supposed, however, that all the stratified rocks always occur in any one portion of the earth's crust in full and complete succession as represented" in Diagram II. "All that is meant is, that such would be their order if every group and formation were present. But whatever number of groups may be present, they never happen out of their regular order of succession; that is, clay-slate never occurs above coal, nor coal above chalk. Thus in London, tertiary strata occupy the surface; in Durham, magnesian limestone; in Fife, the coal measures; and in Perthshire, the old red sandstone and clay-slate; so that it would be fruitless to dig for chalk in Durham, for magnesian limestone in Fife, or for coal in Perthshire. It would not be absurd, however, to dig for coal in Durham, because that mineral underlies the magnesian limestone; or for old red sandstone in Fife, because that formation might be naturally expected to occur under the coal strata of that country, in the regular order of succession."^[8]

Still, after reading all this, we can easily imagine, not so much an air of incredulity taking possession of the countenance of our courteous reader as

a feeling somewhat like this, with which we have often come into contact in those geological classes of young persons which it has been our pleasure to conduct: “Well, all that’s very plain in the book; I see granite lies at the bottom, and pushes itself up to the top very often; and I see in the diagrams that coal and chalk are not found in the same place, and that different localities have their different formations, and the various formations have their different fossils, but I confess that I cannot realize it. I know the earth is round like an orange, a little flattened at the poles—what is called an oblate spheroid; but all this surpasses my power of comprehension; can’t you make it plainer?” Well, let us try; on page [27](#) is a diagram, representing no ideal, but an actual boring into the earth. London is situated on the tertiary formation, in what is called geologically the basin of the London clay, that is almost on the very top of the crust, or external covering that lies on the vast mass of molten and other matter beneath. Here is first a drawing and then a section that may represent this basin:—

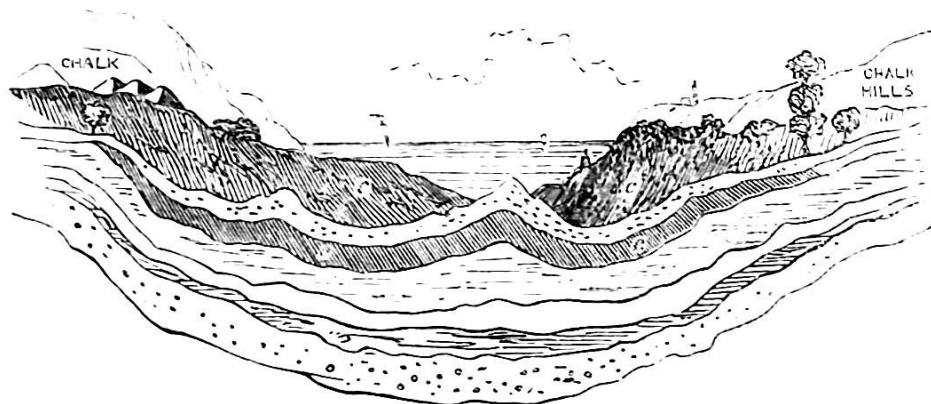


DIAGRAM III.

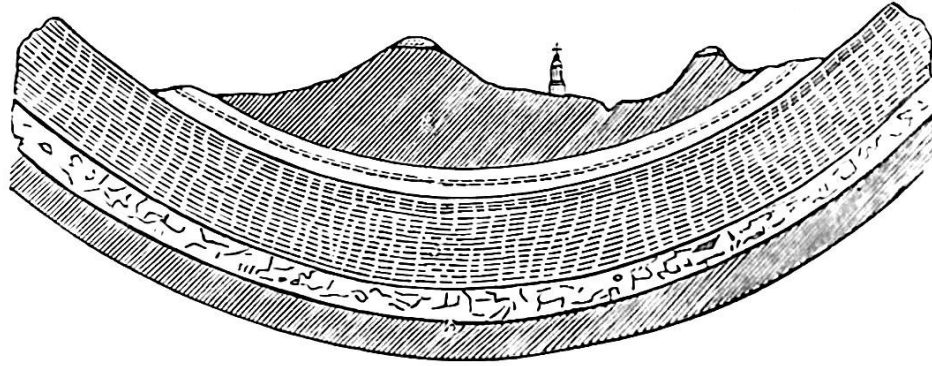


DIAGRAM IV.

The water which falls on the chalk hills flows into them, or into the porous beds adjoining, and would rise upwards to its level but for the superincumbent pressure of the bed of clay above it, *cccc*. Under these circumstances, in order to procure water, Artesian^[9] wells are sunk through the bed of clay, perhaps also through the chalk, but at any rate till the depressed stratum of chalk is reached; and this gives exit to the subterranean water, which at once rises through the iron tubes inserted in the boring to the surface. By these borings through the clay, water is obtained where it would be impossible to sink a well, or where the expense would prohibit the attempt. To explain this matter, here is a diagram (No. V.) which represents the Artesian well at the Model Prison at Pentonville, London, the strata upon which London is built, and which we can apply to the diagram on page [21](#), that the theory of the earth's crust may be the more thoroughly understood before we proceed.

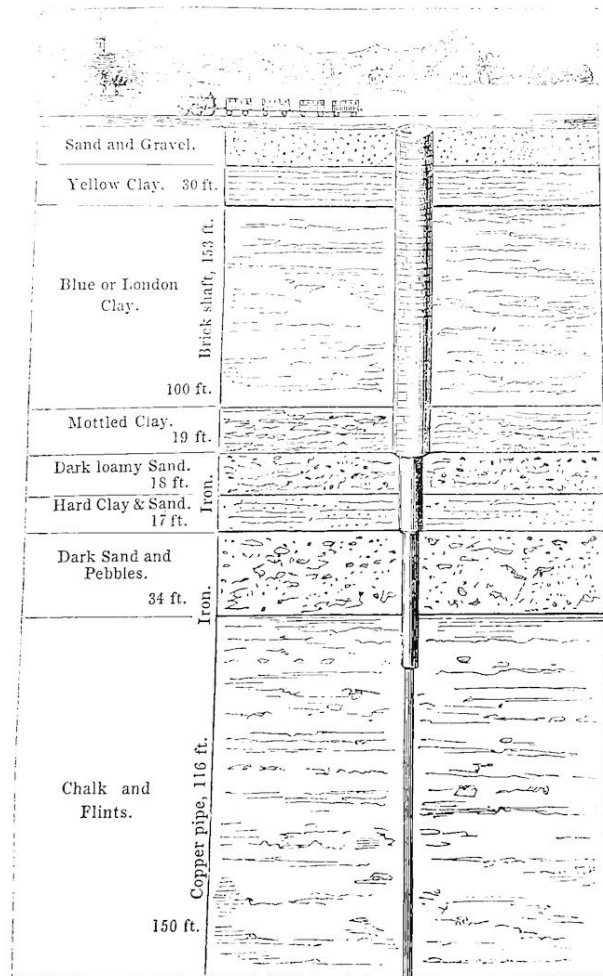


DIAGRAM V.

In the same manner Artesian wells have been sunk in other places, as at Hampstead Water Works, 450 feet deep; Combe & Delafield's, 500 feet deep; and the Trafalgar-square Water Works, 510 feet deep.^[10] Now, the reader has only to take this last diagram, and in imagination to apply it to the one on page 21, in order to see that so far as actual boring and investigation go, the geological theory of the earth's crust is correct; only again let it be observed that this order is never inverted, although it frequently happens that some one or more of the strata may be absent.

Hitherto we have spoken of the earth's crust without reference to that wondrous succession and development of living beings which once had their joy of life, and whose fossil remains, found in the different strata, waken such kindling emotions of the power of Deity, and enlarge indefinitely our conceptions of the boundless resources of His Mind. This

will open before us a new chapter in the history of our planet, already the theatre of such vast revolutions, and which, under the influence of Divine truth, is yet to undergo one greater and nobler than any of these. We have as yet only glanced at the surface page of the wondrous book, now happily opened for us by geologists, to whose names we have already made reference; and as the mind rests with intense pleasure on the discoveries of Champollion, Belzoni, Lane, Layard, Botta, and others who have deciphered the hieroglyphics, in which were written the wars and the chronicles of ancient nations, whose names and deeds are becoming, by books and lectures, and above all by our noble national Museum, familiar even to our children, and a source of help and solace to the hard-toiling artisan; so with profounder interest, as carried back into remoter ages of antiquity, so remote that they seem to lie beyond the power of a human arithmetic to calculate, do we humbly endeavour to decipher the hieroglyphics,^[11] not of Egypt or of Nineveh, but of the vast creation of God, written in characters that require, not only learning and science to understand, but modesty, patience, and triumphant perseverance. He who with these pre-requisites combines reverence for God and His revelation, will always find in Geology material both for manly exercise of thought, and also for reverent adoration of Him who is Himself unsearchable, and whose ways are past finding out.

“We not to explore the secrets, ask
Of His eternal empire, but the more
To magnify His works, the more we know.”—MILTON.

Most happily for Christendom, our noblest men of science are not ashamed of the “reproach of Christ;” and we know not how to conclude this chapter in a strain more accordant with our own thoughts than by quoting the words of an eminent living naturalist:—“I can echo with fullest truth the experience of Bishop Heber; ‘In every ride I have taken, and in every wilderness in which my tent has been pitched, I have found enough to keep my mind from sinking into the languor and the apathy which have been regarded as natural to a tropical climate.’ Nay, I may truly say, I found *no tendency* to apathy or ennui. Every excursion presented something to admire; every day had its novelty; the morning was always pregnant with eager expectation; the evening invariably brought subjects of interest fresh and new; and the days were only too short for enjoyment. They were not days of stirring adventure, of dangerous conflicts with man or with beast, or

of hair-breadth escapes in flood and field; their delights were calm and peaceful, I trust not unholy, nor unbecoming the character of a Christian, who has his heart in heaven, and who traces, even in earth's loveliest scenes, the mark of the spoiler. The sentiments expressed by my friend^[12] and fellow-labourer are those which I would ever associate with the study of science. 'If the sight of nature,' observes Mr. Hill, 'were merely the looking at a painted pageantry, or at a spectacle filling the carnal mind with wonder and delight, the spirit would be overpowered and worked into weariness; but it is admiration at the wisdom, and reverence for the beneficence of Almighty power. He who dwelleth in the light which no man can approach unto, whom no man hath seen, nor can see,' is yet visible in His perfections through the works of His hand, and His designs are made manifest in the purpose of His creatures. Wherever our lot is cast, into whatever scenes our wayward impulses lead us, the mind-illuminated eye gazes on divine things, and the spirit-stirred heart feels its pulses bounding with emotions from the touch of an ever-present Deity. The habit that sees in every object the wisdom and the goodness as well as the power of God, I may speak of, as Coleridge speaks of the poetical spirit, 'it has been to me an exceeding great reward; it has soothed my afflictions; it has multiplied and refined my enjoyments; it has endeared my solitude; and it has given me the habit of wishing to discover the good and the beautiful in all that meets and surrounds me.'

“Great are thy works, Jehovah, infinite
Thy power! what thought can measure thee, or tongue
Relate thee?”^[13]

CHAPTER III.

THE ANCIENT EPOCH.

“Where wast thou when I laid the foundations of the earth?”—*Job xxxviii. 4.*

And now in right earnest let us begin our examination of the earth's crust. Some of the terms we may use will, perhaps, at first sight appear repulsive from their novelty and difficulty; such words we will explain as we proceed, and will only stay the student's course to remark, that there is a necessity for the use of the dead languages in the formation of compound terms that are to become descriptive names, and in their application to newly discovered objects. This necessity arises from the fact that it is only in this way that scientific men of different nations can understand the character of each other's researches, and compare notes with one another. A fossil is found, let us suppose, in the lias formation; it proves to be the organic remains of some very strange and anomalous creature. People go down to Lyme Regis to examine it, and, in doing so, discover others. Comparative anatomists arrange the dislocated parts and give them a name; this must be intelligible to geologists on the Continent as well as in England; and therefore some term descriptive of the animal, once the living possessor of these “dry bones,” must be given, and finally it is called *ichthyosaurus*. Any one in Russia, or Austria, or Italy, who happened to be acquainted with the rudiments of Greek, would know at once the kind of animal referred to by its very name, derived from *ichthus*, a fish, and *sauros*, a lizard. This would indicate to all scientific men the nature of this remarkable animal, of which we shall have to tell some stories by and by as full of wonder as any modern or ancient book of marvels; while, if we had called it *fish-lizard*, only those who understood English would know what we meant. Our object is to simplify as much as possible every difficult term

that may be used; but while we solicit our readers to master each difficulty as it rises, we hope they will not think that, when they have read this little book, they are masters of Geology, our highest ambition being only to impart a *taste* for the science.

To return: our examination commences with the Plutonic rocks, so called in memory of the well-known mythological god of the fiery or infernal regions; and we take *granite*^[14] as a type of these rocks, because it is so familiar to all our readers. There are besides granite, syenite,^[15] greenstone, porphyry, basalt, and others, to dilate upon which would defeat our purpose. Our object is to lay but a little at a time upon the memory, and to let that little be well digested before we pass from the thoroughly known to the unknown. Nothing but actual examination can make the student familiar with the varieties of the rocks of this very ancient epoch in the world's history. Well, everybody knows what granite is; they see it on the kerbstones of the wayside, in the hard paving of the London streets, in the massive slabs of London and Waterloo Bridges, and elsewhere. "Granite!" exclaims the reader, "everybody knows what granite is, and there is an end of it; you make as much fuss about granite as Wordsworth did about his well-known primrose, and the man who could see nothing but a primrose in a primrose."

But there is a poetry and a history about granite upon which we are going to dwell. This piece of granite which I hold in my hand is composed of quartz, mica, and feldspar.^[16] The quartz is white and hard—I can't scratch it with my knife; the mica is in glistening plates or scales; and the feldspar is soft and greyish, and can easily be scratched. Oh, if this granite could speak, what a story could it tell! "To give it, then, a tongue were wise in man." Let us try. "Once upon a time, long, long ages ago, incalculable periods before Adam was placed in possession of Eden, I, the granite, and my contemporaries, came into being. Before us, this planet 'was without form and void.' A dark chaotic period, of which I know nothing, preceded me. When I first emerged into being, at the command of Him who laid the foundations of the earth, this world was a barren, lifeless, uncultivated, uninhabited, untrodden, seasonless waste. Here and there were undulations of land and water, but all was bare, desolate, and silent: not a moss nor a lichen covered the ancient skeleton of the globe; not a sea-weed floated in the broad ocean; not a trace existed even of the least highly organized animal or vegetable; everything was still, and with the stillness of universal

death. The earth was prepared, and the fiat of creation had gone forth; but there was no inhabitant, and no beings endowed with life had been introduced to perform their part in the great mystery of creation.”^[17] And the granite might go on to say—“Man! of three-score years and ten, where wast *thou* when He, my Maker and yours, laid the foundations of the earth? Let me tell you what an important part I have played in the history of your world’s formation. I rise to the highest elevations, and form the sublimest pinnacles on the surface of the globe, and without me your scenery would lose its grandeur and its glory. But for me Albert Smith had never climbed Mont Blanc, nor Humboldt Cotopaxi and Chimborazo; nor would the head of the famed Egyptian Memnon^[18] have been sculptured. You may see me giving to Cornwall, Wales, and Scotland their most valuable minerals and metals. In Europe

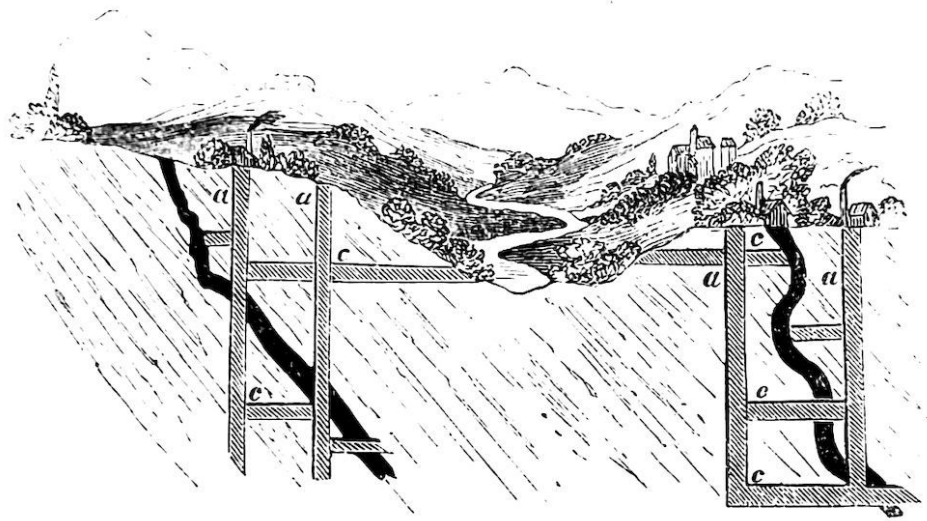
‘I am monarch of all I survey,
My right there is none to dispute.’

The Scandinavians, the Hartz mountains, the Alps, and the Pyrenees are mine; nor is my territory less in Asia, Africa, the great Americas, and in the becoming great Australia; and thus, by my deeply rooted foundations and my vast extension, I constitute the framework, solid and immoveable, of this ‘great globe and all that it inherits.’”

Thus, at any rate, the granite might speak, nor would there be one word of vain boasting in it. Having beard it, or fancied we heard it, which amounts to the same thing, let us soberize ourselves, and put granite into the third person. There are no fossils in granite and the other Plutonic and volcanic rocks; even supposing any forms of life to have been in existence at the period to which we are referring, the action of fire has annihilated all their remains. We should not therefore expect in Cumberland and Cornwall, nor in those parts of Devonshire where granite prevails, to find the fossils peculiar to other formations with which in time we hope to make familiar acquaintance. But though destitute of interest in this respect, how great is its importance and interest in those economic uses which have the geologist for their guide, and the whole family of man for their beneficent operations! “Many varieties of granite are excellent as building stones, though expensive in working to definite forms. Some of the most important public works of Great Britain and Ireland, France and Russia, are of this material. In selecting granite, those varieties in which the constituent minerals and

the scales of mica are superabundant, should be avoided; and, as a practical test, it is wise to notice the country immediately around the quarry, as the sandy varieties rapidly disintegrate,^[19] and form accumulations of micaceous sand. The Hayter or Dartmoor granite, the Aberdeen granite, the Kingstown (Dublin) granite, some beds of the Mourne or county of Down granite, and the Guernsey or Channel Island granite, are well known for their excellence. In some of the quarries the bedding of the granite is more defined than in others; and wherever this is the case, or where marked cleavages or joints prevail, the work is much facilitated. Many old Egyptian works and statues were formed of granite, and it is still used for colossal works, as it takes a fine polish. For example, the great fountain shell, or vase, before the Museum at Berlin, and the pedestal of Peter the Great at St. Petersburg, are of the northern granite, being sculptured from erratic blocks. The splendid Scotch granite columns, in the vestibule of the Fitzwilliam Museum at Cambridge, are beautiful examples of a modern application of this rock to the arts.”^[20]

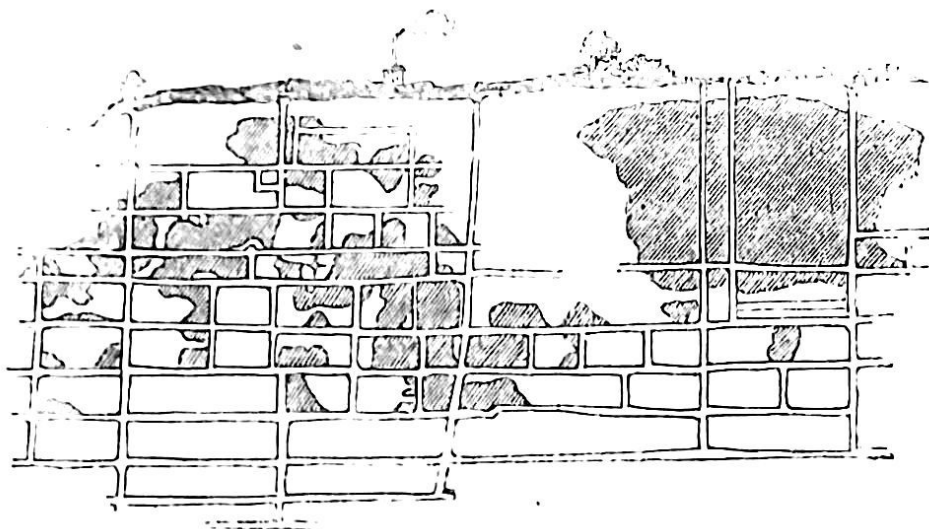
It is also in the Plutonic or igneous rocks that almost all the metals are found; and here we have our first illustration of that order to which we shall frequently call attention; an order as exquisite as can be found in the drawers of a lady's cabinet, forbidding the thought that anything observable at the present time, in the bowels or on the surface of the crust of the earth, can be attributed to the violent diluvial action of the Noachian deluge. The diagram below represents an ideal section of a mining district.



SECTION OF A MINING DISTRICT.

Here the metalliferous vein, we may suppose, has cropped out on the surface of the ground, or, as the miners say, has “come above grass.” Let us now suppose that the position of this vein of ore, copper, lead, or tin, has been ascertained—that is, how it runs, whether from north to south, or from east to west—and also that the “captains” of the mining district around have given their opinion as to the extent and thickness of this underground vein. The next thing is to obtain this mineral wealth. For this purpose shafts (*a a a*) must be sunk, which must reach the vein at a certain depth; then will probably follow cross-cuts (*c c c*), called adit levels (technically an “additt”), driven, as may be seen, at the lowest convenient point above the level of the highest water of the valley; and these, in connexion with the shaft, will serve the purpose of draining the mine and carrying the ore above ground. It will also be seen, by reference to the diagram, that the shafts of a mine do not always correspond; sometimes they are sunk vertically to meet the vein, sometimes they are commenced in the very outcrop itself. On this matter the best geological lesson is a visit to Cornwall, where the student will see that everything depends on the locality of a mine, the nature of the slope of the hill, or the character of the rock in which the vein appears, and so on. “The act of sinking a perpendicular shaft downwards to a depth where it is calculated the lode should be cut, may seem to require little further skill than is necessary to determine correctly the spot on the surface where the work is to commence. But the process in this way is exceedingly tedious; and in a mine at work, where many

galleries already existing are to be traversed, much greater rapidity is desirable. In such a case the shaft is sunk in several *pieces* (see diagram below), or, in other words, the sinking is commenced at the same time in different levels; and no small skill is required to lay out the work, so that the different portions of the shaft thus formed may exactly fit when they are joined together. An exceedingly small error of measurement, in any one of these various and dark subterranean passages, would, in fact, be sufficient to throw the whole into confusion; but such an accident rarely happens, although works of the kind are common in the Cornish mines.”^[21] As an illustration of the immense quantity of water in the mines, we may add—and this is almost as startling as any romantic fiction—that the various branches of the principal level in Cornwall, called “the Great Adit, which receives the waters of the numerous mines in Gwennap, and near Redruth, measure on the whole about 26,000 fathoms, or nearly thirty miles in length; one branch only, at Cardrew mine, extends for nearly five miles and a half, and penetrates ground seventy fathoms beneath the surface. The water flows into a valley communicating with a small inlet of the sea, and is discharged about forty feet above high-water mark.”^[22] In this method about forty millions of tons of water are raised by steam-power out of the mines in Cornwall.



EAST WHEAL CROFLY COPPER MINE, CORNWALL.

Here, then, we have seen two of the economic uses of geology in connexion with *granite* alone; and as we think of these mineral treasures, requiring only the labour and skill of man to bring them out for his service and for the civilization of the world, our boast is in our native land, which, though insular and small, combines within itself everything needful to develop its three sources of national wealth—mining, manufactures, and agriculture—to their highest point. Our boast is not the warrior’s boast, which Shakspeare puts into the mouth of one of his heroes—that this our isle is

“That pale, that white-faced shore,
Whose foot spurns back the ocean’s roaring tides,
And coops from other lands her islanders”—

but rather that, without impropriety or irreverence, the words of Holy Writ may as legitimately be applied to Great Britain as to Palestine. It is a land wherein “thou shalt eat bread without scarceness, thou shalt not lack anything in it; a land whose stones are iron, and out of whose hills thou mayest dig brass.^[23] When thou hast eaten and art full, then thou shalt bless the Lord thy God for the good land which He hath given thee.” (Deut. viii. 9, 10.)

But before we bring this chapter on granite and its kindred rocks to a close, we must glance at one more purpose served by this Plutonic rock. Here is a teacup, and here is a piece of granite: the one comes from Cornwall, the other is made in Staffordshire or Worcestershire. What relation have they to each other? If it were not thought *infra dig.*, we should say the granite is the parent of the teacup. In Cornwall, especially in the neighbourhood of St. Austel, the writer has lately visited what are called the China clay works. “The granite is here in a state of partial decomposition. In some localities, this *growan*” (Cornish for disintegrated granite) “is tolerably firm, when it resembles the Chinese *Kaolin*, and, quarried under the name of China stone, is extensively employed in the potteries. This is ready for the market when cut into blocks of a size convenient for transport; but the softer material, which is dug out of pits, and called *China clay*, or *porcelain earth*, requires a more elaborate preparation for the purpose of separating the quartz, schorl, or mica from the finer particles of the decomposed feldspar. This clay is dug up in *stopes* or layers, which resemble a flight of irregular stairs. A heap of it is then placed upon an

inclined platform, under a small fall of water, and repeatedly stirred with a *piggle* and shovel, by which means the whole is gradually carried down by the water in a state of suspension. The heavy and useless parts collect in a trench below the platform; while the China clay, carried forward through a series of *catch-pits* or tanks, in which the grosser particles are deposited, is ultimately accumulated in larger pits, called *ponds*, from which the clear supernatant water is from time to time withdrawn. As soon as these ponds are filled with clay, they are drained, and the porcelain earth is removed to *pans*, in which it remains undisturbed until sufficiently consolidated to be cut into oblong masses. These are carried to a roofed building, through which the air can freely pass, and dried completely for the market. When dry they are scraped perfectly clean, packed in casks, and carried to one of the adjacent ports, to be shipped for the potteries.”^[24] As furnishing some idea of the extent to which this business is carried on, it may be added that 37,000 tons of this China clay are annually shipped from the south-west of England to the potteries, the value of which is upwards of £50,000, while the number of working men and women thus employed is beyond calculation. This is one of the practical results of geology. This is one of the things which geology, once a neglected and unpopular science, has done for our comfort and welfare. “A hundred years ago, it does not seem that any part of this China clay was made use of, or that this important produce was then of any value whatever.”^[25]

We bring this chapter to a close. Granite and its kindred rocks should stand associated with an actual history and poetry, not inferior to the history and poetry of man’s own handiwork; and we believe geology, so often regarded with dread by the uninitiated, will soon be considered worthy a patient and painstaking investigation. Remembering that geology is still an incomplete science, and that we have much yet to learn concerning the laws of organic and inorganic matter, we should be modest in the maintenance of any theory, while thankful for the acquisition of any fact. “We have yet to learn whether man’s past duration upon the earth—whether even that which is still destined to him—is such, as to allow him to philosophise with success on such matters; whether man, placed for a few centuries on the earth as in a schoolroom, has time to strip the wall of its coating and count its stones, before his PARENT removes him to some other destination.”^[26]

CHAPTER IV.

THE PALÆOZOIC PERIOD.

“In His hand are the deep places of the earth.”—DAVID.

Trench, in his charming little book on the “Study of Words,” says of words that they are “fossil poetry.” He adds, “Just as in some fossil, curious and beautiful shapes of vegetable or animal life, the graceful fern or finely vertebrated lizard, such as now it may be, have been extinct for many thousands of years, are permanently bound up in the stone, and rescued from that perishing which would otherwise have been theirs; so in words are beautiful thoughts and images, the imagination and the feeling of past ages, of men long since in their graves, of men whose very names have perished—these, which would so easily have perished too, are preserved and made safe for ever.”

Geology is the fossil poetry of the earth; such a poetry as those can never dream of who in a pebble see a pebble and nothing more. But to those who walk through this great and beautiful world intent upon finding material for thought and reflection, there is no “picking up a pebble by the wayside without finding all nature in connexion with it;” and the most retired student, in search not simply of the picturesque or of the beautiful, but of anything and everything that can minister to his profounder worship of Him to whom belongeth both “the deep places of the earth and the strength of the hills,” may say of his solitary rambles:—

“There rolls the deep where grew the tree;—
O earth, what changes hast thou seen!
There where the long street roars, hath been
The stillness of the central sea.”[\[27\]](#)

We now enter upon the ancient life, or Palæozoic period of the earth's history, and proceed to examine the oldest forms of life, or the most ancient organic remains found in the crust of the earth. As we do not aim to *teach* geology in this small work, but simply to place the chief geological facts in such a light as to impart a *taste* for the science, the reader will not expect any minute details, which are more likely to perplex than to assist the beginner. Let the reader dismiss from his mind all that he has tried to remember about Upper and Lower Silurian rocks, and the Upper and Lower Ludlow rocks, the Caradoc sandstone and the Llandilo flags, and so on; let us simply say that one part of the crust of the earth, supposed to be between 50,000 and 60,000 feet in thickness,^[28] is called the Silurian system, and constitutes a large and interesting part of the Palæozoic period. The term Silurian was given to this part of the earth's crust in consequence of these rocks being found chiefly in Wales, Devon, and Cornwall—parts of England once inhabited by the Silures, who under Caractacus made so noble a stand against the Romans.

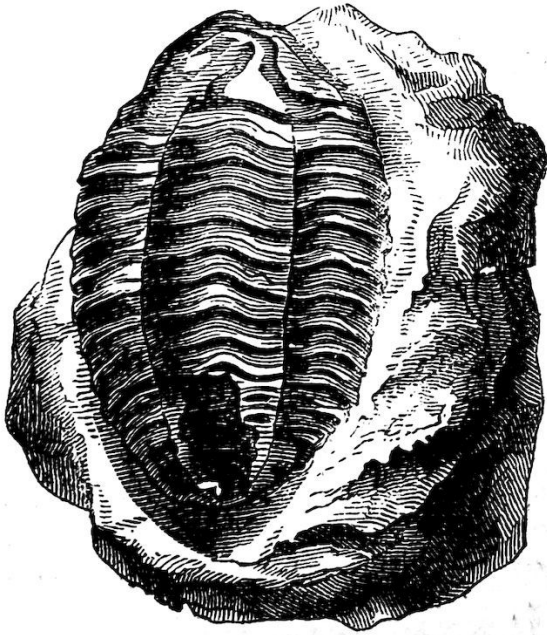
In coming for the first time into contact with the organic remains of pre-Adamite creations, it may be well to entreat the student to mark, as he goes on, the very different and characteristic fossils of the several formations through which we propose to travel. There will be little or no difficulty in doing this, and its mastery will be of invaluable service in our after researches. There is and there can be no royal road to any kind of learning; all, therefore, that we propose to do is to take a few of the big stones, boulders, &c., that have needlessly been allowed to make the road rougher than necessary, out of the way, that thus our companion traveller on this geologic route may feel that every step of ground walked over is a real and solid acquisition. In marking the characteristic fossils of each formation, let us suggest, in passing, the vast amount of pleasure there is in going to a friend's house, and looking at the minerals or organic remains that may be in the cabinet or on the mantel-shelf, and being able to take them up one by one, and to say this is from the Silurian; that is from the Carboniferous; this is from the Cretaceous, and that from the Wealden formations, and so on. Why, it gives a magical feeling of delightful interest to every object we see, and will always make a person a welcome visitor with friends with whom, instead of talking scandal, he can talk geology. Not long since the writer had a very pleasing illustration of this. He had been lecturing on geology in a small agricultural village; there was a good sprinkling of smock-frocks

among the hearers, and he said at the close of one of the lectures, “Now, very likely most of you have got some stones, as you call them, at home on the chimneypiece; perhaps you don’t know their names, or what they were before they became stones; well, bring them next week, and we will do our best to name them for you!” Next week, after the lecture, up came one, and then another, and then a third, and so on; and diving their hands down into the old orthodox agricultural pocket, brought out a variety of specimens, some of them very good indeed, which had been “picked up” by them in the course of their labour, and which, supposed to be “*rather kūrüss*,” had been carefully conveyed home. When these matters were given a “local habitation and a name,” the delight of many was most gratifying.

Now, all this is only just the application of M. Cousin’s words in relation to physical geography: “Give me the map of a country, its configuration, its climate, its waters, its winds, and all its physical geography; give me its natural productions, its flora and fauna, and I pledge myself to tell you *à priori* what the inhabitants of that country will be, and what place that country will take in history, not accidentally, but necessarily; not at a particular epoch, but at all periods of time; in a word, the thought that country is formed to represent.”

These remarks furnish us with a clue. Each formation has its own peculiar and characteristic fossils, and these fossils are arranged with as much care, and preserved as uninjured, as if they had been arranged for a first-class museum. But before proceeding on this fossiliferous tour, we may anticipate a question that may possibly be asked on the threshold of our inquiries, and into which we propose going fully in the sequel of this volume. It may be asked, “Were not these fossils placed in the rocks by the Deluge?” To this, at present, we answer, that so partial and limited was the character of the Deluge, being confined to just so much of the earth as was inhabited by man, and so brief was its duration, compared with the vast geological epochs we shall have to consider, that we do not believe we have one single fossil that can be referred to the Noachian deluge; and before we close, we trust it will have been made evident to every careful reader that fossils, as records of Noah’s flood, are an impossibility; and that the vast antiquity of the globe, taken into connexion with the prevalence of death on a most extensive scale, ages and ages previous to the creation of man, can alone account for our innumerable treasures of the “deep places of the earth.”

1

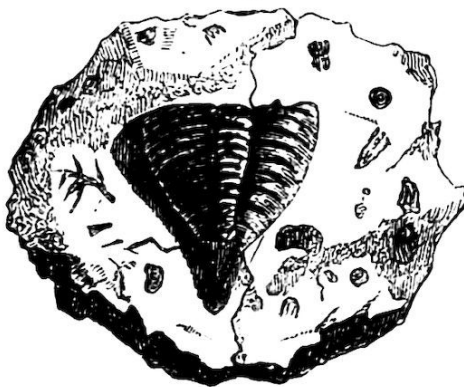


1

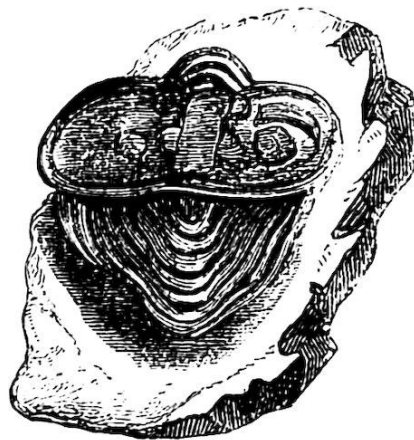
The characteristic fossils of the Silurian system are entirely unique. The trilobite may fairly be regarded as the prominent one; besides which there are orthoceratites, and graptolites, some members of the crinoidean family, with different kinds of corallines, and some other names to be rendered familiar only by future further study. We shall confine ourselves to those that our own recent researches have made us familiar with. First, here is the trilobite. We need not perplex our readers by any of the numerous subdivisions of this remarkable animal's nomenclature; that would defeat the purpose of this book. Any work on geology will do this.^[29]

Here are three trilobites: one (1) by itself; another, (2) imperfect in its bed or matrix, and a third (3) rolled up.

2

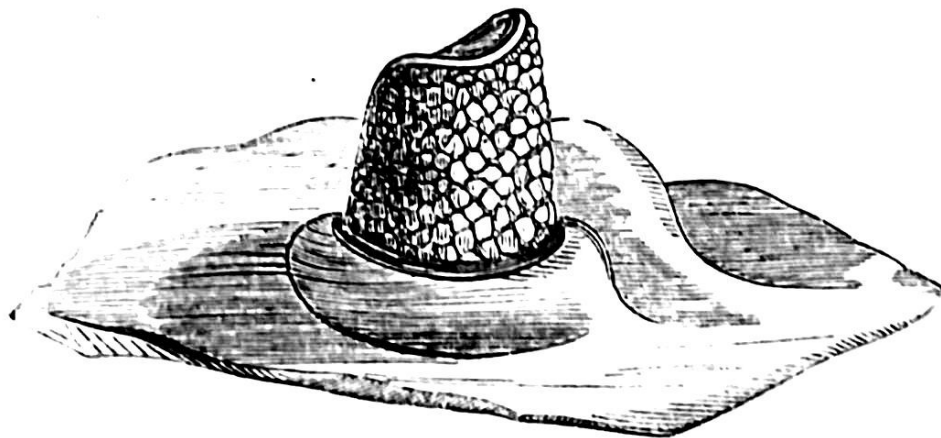


3



This most remarkable crustacean possessed the power of rolling itself up like the wood-louse or the hedgehog; and, reasoning by analogy, we suppose this to have been its defence against its numerous enemies. It is a

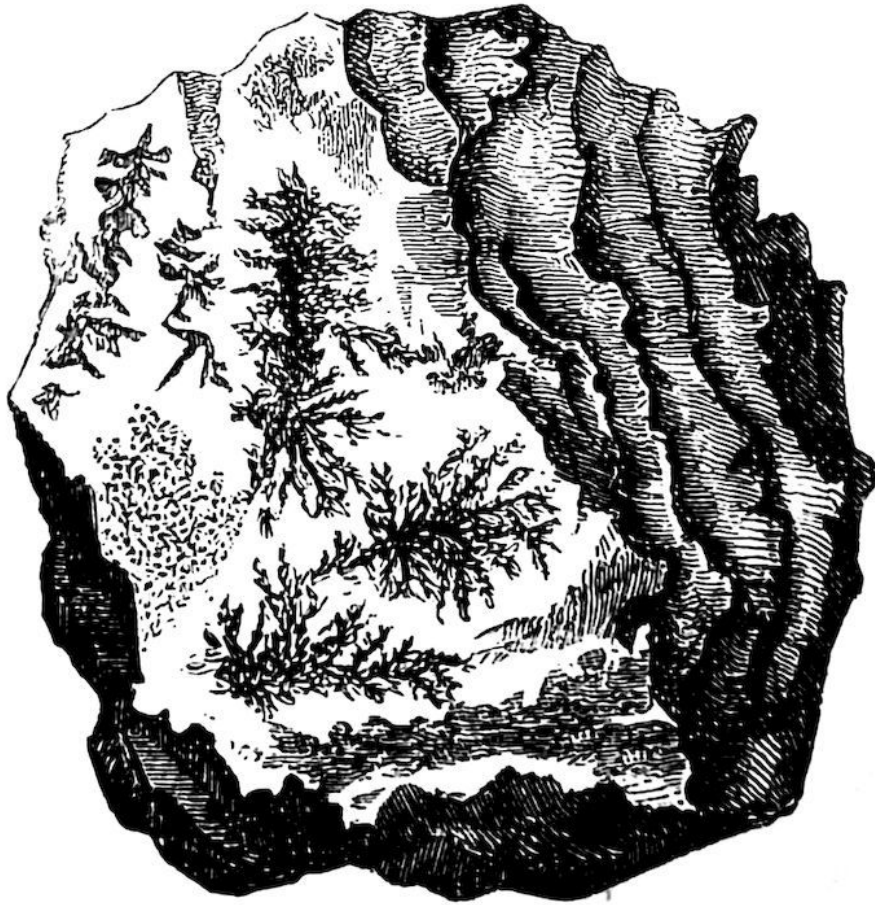
very abundant fossil, found all over Europe, in some parts of America, at the Cape of Good Hope, but never in more *recent strata* than the Silurian. The hinder part of the body is covered with a crescent-like shield, composed of segments like the joints of a lobster's tail; and two furrows divide it into three lobes, whence its name.^[30] Most remarkable are the eyes of this animal, and it is the only specimen in the vestiges of ancient creations in which the eye, that most delicate organization, is preserved; and if, as we believe, this little creature was living and swimming about, now and then fighting with some greater Cephalopodous mollusk, millions and millions of years ago, then in this fact we have the real fossil poetry of science, the romance of an ancient world which geology reveals to our delighted and astonished minds. From Buckland's Bridgewater Treatise we give a drawing of the eyes of the trilobite; and in Buckland's words we add: "This point deserves peculiar consideration, as it affords the most ancient, and almost the only example yet found in the fossil world, of the preservation of parts so delicate as the visual organs of animals that ceased to live many thousands, and perhaps millions of years ago. We must regard these organs with feelings of no ordinary kind, when we recollect we have before us the identical instruments of vision through which the light of heaven was admitted to the sensorium of some of the first created inhabitants of our planet."^[31]



But these are not the only fossils, or organic remains, to be found in the clay, slates, &c., of the Silurian system. Passing by those we have briefly indicated above, there are others of a highly interesting character,

concerning some of which we proceed to give a brief history. Being in Cornwall a short time since, we made a visit to Polperro, a romantic but out-of-the-way town on the south-west coast, for the purpose of procuring some remains of fossil fish considered characteristic of the Silurian system of Murchison, and which have been recently discovered by Mr. Couch, an eminent local naturalist, in the cliffs east and west of that town. We did not see Mr. Couch, but found our way to a coast-guardsmen, also a naturalist, whom we found to be a most skilful bird and fish stuffer, and a ranger for objects of natural history among the surrounding clay-slates and other rocks. William Loughrin's collection of Cornish curiosities will well repay any traveller going out of the way twenty or thirty miles, and they will find in him a fine specimen of an intelligent and noble class of men. Below we give some specimens from the Polperro slate. No. 1 might be taken for impressions of sea-weed, so remarkably does it resemble the sea-weed thrown up on our beaches; but it is generally conceded that this is merely a crystallization of oxydized matter, such as may often be found in connexion with manganese.

No. 1.

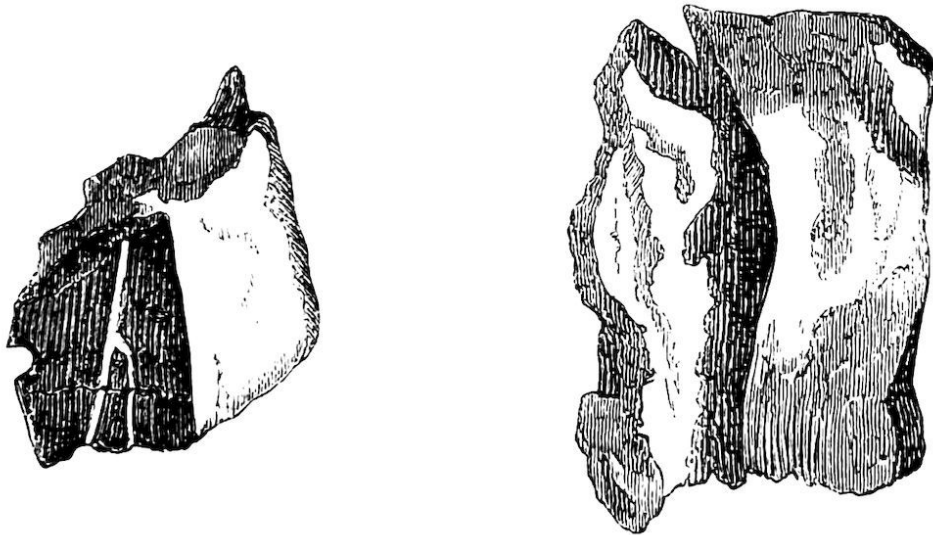


No. 1.

No. 2 is the Bellerophon,^[32] a shell which we shall afterwards find in the mountain limestone, but which is rare in connexion with the Silurian rocks.

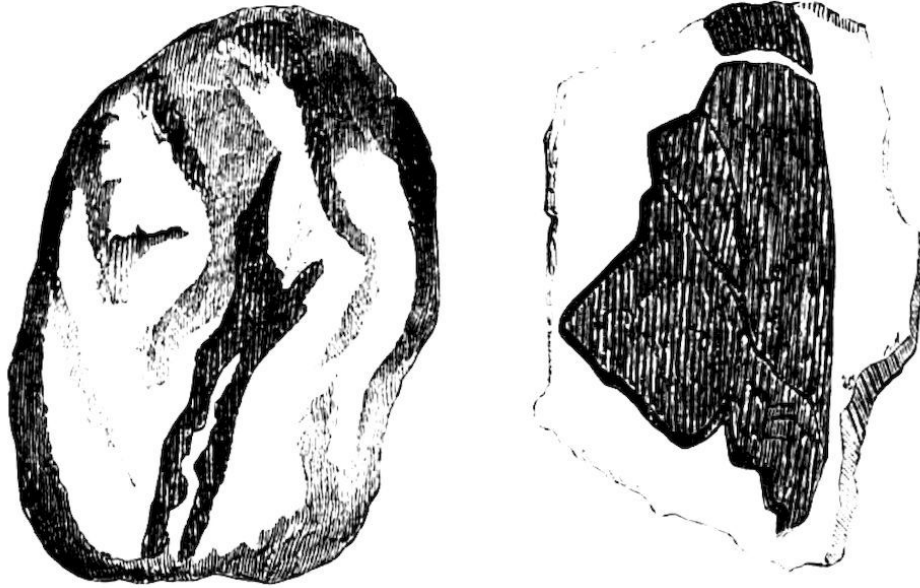


No. 2. BELLEROPHON, a shell which seems to have been abundant.

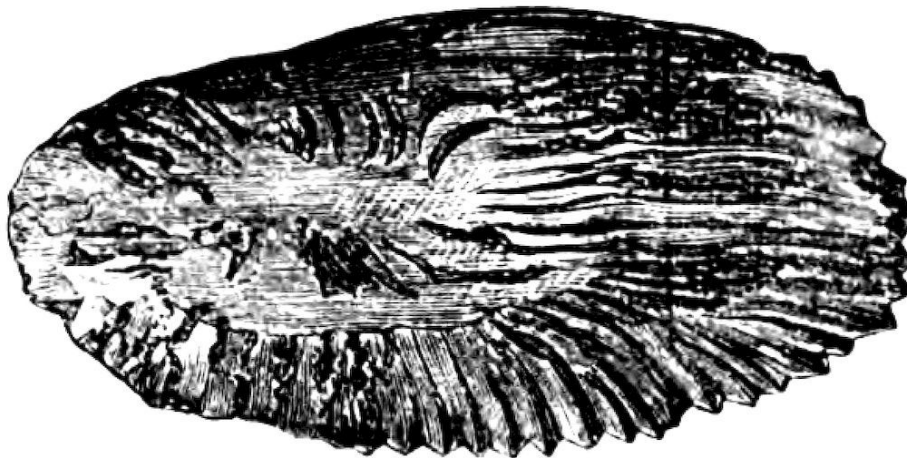


No. 3. REMAINS OF VEGETABLE TEXTURE.

No. 3 we know not how to describe. We are not certain what organic remains these are; so far as we have been able to examine them, they appear to us the remains of succulent vegetables, (?) probably the thick, soft stems of sea-weed, that may once have reposed in quiescence on the mud of which these slates are composed, and afterwards have been crushed by the superposition of mud and shale, until in the course of ages, by upheaval and depression, they have become a second time visitants of our atmosphere, and now expose themselves to our study and speculations.



No. 3. (Portion magnified two natural sizes.)



No. 4. CORALLINE. (Natural size.)

Here is one more form of life of this ancient period; it is evidently a coralline, which we also procured at Polperro.

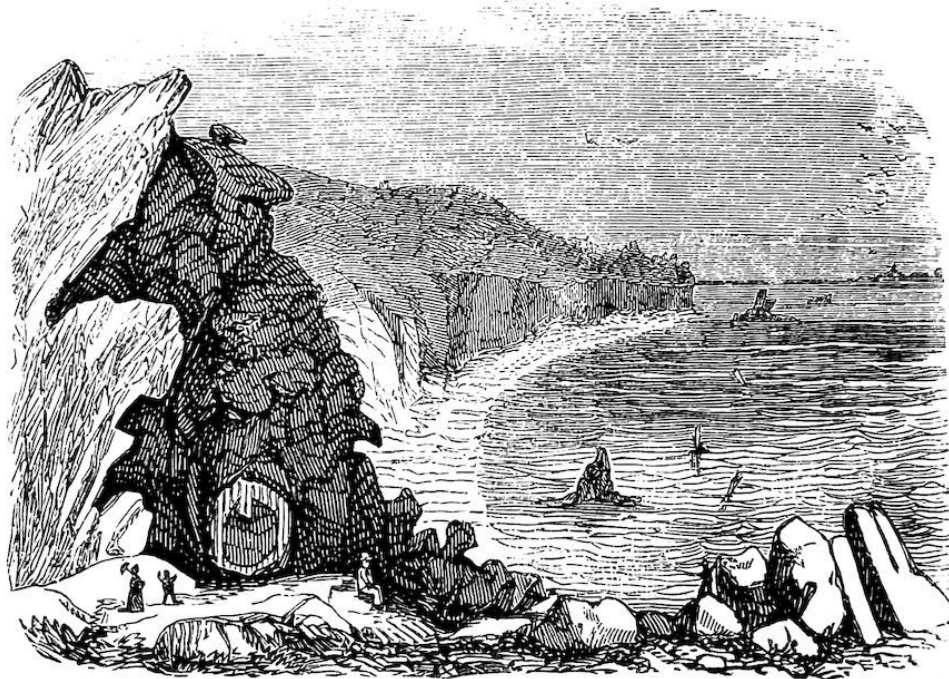
Let us suppose our readers to have made themselves familiar with these organic remains, simply as characteristic and illustrative of this formation; they will easily find their way into other traces and remnants of ancient life in the Silurian epoch. How absurd must seem the development hypothesis to those who rightly ponder these old, old vestiges! It seems to us a very idle idea to suppose that a trilobite could develop itself into a bird, or a

monkey, or by any series of happy accidents, could become a man;^[33] yet such has been the theory of those who overlook what some writer on geology, whose name we forget, has expressed strongly in these words: “There is no fact which has been demonstrated more completely to the satisfaction of every man of real science, than that there is no known power in nature capable of creating a new species of animal, or of transmuting one species into another.”

We close this chapter on the Silurian system in the eloquent words of Professor Sedgwick: “The elevation of the faunas of successive periods was not made by transmutation, but by creative additions, and it is by watching these additions that we get some insight into nature’s true historical progress. Judging by our evidence—and what else have we to judge by?—there was a time when Cephalopods were the highest type of animal life. *They* were then the Primates of this world, and, corresponding to their office and position, some of them were of noble structure and gigantic size. But these creatures were degraded from their rank at the head of Nature, and *Fishes* next took the lead; and they did not rise up in nature in some degenerate form, as if they were only the transmuted progeny of the Cephalopods, but they started into life in the very highest *ichthyic* type ever reached.

“Following our history chronologically, *Reptiles* next took the lead, and, with some evanescent exceptions, they flourished during the countless ages of the secondary period as the lords and despots of the world: and they had an organic perfection corresponding to their exalted rank in Nature’s kingdom; for their highest orders were not merely great in strength and stature, but were anatomically raised far above any forms of the Reptile class now living in the earth. This class, however, was in its turn to lose its rank. *Mammals* were added next (near the commencement of the tertiary period), and seem to have been added suddenly. Some of the early extinct forms of this class, which we now know only by ransacking the ancient catacombs of Nature, were powerful and gigantic, and we believe well fitted for the place they filled. But they in turn were to be degraded from their place in Nature, and she became what she now is by the addition of man. By this last addition she became more exalted than before. Man stands by himself, the despotic lord of the living world; not so great in organic strength as many of the despots that went before him in Nature’s chronicle, but raised far above them all by a higher development of brain, by a

framework that fits him for the operations of mechanical skill, by superadded reason, by a social instinct of combinations, by a prescience that leads him to act prospectively, by a conscience that makes him amenable to law, by conceptions that transcend the narrow limits of his vision, by hopes that have no full fruition here, by an inborn capacity of rising from individual facts to the apprehension of general laws, by a conception of a cause for all the phenomena of sense, and lastly, by a consequent belief in the God of nature:—such is the history of nature.”^[34]



LANDS END, CORNWALL.

CHAPTER V.

THE OLD RED SANDSTONE.

“The fishes of the sea shall declare unto thee.”—JOB.

Lord Bacon remarks, “Some men think that the gratification of curiosity is the end of knowledge, some the love of fame, some the pleasure of dispute, and some the necessity of supporting themselves by knowledge; but the real use of all knowledge is this, that we should dedicate that reason which was given us by God to the use and advantage of man.” The historian of the old red sandstone, Hugh Miller, to whose researches not only we, but such men as Murchison, Lyell, Ansted, Agassiz and others, are so exclusively indebted, is a philosopher in this last category. He does not hesitate to tell us, how, as a Cromarty quarryman “twenty years ago,” he commenced a “life of labour and restraint,” a “slim, loose-jointed boy, fond of the pretty intangibilities of romance, and of dreaming when broad awake;”^[35] and how, as a quarryman, he ever kept his eyes open, to observe the results of every blow of the hammer, stroke of the pick, or blast of the powder; and finding himself in the midst of new and undreamt-of relics of an old creation, preserved in “tables of stone,” he adds *his* testimony to that of the great father of inductive philosophy, “that it cannot be too extensively known, that nature is vast and knowledge limited, and that no individual, however humble in place or acquirement, need despair of adding to the general fund.”^[36]

We here enter upon a marvellous field of discovery. Hitherto the forms of life we have met with have all been invertebrate. The trilobite, something between a crab and a beetle, once revelling, in untold myriads, probably on the land as well as in the water, and of which two hundred and fifty species have been brought to light, is the highest type of life with which our

researches have made us familiar. We are now to begin the study of fossil fish, and to their discovery, strange forms, and characters, this chapter will be specially devoted. It was once a generally received opinion among even the most learned geologists, that the “old red sandstone,” or the “Devonian system,” was particularly barren of fossils, but the labours (literally such, “mente, manu, malleoque”^[37]) of Hugh Miller have proved the contrary. “The fossils,” he says, “are remarkably numerous, and in a state of high preservation. I have a hundred solid proofs by which to establish the proof of my assertion, within less than a yard of me. Half my closet walls are covered with the peculiar fossils of the lower old red sandstone; and certainly a stranger assemblage of forms have rarely been grouped together; creatures whose very type is lost, fantastic and uncouth, and which puzzle the naturalist to assign them even their class; boat-like animals, furnished with oars and a rudder; fish plated over like the tortoise, above and below, with a strong armour of bone, and furnished with but one solitary rudder-like fin; other fish less equivocal in their form, but with the membranes of their fins thickly covered with scales; creatures bristling over with thorns, others glistening in an enamelled coat, as if beautifully japanned, the tail in every instance among the less equivocal shapes, formed not equally as in existing fish, on each side the central vertebral column, but chiefly on the lower side, the column sending out its diminished vertebræ to the extreme termination of the fin. All the forms testify of a remote antiquity—of a period whose fashions have passed away.”^[38]

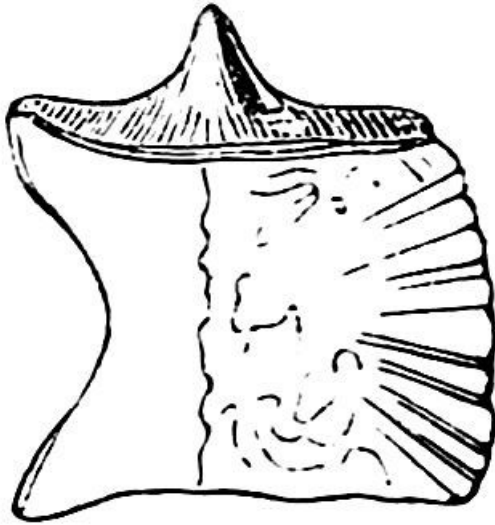
The old red sandstone formation prevails in the north of Scotland, Herefordshire, north of Devonshire, part of Cornwall, and in Worcestershire and Shropshire. Our attention will be principally confined to Cromarty, whose romantic bay and high hills have long arrested the admiring gaze of the traveller. This was the scene of Hugh Miller’s labours and discoveries; this the great library in which he read the history of pre-Adamite ichthyolites^[39] exposed not only to the light of day, but for the first time to the inspection of human eyes, by the sweat-of-brow toil of one of Scotland’s noble sons. Before we get into the hard names that must be connected with this chapter, let us hear Mr. Miller describe this library of God’s books that was so long his wonder and his study in Cromartyshire. “The quarry in which I wrought lay on the southern shore of a noble inland bay, or frith rather, with a little clear stream on the one side, and a thick fir-wood on the other. Not the united labours of a thousand men for a thousand

years could have furnished a better section of the geology of this district than this range of cliffs; it may be regarded as a sort of chance dissection on the earth's crust. We see in one place the primary rock, with its veins of granite and quartz, its dizzy precipices of gneiss, its huge masses of hornblend; we find the secondary rock in another, with its beds of sandstone and shale, its spars, its clays, and its nodular limestones. We discover the still little known, but very interesting fossils of the old red sandstone in one deposition; we find the beautifully preserved shell and lignites of the lias in another. There are the remains of two several creations at once before us. The shore, too, is heaped with rolled fragments of almost every variety of rock,—basalts, ironstones, hyperstenes, porphyries, bituminous shales, and micaceous schists. In short, the young geologist, had he all Europe before him, could hardly choose for himself a better field. I had, however, no one to tell me so at the time, for geology had not yet travelled so far north; and so, without guide or vocabulary, I had to grope my way as best I might, and find out all its wonders for myself. But so slow was the process, and so much was I a seeker in the dark, that the facts contained in these few sentiments were the patient gatherings of years.”^[40]

Now with regard to the hard names to which we have just made allusion—names that, apart from their etymology, which is nothing more than “sending vagrant words back to their parish,” are enough to startle any one; names such as heterocercal, homocercal, cephalaspis, pterichthys, coccosteus, osteolepis, &c. &c.—why, they will all presently become plain, and, we hope, familiar to our readers. “They are,” says Hugh Miller, “like all names in science, unfamiliar in their aspect to mere English readers, just because they are names not for England alone, but for England and the world. I am assured, however, that they are all composed of very good Greek, and picturesquely descriptive of some peculiarity in the fossils they designate.”^[41]

The rest of this chapter will be occupied with an account of the four most remarkable and characteristic fishes of this formation, to understand which a few preliminary remarks are necessary. Cuvier divided all fish into two groups, the *bony* and the *cartilaginous*; and these two groups he subdivided into two divisions, characterised by differences in their *fins*, or organs of locomotion, one of which he called Acanthopterygian,^[42] (thorny-finned,) and the other, Malacopterygian,^[43] (or soft-finned.) This concise arrangement did not, however, meet all the wants of the fish-students, and it

was often practically difficult to know under which class to arrange particular specimens. More recently M. Agassiz has arranged fish, not according to their *fins*, but according to their *scales*; and simple as this classification may seem, it is one of the greatest triumphs of genius in modern times, inasmuch as all fishes extinct and existing, that have inhabited or are inhabiting the “waters under the earth,” may be grouped easily under the following four divisions:—



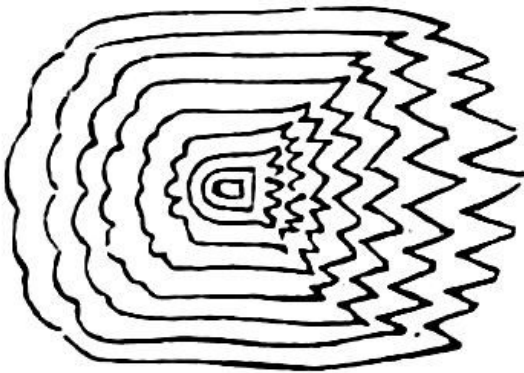
1. GANOID SCALE; as bony pike.^[44]

One more preliminary remark, and we will proceed to look at the four fishes already alluded to. Neither the teacher nor the student of any science can skip definitions, axioms, postulates, and so on; they must just be mastered, and their mastery is a real pleasure. In addition to a marked difference in the *fins*, a difference was observed also in the *tails* of fossil (extinct) and living pieces of fish. This difference between the tails of fish has been happily described in two words, heterocercal and homocercal, of which the figures below will give

a better idea than a lengthened description.



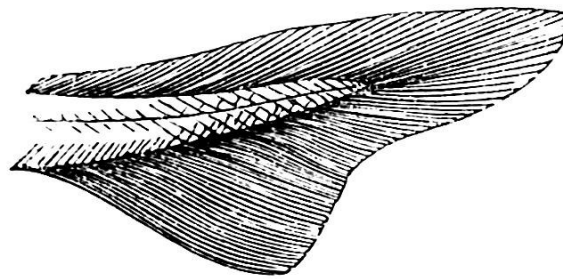
2. PLACOID SCALE.^[45]



3. CTENOID SCALE; as sole or perch.
^[46]



4. CYCLOID SCALE; as herring.^[47]

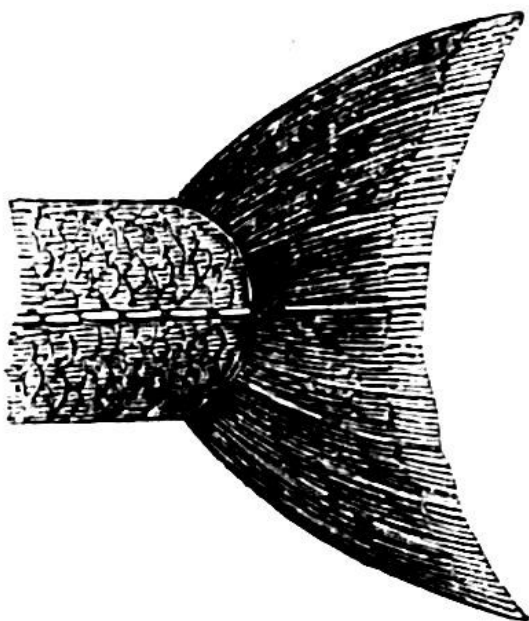


1. HETEROCERCAL.

The *heterocercal* fish, it will be seen, are unequally lobed, that is, the spinal vertebræ are prolonged into the upper lobe of the tail, as seen in the shark, and of which our own dog-fish is an example; while the *homocercal* fish are equally lobed, and the spine does not extend into either.

The fossil fish of the old red sandstone belong almost, if not entirely, to the classes of fish that have ganoid or placoid scales, and heterocercal tails; and of these fish we will now say a few words of the four most remarkable specimens of the one thousand and upwards fossil species that have been discovered, and which can only be known familiarly by accomplished geologists in the ichthyolite department.

1. Here is a drawing of the *Cephalaspis*,^[48] or buckler-headed fish. What an extraordinary looking creature this is! Like the crescent shape of a saddler's knife without the handle—broad and flat, with points on each side running down, ever fixed in warlike attitude against its enemies—it reminds



2. HOMOCERCAL.

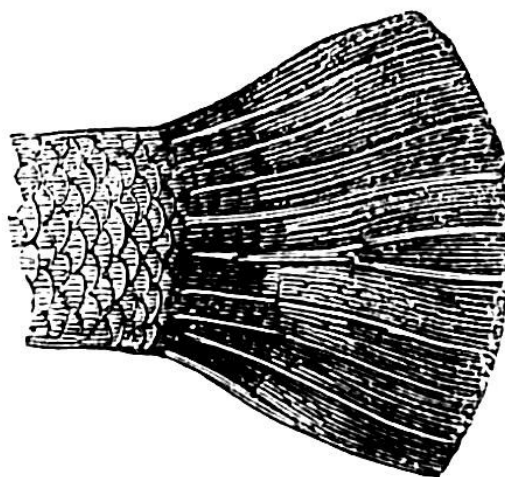
his long extinct neighbour. Hugh Miller's description is the best, and as he was its discoverer, let us give it.

"The figure of the *Coccosteus* I would compare to a boy's kite; there is a rounded head, a triangular body, a long tail attached to the apex of the triangle, and arms thin and rounded where they attach to the body, and spreading out towards their termination, like the ancient one-sided shovel which we see sculptured on old tombstones, or the rudder of an ancient galley. A ring of plates, like the ring-stones of an arch, runs along what we may call the hoop of the kite. The form of the key-stone plate is perfect; the shapes of the others are elegantly varied, as if for ornament; and what would be otherwise

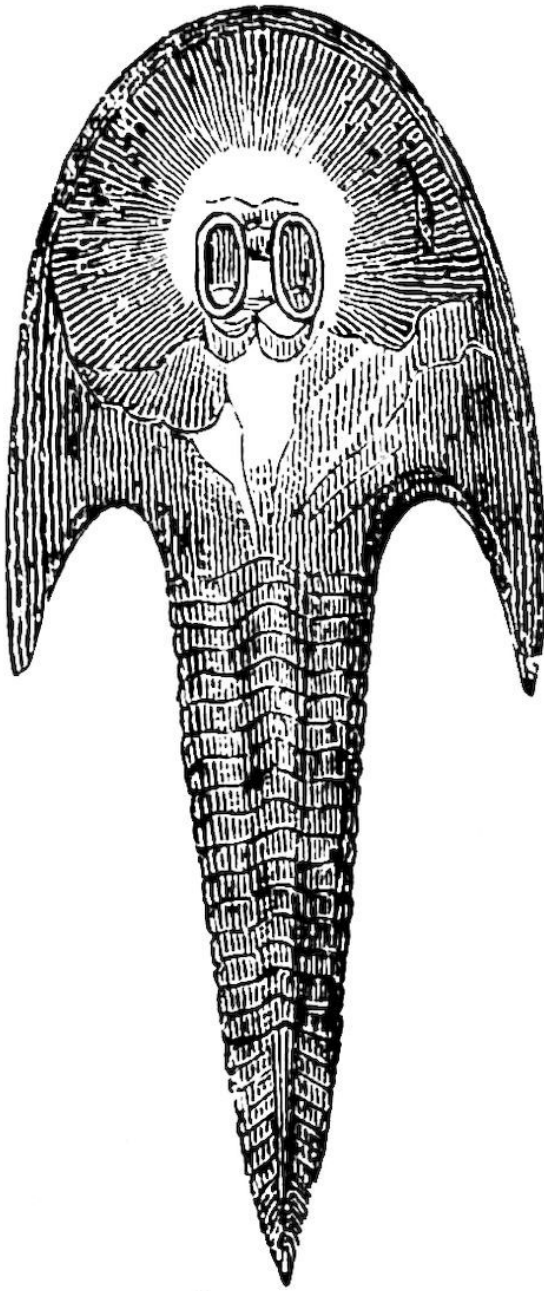
one of an extinct trilobite, and of a living sole or ray, at the same time; and one can easily fancy how hard it must have been for its ancient foes to swallow down so singular and so knife-like looking a creature. This is one of the curious organisms of old life discovered in Cromarty, Herefordshire, and in Russia, the original of which, restored in the drawing, seldom if ever exceeded seven inches.

Let us look now at another curiosity from the same quarter.

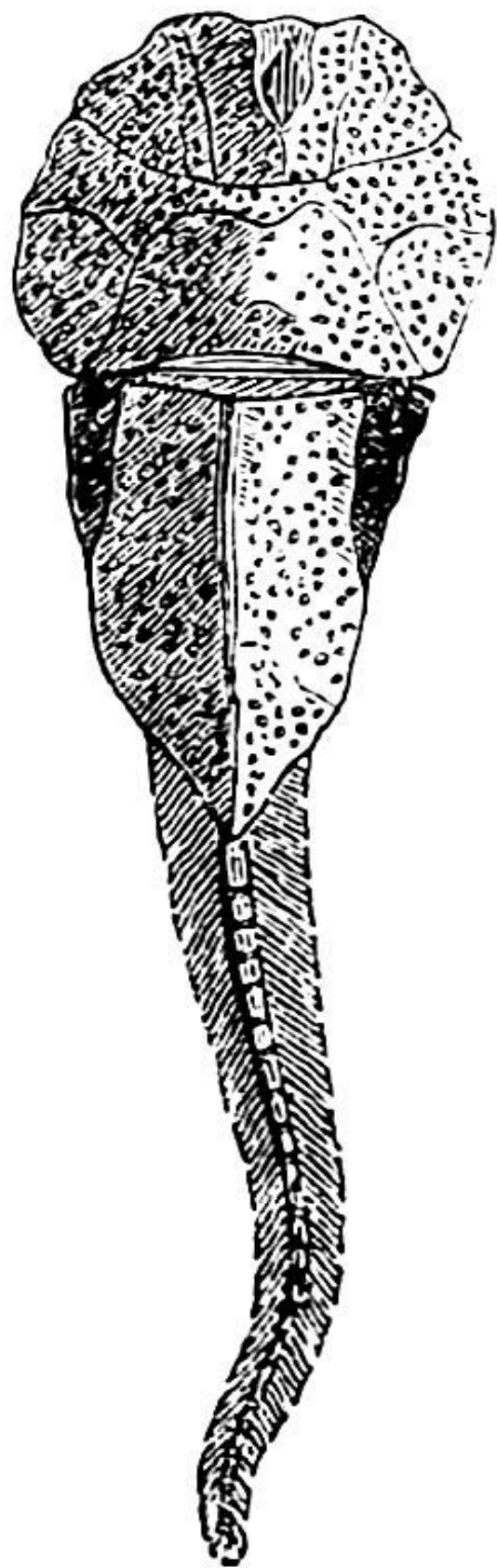
2. Here is a drawing of the *Coccosteus*,^[49] or berry-boned fish. This creature is equally singular with

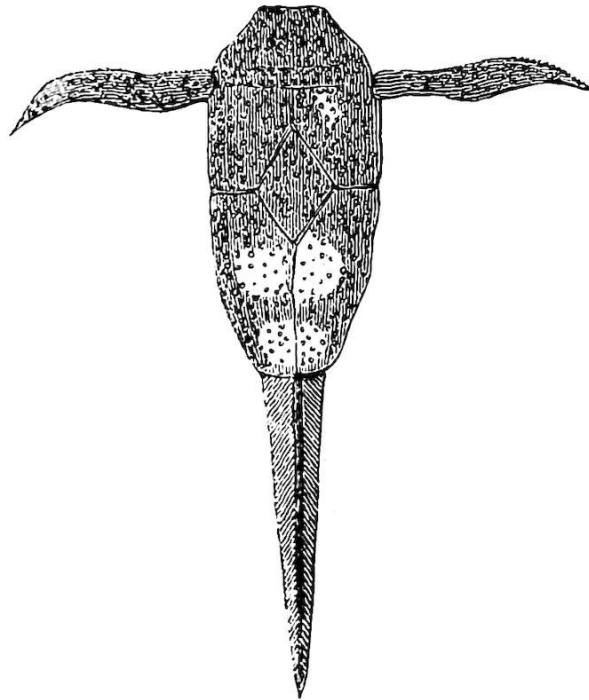


3. HOMOCERCAL.



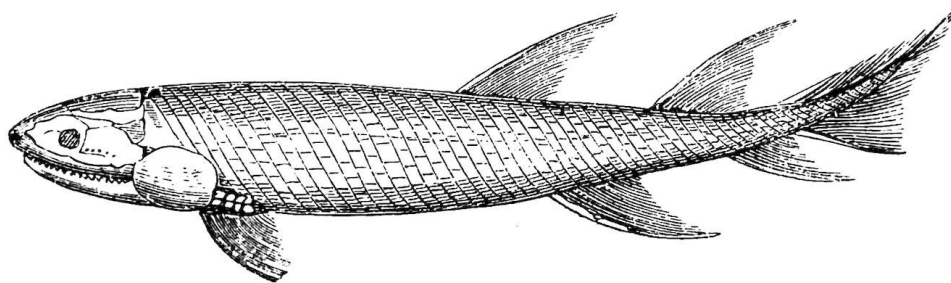
the opening of the arch is filled up with one large plate of an outline singularly elegant.”^[50]





3. Above is the Pterichthys,^[51] or winged fish. We have here a fish more strikingly different to any existing species than either of the other two just passed under review. “Imagine,” says Miller, “the figure of a man rudely drawn in black on a grey ground; the head cut off by the shoulders; the arms spread at full, as in the attitude of swimming; the body rather long than otherwise, and narrowing from the chest downwards; one of the legs cut away at the hip-joint; the other, as if to preserve the balance, placed directly in the centre of the figure, which it seems to support. Such, at the first glance, is the appearance of the fossil.”^[52]

We will now turn to the fourth and last of the singular fishes of this formation.



4. The Osteolepis,^[53] or bony scaled fish. Here we have in the old red sandstone the first perfect specimen of a fish with pectoral, abdominal, and caudal fins, ending as the others do in the heterocercal tail. The vertebral column seems to have run on to well-nigh the extremity of the caudal fin, which we find developed chiefly on the under side. The tail was a one-sided tail. Take into account with these peculiarities such as the naked skull, jaws, and operculum,^[54] the naked and thickly set rays, and the unequally lobed condition of tail, a body covered with scales that glitter like sheets of mica, and assume, according to their position, the parallelogramical, rhomboidal, angular, or polygonal form, a lateral line raised, not depressed, a raised bar on the inner or bony side of the scales, which, like the doubled up end of a tile, seems to have served the purpose of fastening them in their places, a general clustering of alternate fins towards the tail—and the *tout ensemble* must surely impart to the reader the idea of a very singular little fish.^[55]

Most hasty and superficial is this glance through the wonders of the old red sandstone. On the economic uses of this formation, as *tile-stones* and *paving-stones*, we need not dwell; apart from this, these singular inhabitants of the seas of past ages, the mud of which, elevated and hardened, has become solid rock, tell us stories of that long since ancient time to which no poetry could do justice. Carried away from the present into those remote eras, our minds revel in the realization of scenery and inhabitants, of which now we possess only the fossil pictures. At the British Museum, we gaze with feelings approaching to repulsion on the stiff and unnatural forms of Egyptian mummies, but with what feelings of profound wonder do we look on these small fishes, so numerous that the relics of them, found in the Orkneys, may be carried away by cartloads! No number of creations can exhaust God, for in Him all fulness dwelleth. The God in whom we now live, and move, and have our being, is the same God who gave to these pre-Adamite fish their marvellous structures, minutely but fearfully and wonderfully made, and who, when their joy of life and functions of life had ceased, consigned them to a calm and peaceful grave. He is the same God who now upholds all things by the word of his power, and whom we desire to honour by the attentive and reverent perusal of his manifold works. We are tautologists; we say and do the same thing over and over again. God never repeats himself: each successive creation—and how many, extending through countless ages, does geology disclose!—only reveals some new aspect of wisdom, love, and beneficence. To the mind that cannot repose in

God, we say, Study God, in his works and in his word; yea, come back to this remote sandstone era and ask of the “fishes, and they shall declare unto thee” the might and majesty, the skill and contrivance of the Almighty; and though you and I were not there, nor had Adam yet trod this blessed earth,

“Think not, though men were none,
That heaven could want spectators, God want praise;
Millions of spiritual creatures walked the earth,
And these with ceaseless praise His works beheld.”

CHAPTER VI.

THE CARBONIFEROUS SYSTEM.

“As for the earth, out of it cometh bread, and under it is turned up as it were fire.”—JOB.

Suppose this lump of coal could speak, what would it say? Would it not say something like this: “To get me up out of the earth involves dirt, danger, slush, and much tallow-candle; but now you have me, let me tell you my story, for though black I am comely, and but for me—but I anticipate. Now and then I make a dust in your libraries, and inadvertently shoot out sparks and firestones, but nevertheless I am of more use to man than the old granite or the proudest Parian marble; you may get a long way in philosophy, but you will never get beyond coal. I am the real Koh-i-noor of the British empire; and though I can’t, like my namesake, put on a white dress, I am nevertheless worth the soiling of your whitest gloves. *Chemistry* makes no discoveries without me: I light the fire of the laboratory, and furnish man with the means of every crucial test. *Civilization* wants me every day on land and sea, and though in one sense my labours end in smoke, in another they end in commerce, progress, national brotherhood, and interchanging productions of every clime. The *poor student* needs me, for I light his lamp, warm his feet, and cook his food while he is doing sweat-of-brain work for others. And best of all, the *poor man* is a rich man when he has me; he knows that next of kin to good food is good fuel, and man by my help is making such progress, that the day will come when every man will sit by his own blazing fire, instead of seeking joy elsewhere amidst false and pernicious excitements.”

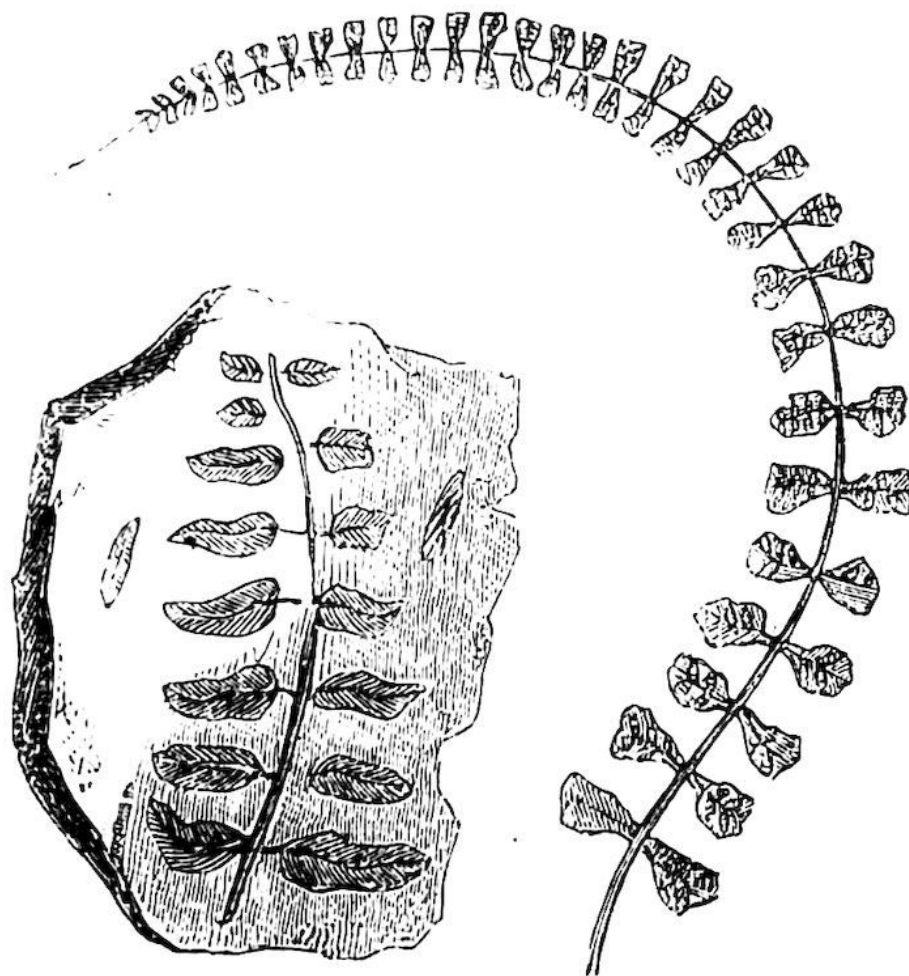
Something like this our friend Coal would be sure to say; and that Coal may not complain of any aloofness on our parts, let us proceed to an examination of the carboniferous system.

“’Tis very pregnant,
The jewel that we find, we stoop and take it,
Because we see it; but what we do not see,
We tread upon, and never think of it.”

Measure for Measure.

The carboniferous system is not all coal; underlying, and often overlying, the coal measures, for the most part, is the mountain limestone, a formation pre-eminently rich in marine fossils. During the tremendous convulsions experienced by the earth immediately after the deposition of the old red sandstone, a vast sea of lime, thick, muddy, and hot, seems to have been poured out over a large portion of the British islands and elsewhere. This flow of liquid lime covered and encased many then existing animals, and we now find it full of fossils of the crinoidean family, a few molluscs, and traces of fish. We shall not, however, stay to examine these now, as we shall meet with them again in the Oolite; our attention will be limited to that part of the carboniferous system which includes only the coal measures, properly so called.

Coal is a vegetable that, by *chemical change* and by *mechanical pressure*, has become a bituminous mineral; and this will render it needful to say a word or two on the ancient vegetable kingdom. The vast quantities^[56] of remains of leaves, ferns, and stems of trees, found in the coal measures, are not in themselves evidence sufficient of the vegetable origin of coal; we arrive at that conclusion in consequence of the researches of modern philosophers, who having applied the powers of the microscope to the internal structure of coal, have discovered the cellular and reticular construction of vegetable life beautifully preserved, and thus previous convictions have become certainties. The examination of the ancient vegetable kingdom is, however, attended with much difficulty, in consequence of the total destruction in most cases of the stems and trunks of the plants, and the entire absence, in consequence of pressure, of all fructification on the fronds of the ferns. If we take an existing species of fern, say the rare and delicate “maiden-hair fern,”^[57] one of the smallest and most elegant ferns of England, we find the fructification very distinct on the under side, and the different methods in which this fructification is arranged is now the principal guide in the classification of ferns. But if we take a fossil fern, say the pectopteris, found in the coal measures, we shall see that there has been so much dislocation and crushing, that all appearance of seed-vessels has disappeared. The following sketch will explain this.



EXTINCT FERN, AND MAIDEN-HAIR FERN.

“Nothing,” says Professor Ansted, “however, is more certain than that all coal was once vegetable; for in most cases the woody structure may be detected under the microscope, and this, if not in the coal in its ordinary state, at least in the burnt ashes which remain after it has been exposed to the action of heat, and has lost its bituminous and semi-crystalline character. This has been too well and too frequently proved by actual experiment to require more than the mere statement of the fact.” And here let us say a few words, which to a few perhaps may have the charm of novelty, about the economic history of coal; for as Cowper says that the first curse, “labour,” has, by God’s blessing on it, been “softened into mercy,” so do we add also, in his words, heartily subscribing to their truth,—

“Thus studied, used, and consecrated thus,
On earth what is, seems formed indeed for us;

Not as the plaything of a froward child,
Fretful unless diverted and beguiled;
But as scale, by which the soul ascends
From mighty means to more important ends;
Securely, though by steps but rarely trod,
Mounts from inferior beings up to God;
And sees, by no fallacious light or dim,
Earth made for man, and man himself for Him."

How long coal has been known and used, we cannot certainly tell, but a writer in Lardner's Cabinet Cyclopædia states that the first mention of coal is in the pages of one Theophrastus, who was, it seems, a pupil of Aristotle. He says, "Those fossil substances that are called coals," (Greek, ἄνθραξ) "and are broken for use, are earthy; they kindle, however, and burn like wood coals; they are found in Liguria and in the way to Olympias over the mountains, and are used by the smiths." Cæsar, although he speaks of the metals of the British isles, does not once mention its coal; but it seems more than likely that it was both known and used by the Romans during their occupation of Britain. Horsley, in his "Britannia Romana," says of Benwell, a village near Newcastle-on-Tyne, "There was a coalry not far from this place, which is judged by those who are best skilled in such affairs to have been wrought by the Romans; and, in digging up the foundations of one of the Roman walled cities, coal cinders very large were dug up, which glowed in the fire like other coal cinders, and were not to be known from them when taken up."

During the time of the Saxons, we find ourselves on less doubtful ground. In a grant made to the monks of Peterborough Abbey for one night's annual entertainment, those good old souls had, we find, "ten vessels of Welsh ale, two vessels of common ale, sixty cartloads of wood, and twelve cartloads of *fossil coal*," (*carbonum fossilium*.)

The Danes had so much fighting on hand, that they troubled themselves neither with coal nor civilization; and we know little of our English diamond until we come to Henry the Third's reign, when, in 1239, a charter was granted to the inhabitants of Newcastle-on-Tyne to dig coals, and we find the coal called for the first time "*carbo maris*," or sea-coal, a term retained through all the succeeding centuries. About this time chimneys came into fashion. As long as people burnt wood they scarcely needed chimneys, but coal introduced chimneys, to say nothing of steamboats and railroads. The Archbishop of Canterbury, who at that time used to reside alternately at Croydon and at Lambeth, had by royal permission thirty

cartloads of “sea-borne coal” annually delivered at his archiepiscopal palace, because, says the historian, “for his own private use in his own chamber he now had the convenience of chimneys.”

The smoke nuisance of that day deserves a passing notice. Smoke was then with many a grand luxury. Old Hollingshed says, “Now we have many chimneys, yet our tenderlings do complain of rheums, and catarrhs, and poses. Once we had nought but rere-doses,^[58] and our heads did never ake. For the smoke of those days was a good hardening for the house, and a far better medicine to keep the good man and his family from the quack or the pose, with which then very few were acquainted. There are old men yet dwelling in the village where I remain, who have noted how the multitude of chimneys do increase, whereas in their young days, there was not above two or three, if so many, in some uplandish towns of the realm, and peradventure in the manor places of some great lords; but each one made his fire against a rere-dose in the hall, where he dined and dressed his meat. But when our houses were built of willow, then we had oaken men; but now that our houses are made of oak, our men are not only become willow, but a great many altogether men of straw, which is a sore alteration.”

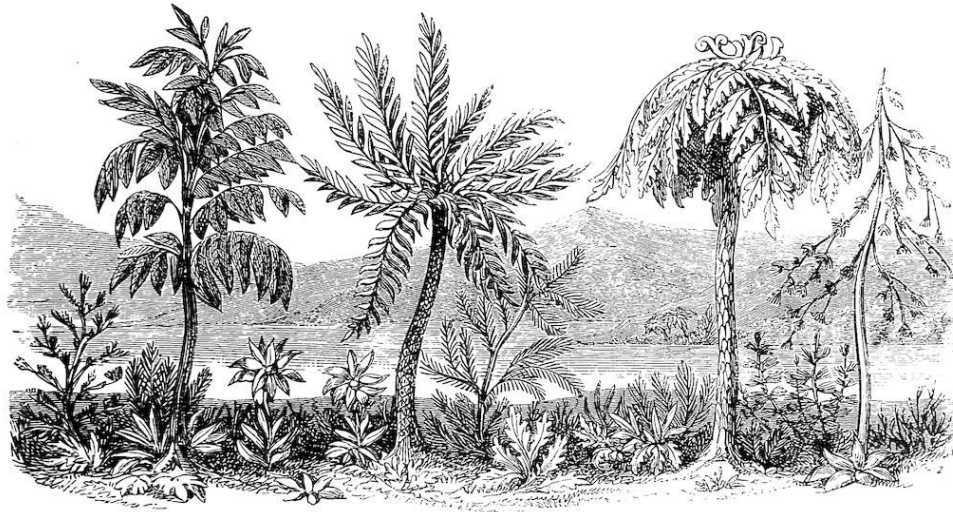
Leaving this digression, let us try and get a bird’s-eye view of the coal-fields of the British Isles. If we commence in Devonshire, we find there the Devonian culms, or Bovey Tracey coal, lying near the surface of the ground, and of little except local use. Crossing over the Bristol Channel, we come into Pembrokeshire, to the Welsh basin, remarkable because thence we mostly get our anthracite coal. Thence we pass on to the Derbyshire coal-fields, that go with little interruption into Scotland, averaging 200 miles in length, and about 40 in width,—once mighty tropical swamps, jungles, and forests, now become chief minerals of commerce. Included in this last immense field is the great Newcastle coal district, the most celebrated of any, supplying almost all the south of England, and nearly all London, with their best coals; and the Scotch carboniferous system, celebrated for its numerous fossils, and for its general base of old red sandstone. In addition to which there is the Irish carboniferous system, occupying as much as 1,000 square miles, but of an inferior quality, and not likely to be of any great economical importance.

In the words of Professor Ansted, we add: “This account of the coal-beds gives a very imperfect notion of the quantity of vegetable matter required to

form them; and, on the other hand, the rate of increase of vegetables, and the quantity annually brought down by some great rivers both of the eastern and western continents, is beyond all measure greater than is the case in our drier and colder climates. Certain kinds of trees which contributed largely to the formation of the coal, seem to have been almost entirely succulent,^[59] and capable of being squeezed into a small compass during partial decomposition. This squeezing process must have been conducted on a grand scale, and each bed in succession was probably soon covered up by muddy and sandy accumulations, now alternating with the coal in the form of shale and gritstone. Sometimes the trunks of trees caught in the mud would be retained in a slanting or nearly vertical position, while the sands were accumulating around them; sometimes the whole would be quietly buried, and soon cease to exhibit any external marks of vegetable origin.”^[60]

There are various kinds of coal on which we may bestow a few words. There is *anthracite*, or non-bituminous coal, and which, therefore, burns without flame or smoke, and is extensively used in malting; and *sea-coal*, which is highly bituminous, and which gives forth so much flame and smoke, that in the good old times of 1306, Parliament forbade its use in London by fine and by demolition of all furnaces in which it was burnt, because “this coal did corrupt the air with its great smoke and stink;” and *cannel-coal*, the etymology of which, they say, is from the word candle, because in many parts of Lancashire the poor use it in place of oil or tallow for lights; and *jet*, sometimes called black amber, which in France employs about 1,200 men in one district, in making earrings, rosaries, and other ornaments; and last of all, there is wood passing into coal called *lignite*, found only in the Devonshire culms.

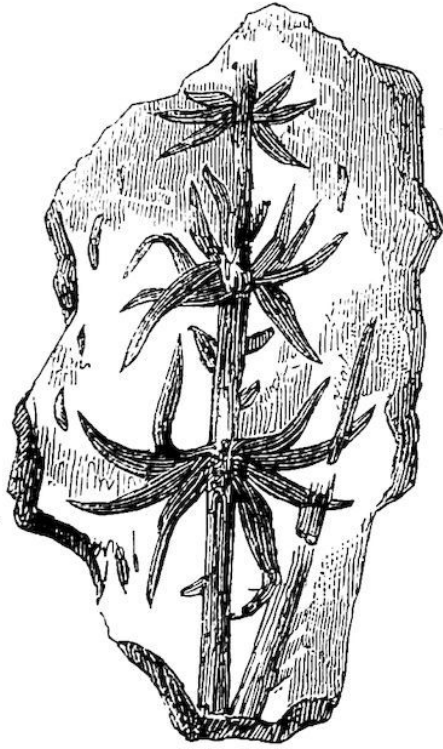
Having thus glanced at the natural history and varieties of coal, we may here try and realize the flora of the carboniferous era. An examination of the fossils of this period enables us to come to undoubted conclusions concerning the trees and plants of that era, so that it is no mere dream to look upon a picture like the following, and see in it a landscape of the coal-forming time of the British islands.



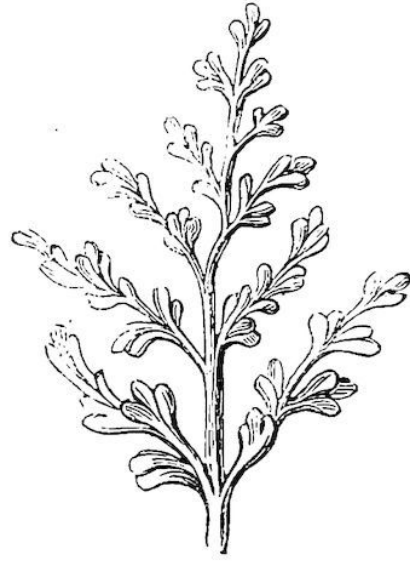
FLORA OF THE COAL MEASURES RESTORED.

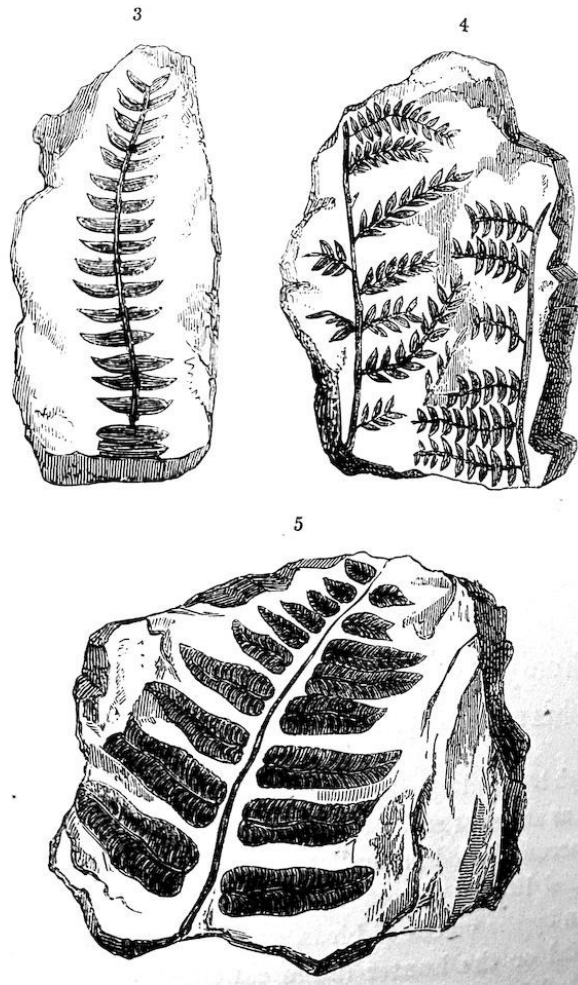
The sun then poured down his golden beams of heat and light, and a tropical climate prevailed in our now cold and humid England. The mountain tops were gilded with his rays; a vast ocean studded with islands, and these crowned with gigantic palms and ferns, then covered our northern hemisphere. In that ocean but few fish were to be found, though many rare molluscos animals swam to and fro, enjoying their brief term of life, and discharging all their appropriate functions. Mountain streams discharged their muddy waters into this ocean, leaving along their margin course broken trees, vegetables, grasses and ferns. The giant *Lepidodendron* looked like a monarch of the ancient world, while around him smaller ferns, vying with each other in beauty and grace, grew, “first the blade” and then the ripened frond, until, in obedience to the great law of organic life, they died and decayed, and became material for the coming man’s future use. But amidst all this prodigal luxuriousness of the vegetable world, there appears to have been neither bird nor beast to break the monotony of the scene; all was silent as the grave—rank, moist verdure below; magnificent ferns and palms above, and the stillness of death on every side.^[61]

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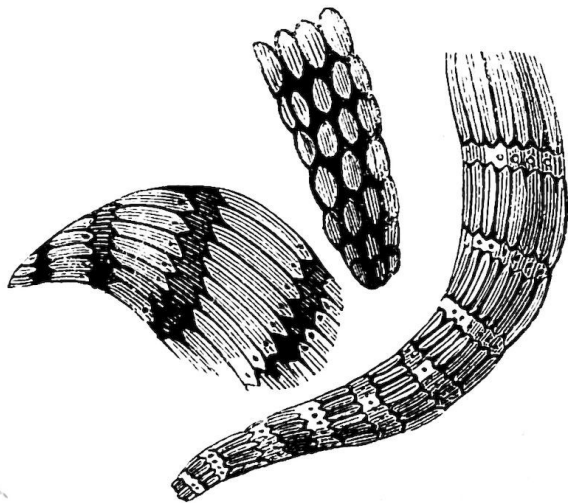




Let us, however, glance at the principal ferns, whose fossil remains we have often found at the mouth of many a coalpit thrown out among the waste. The uncouth names given to them, uncouth only in appearance, must not deter the reader from his acquaintance with their peculiarities; for are not the names of botanical science almost, if not quite, as repellent at first? This star-shaped beauty, (1) the *asterophyllite*, (from *aster*, a star, and *phyllon*, a leaf,) was a common one; this (2) is the *sphenopteris* (from *sphēn*, a wedge, and *pteron*, a wing), so named from a fancied resemblance of the petals of the frond to a wedge; the next (3) is the *pecopteris* (from *pekos*, a comb, and *pteron*, a wing), from a resemblance of the frond to the teeth of a comb; the next (4) is the *odontopteris* (from *odous*, a tooth, and *pteron*, a wing), and in this the frond is something like the jaws of a shark bound together by a central stem, from which they diverge; and the last (5), our favourite, is the *neuropteris* (from *neuros*, a nerve, and *pteron*, a wing),

on account of the exquisite beauty with which the fibres, like nerves, distribute themselves.

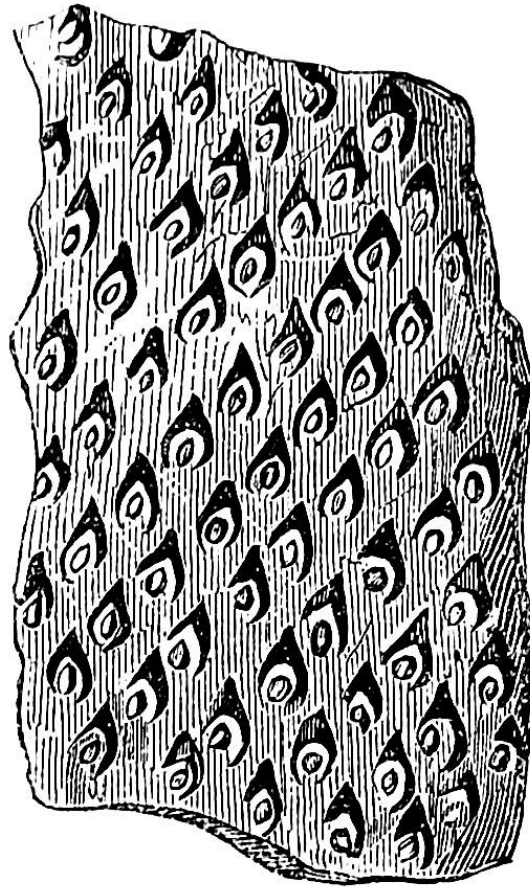
“Besides the ferns, then growing to a great size, there were other plants whose modern representatives are uniformly small; but as the resemblance in this case is simply one of general form, and the great majority of other trees seem to possess no living type to which they can be referred, it is by no means impossible that these also may be completely lost. One example of them is seen in a plant, fragments of which are extremely common in the coal measures, and which has been called *calamite*.^[62] The remains of calamites consist of jointed fragments, which were originally cylindrical, but are now almost always crushed and flattened. They resemble very closely in general appearance the common jointed reed, growing in marshes, and called *equisetum*, or mare’s tail; but instead of being confined to a small size, they would seem to have formed trees, having a stem more than a foot in diameter, and jointed branches and leaves of similar gigantic proportions. They were evidently soft and succulent, and very easily crushed. They seem to have grown in great multitudes near the place where the coal is now accumulated; and though often broken, they seldom bear marks of having being transported from a distance.”^[63] The fossils of the carboniferous system here figured we found not long since in the neighbourhood of Stockport.



CALAMITES.



CALAMITE.



STIGMARIA FICOIDES.

This chapter on the carboniferous system must not be further lengthened. We do not aspire to teach the science of geology; we aim only to impart such a taste for it as shall lead the reader to consult our master works on this subject, and if we succeed in this humble but useful aim, our purpose will be fully answered. Only in reference to the economic uses of coal, we will quote the following, copied, we believe, from the "Athenæum" some time since, but unfortunately copied without reference to its original; a lesson for common-place-book keepers. The writer in speaking of coal-gas says: "The consumption of gas is enormous. The following statistics give us an insight into the extent which this branch of industry has attained. In England 6,000,000 tons of coals are annually employed for the manufacture of gas, and from 12,000,000 to 15,000,000 pounds sterling expended in its

production. In London alone 500,000 tons of coals are annually used, producing 4,500,000,000 cubic feet of gas, and 500,000 chaldrons of coke; of the latter, 125,000 chaldrons are consumed in manufacturing the gas, and the remainder sold for fuel. Upwards of half-a-million houses in London burn gas, and the length of the main arteries for conveying it is 1,600 miles. The capital employed in the metropolis is 4,000,000*l*. The manufacture of coal-gas for the purpose of illumination affords one of the most striking instances of the triumphs of science when enlisted in the divine cause of civilization. Looking at it as a whole, and regarding the ingenuity evinced in the construction of apparatus, the chemical skill and beauty displayed in the process, and the very valuable purposes to which it is applied, it forms one of the most beautiful, curious, and useful of our manufactures; and probably there is no subject of a manufacturing character in the present day which more engages public attention, coal-gas having now become not a mere luxury, or even convenience, but an absolute necessary. In the words of my late colleague and friend, Dr. Hofmann, ‘The extent to which the use of gas has affected the arts and manufactures in this country, can only be conceived by those who are aware of its innumerable applications in the double capacity of giving light and heat. To our experimental chemists the benefits afforded by gas cannot be overrated, more especially in England, where the price of spirits of wine is so exorbitant. But for the use of gas in the laboratory, the progress of chemistry in this country must have been greatly retarded.’

“In speaking of the general influence of the manufacture of coal-gas, it is impossible to leave unnoticed the number of hands daily engaged in raising whole strata of coal, in loading and navigating the fleets employed in conveying it, not only to the different parts of this kingdom, but to foreign countries, which consume a larger quantity of English coal for the production than is generally known. The extension of the gas enterprise produced a sensible effect on the ironworks, by the vast number of retorts, the stupendous gas-holders, and endless pipes required for generating, storing, and conveying it.

“Several other branches of trade were also forced into increased activity, and even new trades sprung up in consequence of the extended use of gas. The substances produced in the purification of gas naturally attracted the attention of the gas manufacturer; and chemistry soon pointed out valuable purposes to which they might be applied. The oily matter, which separates

as a secondary product in the distillation of coal, yielded, when purified in its more volatile portion, the most convenient solvent for caoutchouc; another part of it was found to be an efficient preservative of timber, and the pitchy residue formed the chief ingredient of an excellent substitute for the flag stones of our pavements; while the ammoniacal liquors were found useful in improving the fertility of land. Thus, after the lapse of countless ages, was the nitrogen of petrified fern forests resuscitated in the ammoniacal liquors of the gas-works, to vegetate once more and increase the produce of our corn fields.”



SIR HUMPHREY DAVY'S LAMP, AND MINER AT WORK.

“All nature feels the secret power,
And through eternal change obeys
Up from the deepest region creeps
The trace of life of former days.”

Faust.

CHAPTER VII.

SECONDARY FORMATIONS.

No. 1. The New Red Sandstone.

“There is a path which no fowl knoweth, and which the vulture’s eye hath not seen; the lions’ whelps have not trodden it, nor the fierce lion passed by it.”—JOB.

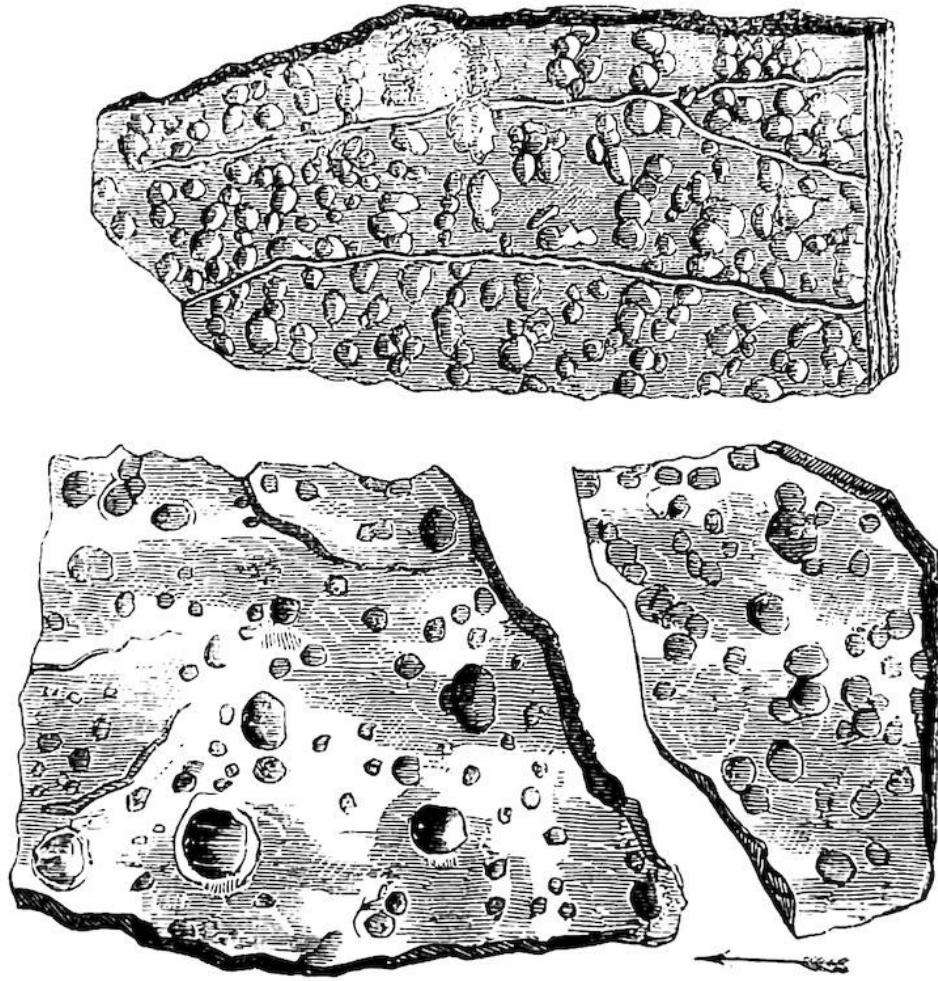
We now take our leave of the Palæozoic period, and enter upon the investigation of other and more recent geological epochs in the history of the crust of our planet. This division is known by the names Secondary or Mesozoic,^[64] and is inclusive of the New Red Sandstone, Oolitic, Wealden, and Cretaceous groups. If, in our previous survey, we have had our minds filled with wonder as we looked at the disinterred relics of past creations, and have gazed at these fossil forms of ancient life with almost a loving interest in their still remaining beauty; so, as we now study higher types of life, and behold how “other wonders rise, and seize the soul the prisoner of amaze,” we shall find reason upon reason for the penetration of our minds with the profoundest adoration of Deity. No man turning up a tumulus, and there finding coins, weapons, beads, vases, or other such historical relics, would venture to say such things were created there; on the contrary, he would acknowledge that they were Roman, and that he had come to that conclusion by perceiving their resemblance to other and similar ancient Roman relics, discovered where there could be no doubt of their origin and history. Or if a traveller were to visit the cities of Herculaneum and Pompeii, and there find buried beneath the overwhelming torrent of once burning lava, all possible kinds of human memorials, not only in human works, but also in the skeleton remains of human *beings*, would he not come to the conclusion that these were indubitable evidences of those cities having once been inhabited by man, and that these skeletons were once

covered with warm flesh, and that they had lived, and moved, and had their being, even as we do now, amidst the activities and enjoyments of actual life? We apply this to geology. There are persons who never judge by *evidence*, (though what else have we to judge by?) but rashly jump to conclusions about geological facts, that have not a particle of common sense to sustain them. They never think that every rounded pebble they meet with has been so rounded by the action of water; they imagine sand to have been created as sand, instead of taking the geologist's proof, that all sand has been produced by the action of moving water on solid rock. They believe that fossils were created, and that God put encrinital remains, and dead ammonites, and bones of saurians, and teeth and bones of great mammals, in the earth, just as we find them in the cliffs and caves of this and every country; and they imagine that thus to account for the wonders of creation redounds to the glory of that God whom thus they ignorantly worship. Even our great publishing society in Paternoster Row,^[65] that has published about everything in natural history but geology, has acknowledged to me that it declines to undertake a work on this science, because of the *theological difficulties* connected with the subject. Why, what is this but the very way to breed infidelity? The man who studies nature and who studies his Bible, is not ashamed to say he believes them both; though two books, they are both given by inspiration of God. Man may be a liar, but neither nature nor the Bible can lie; and while one tells us the history of man, the other reveals to us the history of the creation, and succession of those beings which preceded the advent of man.

We now come to the New Red Sandstone, which must occupy our attention both on account of the unique fossil remains found in it, and also on account of its economic use and value in commerce. Few formations, small as it is, possess so many points of interest to the beginner as the new red sandstone; for, lying just above the carboniferous, and between it and the oolitic group, we find in it certain curiosities of very olden time, that are full of marvellous power to fill us with amaze. Every one remembers Robinson Crusoe's surprise at finding "the print of a man's naked foot on the shore, which was very plain to be seen in the sand," and how he "stood like one thunder-struck, or as if he had seen an apparition;" and then how he "went again to see if it might not be his fancy, but there was no room for that, for there was exactly the very print of a foot, toes, heel, and every part

of a foot;” and then how, after “innumerable fluttering thoughts, and out of himself,” he went home terrified to his fortification.

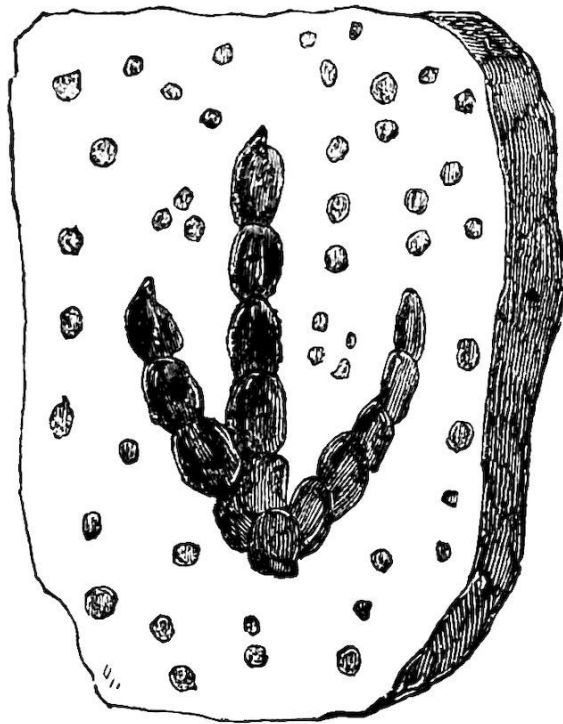
Equally surprising are the discoveries made in the old red sandstone. Large slabs of this rock have been discovered in England, in Scotland, and in the United States, on which are left, as Robinson left the impression of his foot, the undisturbed footmarks of *pre-Adamite* animals; the ebb and flow of the tide of those distant ages; the ripple-mark showing the direction of the wind; and casts of the rainprints made by showers, long long ages ere man had taken possession of the “deep places of the earth.” “Romantic nonsense!” says a grave friend; “let us go to something practical, instead of losing ourselves in such idle speculations.” Now, you are just the person whose ear we want to catch; and to you we say, just listen to the evidence of these assertions. “The casts of rainprints below project from the under side of two layers; the one a sandy shale, and the other a sandstone presenting a warty or a blistered surface, and affording evidence of cracks formed by the shrinkage of subjacent clay on which rain had fallen. The great humidity of the climate of the coal period had been previously inferred from the nature of its vegetation, and the continuity of its forests for hundreds of miles; but it is satisfactory to have at length obtained such positive proofs of showers of rain, the drops of which resembled in their average size those which now fall from the clouds. From such data we may presume that the atmosphere of the carboniferous period corresponded in density with that now investing the globe and that different currents of air varied then as now in temperature, so as to give rise, by their mixture, to the condensation of aqueous vapour.”^[66]



CASTS OF RAIN-PRINTS.

Again, let us hear the words of Professor Ansted. "It may appear at first sight that nothing can be more fleeting, or less likely to be handed down to future ages, among the fossils of a bed of sandstone, than the casts of the impressions of the footsteps of an animal, which by chance may have walked over that bed when it existed in the condition of loose sand forming a seashore. A little consideration, however, will show that it is in fact a very possible occurrence, as, if the wet sand should be immediately covered up with a thin coating of marl, and another layer of sand be superimposed, such an impression will be permanently preserved. In after ages, also, when the soft sands have become sandstones, and are elevated above their former level, the stones split asunder wherever a layer of different material occurs; and thus it happens that the casts of the footsteps may be preserved and

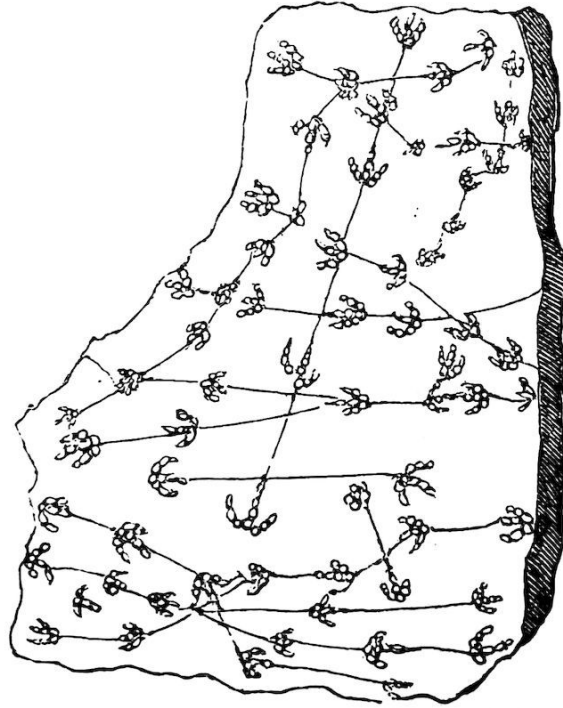
exhibited, although all other traces of the former existence of the animal have been lost.”^[67]



FOOTPRINTS OF A TRYDACTYLE BIRD,
AND IMPRESSION OF RAIN.
(Nat. size.)

If we go to the British Museum, on the north wall of room No. 1. we shall find slabs of sandstone containing footprints of animals, apparently bipeds and quadrupeds, of which we find the following notice in the catalogue of the Museum; and when this description is compared with the three drawings that follow, we make no doubt of carrying the conviction of the reader along with our own, as to the origin of these extraordinary *ichnites*,^[68] as such petrified prints are termed:—“The slabs of sandstone on the north wall of this room, with the supposed tracks of an animal called *Cheirotherium*, are that on the left from the quarries of Hildburghausen in Saxony, and that in the centre from those of Horton Hill, near Liverpool, (the latter presented by J. Tomkinson, Esq.) On the right hand are placed slabs from the same new red sandstone formation, with equally enigmatical imprests of various dimensions, called *Ornithichnites*,^[69] being very like footmarks of birds; they occur in the sandstone beds near Greenfield,

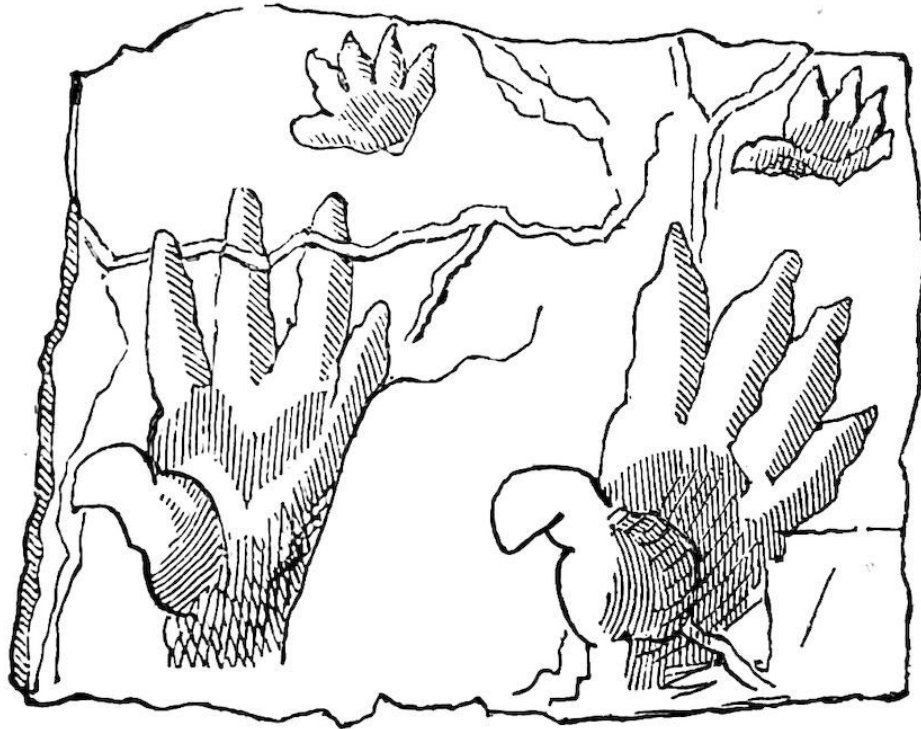
Massachusetts, at a cataract in the Connecticut River known by the name of Turner's Falls."



FOOTPRINTS OF BIPEDS (BIRDS?) FROM
TURNER'S FALLS.
(Size of slab, 8 ft. by 6.)

The lines in this drawing are merely to indicate the direction, the line of progress, of these bipeds, and the reader by following the lines will find the illustration all the more interesting.

But the most remarkable footprints preserved on slabs of sandstone are those of a quadruped, whose hinder feet were much larger than his fore feet. Some of our marsupial^[70] quadrupeds, such as the opossum and kangaroo, and many species of batrachian^[71] reptiles, are distinguished by the same peculiarity. Below is a copy of this slab, which is in the window recess of the same room of the British Museum.



The animal that left these impressions on the soft sandy shore, that are now converted into hard stone, was originally named the Cheirotherium,^[72] and, indeed, this name is still retained by many writers, the hand-like footprints being quite a sufficient reason for so appropriate a name; but latterly the teeth of a fossil animal, supposed to be the same as the Cheirotherium, having been examined, and disclosing a peculiarly labyrinthine character, the animal has been called Labyrinthodon.^[73] Professor Owen, the great comparative anatomist of geology, has fairly established the real character of this animal. He says it is a huge frog, a gigantic batrachian, with hinder feet at least twelve inches in length, combining a crocodilian with a frog-like structure; and although the actual shape and proportions of such an animal must remain greatly an enigma, it is one of the wondrous marvels of geology to pause over these extinct huge creatures, and mark in them the exhaustless resources of creative power.

“So reads he nature, whom the lamp of truth
Illuminates,—thy lamp, mysterious Word!
Which whoso sees, no longer wanders lost,
With intellects bemazed in endless doubt,
But runs the road of wisdom. Thou hast built
Worlds that never had been, hadst thou in strength
Been less, or less benevolent than strong.”

In Professor Ansted's remarkable prose poem on geology, called, "The Ancient World," we have the following picture of the new red sandstone period, which we quote for its vivid but faithful colouring:—"We may imagine a wide, low, sandy track by the sea-side; the hills and cliffs of limestone, which still rise boldly on the shores of the Avon, and in Derbyshire and Yorkshire, having then been recently elevated, and forming a fringe to the coast line. In some places, where footprints are found in successive beds and at different levels, local elevation was probably going on, and the line of coast was occasionally shifting. The sandy flats thus laid bare, and not reached by the ordinary level of high water, were of course traversed by the ancient animals of that period; but only a few faint records of them have been handed down for our observation. Amongst these, however, we are able to enumerate turtles and tortoises, a little lizard having a bird-like beak, and probably a bird's foot,—birds themselves, some larger than an ostrich, others as small as our smaller waders. In some parts of the world there were also large reptiles with powerful tusks, not surpassed in the amount of their departure from the ordinary structure of reptiles by any known aberrant forms of that strange and varied tribe.

"Amongst the most striking of these objects, at least on our own shores, would be the numerous and gigantic Labyrinthodons. We may imagine one of these animals, as large as a rhinoceros, pacing leisurely over the sands, leaving deep imprints of its heavy, elephantine hind foot, strangely contrasting with the diminutive step of its short fore extremities. Another, a small variety, provided like the kangaroo, not only with powerful hind legs, but also with a strong tail,^[74] also leaves its impress on the sand, although itself, perhaps, soon fell a victim to the voracity of its larger congener. These and others of their kind, passing over the sands, and marking there the form of their expanded feet, marched onwards in their course, fulfilled their part in nature, and then disappeared for ever from the earth, leaving, in some cases, no fragment of bone, and no other indication of their shape and size than this obscure intimation of their existence.

"It is strange that in a thin bed of fine clay, occurring between two masses of sandstone, we should thus have convincing evidence preserved concerning some of the earth's inhabitants at this early period. The ripple mark, the worm track, the scratching of the small crab on the sand, and even the impression of rain drops, so *distinct as to indicate the direction of the wind* at the time of the shower,—these and the footprints of the bird and the

reptile are all stereotyped, and offer an evidence which no argument can gainsay, no prejudice resist, concerning the natural history of a very ancient period of the earth's history. But the waves that made that ripple mark have long since ceased to wash those shores; for ages has the surface then exposed been concealed under great thicknesses of strata; the worm and the crab have left no solid fragment to speak to their form or structure; the bird has left no bone that has yet been discovered; and the fragments of the reptile are small, imperfect, and extremely rare. Still, enough is known to determine the fact, and that fact is the more interesting and valuable from the very circumstances under which it is presented.”^[75]

But reminding ourselves of one part of the title of our book, which professes not only to describe the crust of the earth, but also to point out its uses, we must add a few words on the economic value of this small but interesting formation. In this same new red sandstone are found the salt mines of Cheshire, and the brine pits of Worcestershire, which supply all the rock and table salt consumed in England, besides vast quantities for exportation. The rock salt of Cheshire was first discovered near Northwich, while searching for coal; but the largest mine, called the Wilton Mine, is at Nantwich, and still yields about 60,000 tons of salt annually. The salt is generally found from twenty-eight to forty-eight yards beneath the surface, in thick strata varying from fifteen to thirty-five yards in thickness. Besides these beds of salt, there are brine springs from twenty to forty yards in depth. Our common table salt is almost exclusively derived from these springs, which is produced by evaporating the water, and allowing the salt to settle at the bottom of the pans, where, after being washed, it is placed in moulds like the China clay, and comes to our grocers' shops in the blocks we frequently see. “So far as observation has yet gone, the English supply is practically inexhaustible; no limit is known to the extent of the beds or the springs; and it ought to be regarded as one of the blessings which we owe to the mineral wealth of our country, that the beautiful table salt of England may be obtained at such an extremely low price as that now charged for it.”^[76]

To this formation, with its fossil footprints, we owe doubtless the fine fancy of Longfellow, in one of his sweet minor poems; and we shall bring this chapter to a close by quoting the last three verses of this lyric. If we can fulfil such a mission, we had better be frail and erring men than huge Labyrinthodons:—

“Lives of great men all remind us,
We can make our lives sublime,
And departing, leave behind us
Footprints on the sands of time.

“Footprints, that perhaps another,
Sailing o’er life’s solemn main,
A forlorn and shipwreck’d brother,
Seeing, shall take heart again.

“Let us, then, be up and doing,
With a heart for any fate;
Still achieving, still pursuing,
Learn to labour and to wait.”

CHAPTER VIII.

SECONDARY ROCKS.

No. 2. The Oolitic System.

“Hast thou perceived the breadth of the earth? declare if thou knowest it all.”—JOB.

The next division of the secondary rocks is termed the Oolitic system or group. This is a term rather of convenience than of scientific accuracy. In this title it is intended to include the Lias, the Oolite proper, and the Wealden formations. This chapter will be limited to a consideration of the first two series of rocks just named, and in each we shall find abundant material for thoughtful contemplation and intelligent wonder. The English student of geology possesses this great advantage over the student of geology in other lands: this little “corner of Europe,” called England, contains types of almost all the European rocks, and not a few of those that are found in Asia, Africa, and America. To this fact Professor Whewell alluded when he said, “As if nature wished to imitate our geological maps, she has placed in the corner of Europe our island, containing an index series of European formations in full detail.” Out of this circumstance, though little thought of by any except the geologist, arise our threefold sources of wealth. But for the varied distribution of rocks through our country, neither mining, nor manufacturing, nor agricultural operations could be carried on to that extent, and with that success, which have made this country the envy and the admiration of the civilized world. In the warlike age, when Shakspeare wrote, we expect that his praise of England will be on account of her internal security from foreign invasion, and in the hardy prowess of her sons; as when he says, and we are not insensible to the patriotic emotions wakened up by such noble words,—

“This royal throne of kings, this sceptred isle,
This earth of majesty, this seat of Mars,
This fortress built by nature for herself,
Against infection and the hand of war;
This blessed plot, this earth, this realm, this England.”

King Richard II.

But, we confess, we rather dwell on other features in our physical and social history, as affording the best proof of our real greatness, and the best illustration of our untiring Anglo-Saxon energy. We would rather record such facts as the following, than announce any “famous victory;” we would rather turn fondly to considerations like these, than contemplate

“Our sands that will not bear her enemy’s boats,
But suck them to the top-mast.”

“Though animal organization is beyond the constructive skill of man, he takes the elements existing in nature, and by new combinations gets new power. He keeps adding to the qualities of his noblest coursers, his fleetest dogs, and his goodliest beeves. He year by year develops the resources of the soil, reclaims the marsh from wild fowl, the heath from rabbits, and the flinty hillside from briars and thistles. He goes on multiplying the blades of grass and grains of corn, and compels an equal area to yield a twofold substance. He discovers in his raw materials unsuspected properties, until soda and sand are converted into a Crystal Palace, and water, coal, and stony ore into a train, which rushes with the might of an earthquake and the velocity of the wind. He devises fresh applications of machinery, and in the creations of his ingenuity finds a servant and a master. The broad result to England is quickly told: fifty years have doubled the population, and employment and subsistence have been doubled likewise. An engine is contrived which economises labour, and threatens starvation to the labourer; but the issue proves that the work it makes is more than it saves. Annihilate all the cranks and wheels constructed in the interval, and return our counties with their present population to the condition in which they were when the century began, and there would be nothing but famine in the land. A government wiser than man’s has provided, in the constant exertion of talent, for the increase of our race, and maintains a proportion between our wants and our progress. Every round we rise in the ladder leads to a higher; but our step is limited, or we should outstrip our needs by too prodigious a stride, and encroach on the rights of a future age.”^[77]

There is no turning over a leaf in the many-paged book of geologic investigation, without finding the frequent application of thoughts like these. Every part of the crust of the earth has its uses, and uses, too, that are peculiar to *it*; and as we have endeavoured hitherto to point out the economic uses of each formation in the great onward progress of humanity, we shall not find ourselves at a loss in this respect, now that we enter the second division of the secondary rocks.

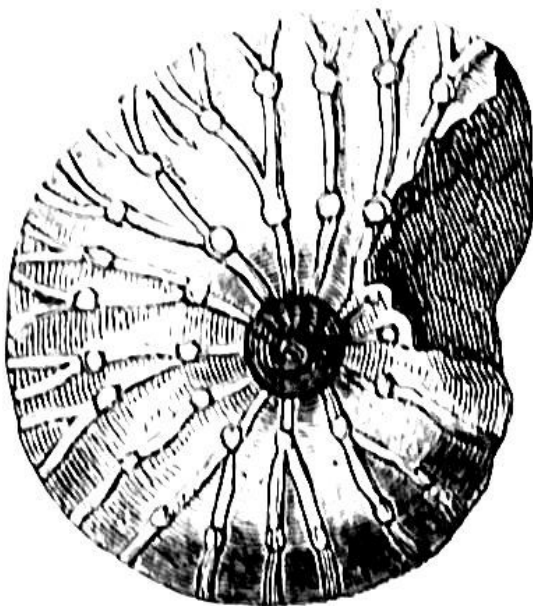
We commence with the Lias.^[78] During the new red sandstone period, clay and marl were being deposited at the bottom of the seas and lakes then in existence. These were the natural degradations of existing rocks; into these soft deposits sunk various pre-Adamite remains, finding in the soft argillaceous beds ready to receive them the “possession of a burying-place” provided for them by the infinite Creator. Here they remained until in process of time, at the close of the new red sandstone period, these beds and their contents were upheaved from beneath the ocean, apparently without much violence; and becoming hardened by the chemical action of sun and wind, present us with the formation we are now studying, rich in its peculiarly characteristic fossils. The name *lias*, or *layers*, indicates the finely stratified condition of the rocks, and affords proof of the tranquil method of their deposit and upheaval. They stretch in a north-easterly direction from Lyme Regis, in Dorsetshire, where it may be seen on the open coast cliffs for about four miles, on to Whitby, in Yorkshire, where also it lies open to the sea, in cliffs of considerable elevation, and lying conformably with other strata, and is thus particularly favourable to geologic examination. It is in the shales of the *lias* at Whitby, and at Lyme Regis, that most of the extraordinary and remarkable fossils have been met with that we are about to describe, and for which this formation is so justly renowned. Indeed, so far as palæontology, or the knowledge of ancient beings, is concerned, there is no formation more full of interest to the student.

Here we meet for the first time with the ammonite. We will introduce him first in his fictitious, and then in his real character, and this we do to show how science dispels the follies of ignorance and superstition. The ammonites were once supposed to be petrified snakes—indeed they are even now called by the ignorant, “*snake-stones*;” and the pleasant little legend about these snake-stones was this, that St. Hilda, who once resided near Whitby, was very much annoyed, as any matron would be, especially if

she kept an establishment for young ladies, as St. Hilda is alleged to have done, by the multitude of snakes that infested the place, and disturbed her equanimity. Accordingly, she set to work, and having first prayed their heads off, then prayed the snakes into stone. In Scott's "Marmion" the legend reads thus:—

"And how the nuns of Whitby told
How of countless snakes, each one
Was changed into a coil of stone
When holy Hilda prayed,
Themselves within their sacred bound,
Their stony folds had often found."—CANTO 2.

Richardson, in his *Geology*, relates the "instance of a dealer who having been requested by his customers to supply them with some of the creatures which had escaped decapitation, contrived to manufacture some heads of plaster of Paris, and affixed them to the specimens; thus he pursued a thriving trade, until some remorseless geologist visiting the place, not only beheaded the reptiles, but showed that they were in reality fossil shells." With the figure of the ammonite every reader of geological books is familiar; it is, perhaps, the best known and most beautiful of all our fossils. We give below a representation of four different kinds, found in the lias and oolite.



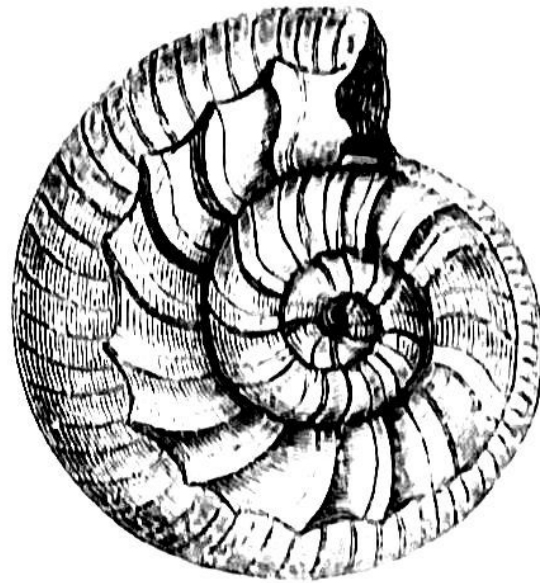
In general outline it will be seen that the ammonite somewhat resembles the nautilus, and yet there are characteristic differences that are so striking as to mock the development hypothesis. In the first place, the shell of the ammonite, though of the same flat discoidal form as that of the nautilus, appears to have been much thinner; secondly, it will be seen that the whorls of the ammonite are rounder and more in number than those of the nautilus; and lastly, the siphuncle, of which more presently, runs *round* the chambers of the ammonite, but

1. AH. BECHEI. LIAS

of these peculiarities; and to aid us, we give below two drawings of sections of ammonites, and two drawings of a nautilus.

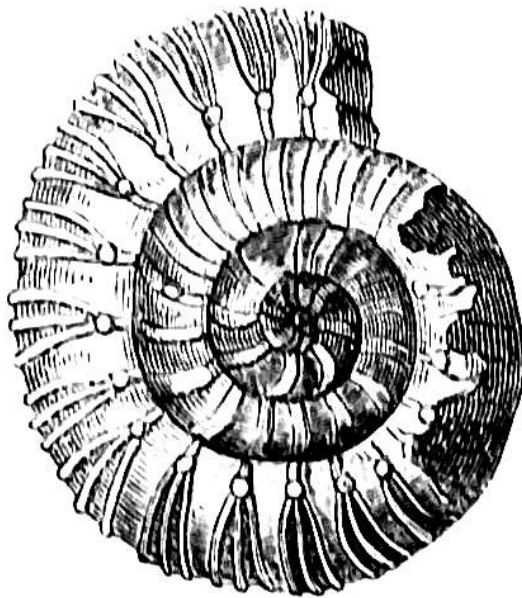
The shell of the ammonite is a continued arch, having transverse arches or ribs crossing the main arch, giving to *some* particular forms of beauty, and to *all* the peculiar symmetry of a series of spiral curves. But, to compensate for the thinness of the shell, a peculiar adaptation is provided; it consists in the flutings which are seen in the surface, occasioned by the transverse ribs. A pencil-case made of a thin

through the chambers of the nautilus. Let us look a little at each



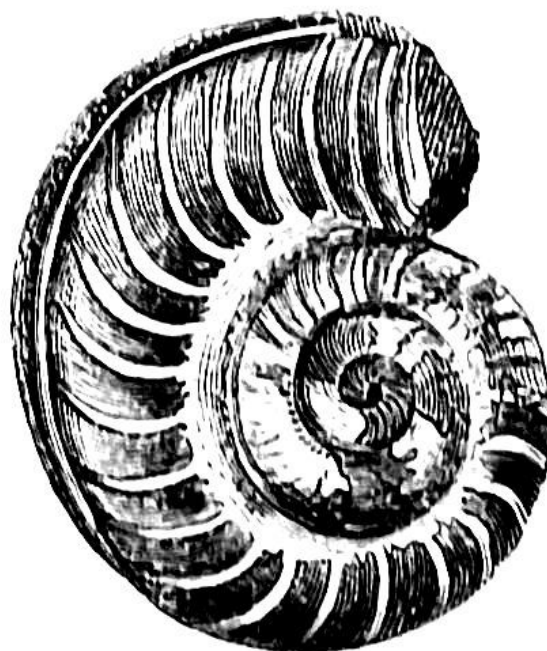
2. AM. BRODIEI. OOLITE.

plate of silver is all the stronger for being fluted, and the zinc roof of a railway station is fluted or corrugated, on the same principle. It is thus that strength is combined with economy of material and elegance of form. In the ammonite we see this recent invention anticipated by the Creator, long ages ere man had appeared. In addition to this, those round knobs or bosses studding some of the ammonites (*e.g.* 1 and 3), like gems upon a diadem, add strength as well as beauty to their frail forms, and thus served the same purpose as the groin work in gothic architecture, a beautiful illustration of which may

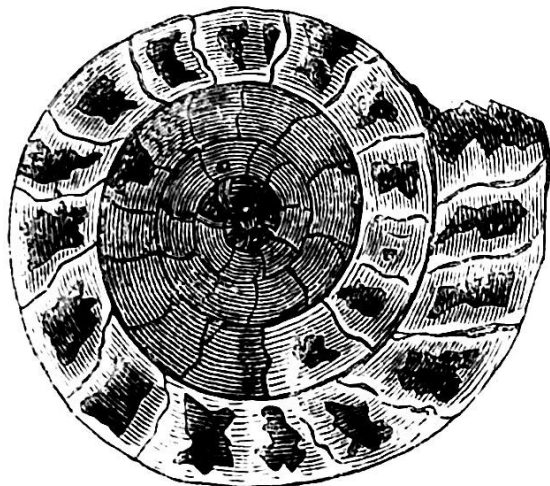


3. AM. HUMPHRIESIANUS. LIAS.

be seen in the roof of Salisbury Cathedral. Then, looking at the chambers of the shell in the sections, we find that some were for living in, while others were mere empty air-cells, used for purposes of elevation or depression, according as the animal wished to rise to the top or sink to the bottom of the sea,—these front chambers being the drawing-rooms in which the aristocratic ammonite lived. Running round them is an hydraulic instrument, called the siphuncle, or air-tube, by means of which singular mechanism this curious animal altered his specific gravity for purposes of sinking or swimming. “The



4. AM. WALCOTTI. LIAS.

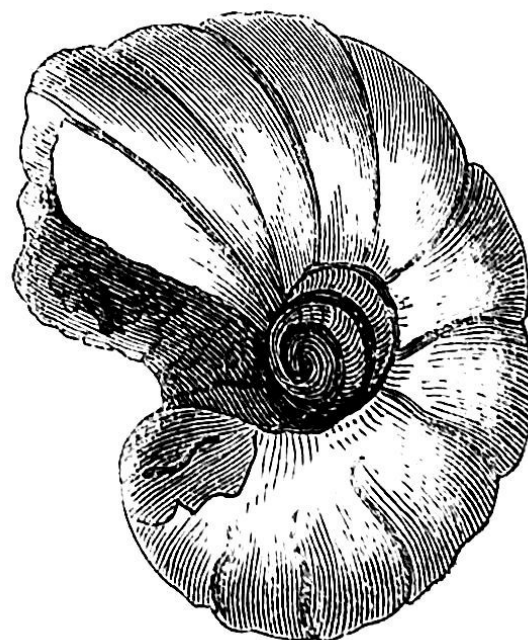


AM. KÆNIGI LIAS.

universal prevalence of such delicate contrivances in the siphuncle, and of such undeviating and systematic union of buoyancy and strength in the air chambers throughout this entire family, are amongst the most prominent instances of order and method that pervade these remains of former races that inhabited the ancient seas; and strange indeed must be the construction of that mind, which can believe that all this order and method can have existed without the direction and agency of some commanding and controlling

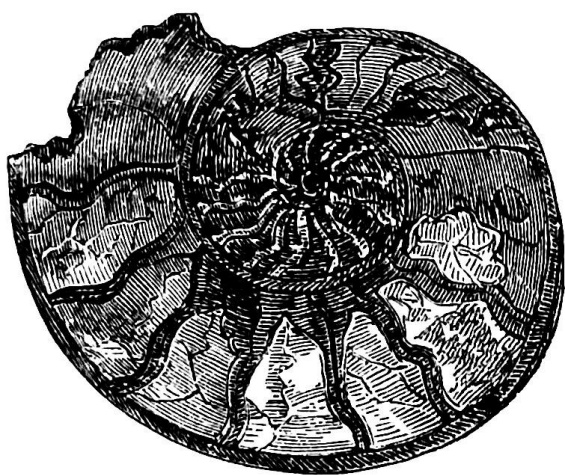
Mind,”^[79] These are what Cowper finely calls “*the unambiguous footsteps of the God;*” and in tracing them our minds are elevated into exalted ideas of Him, whose wisdom is unsearchable, and whose ways are past finding out. With regard to the sections of the ammonite, as seen in the two

previous figures, dimly indeed compared with the beautiful specimens from which they were copied, we can only add, in the words of the treatise just quoted, “Nothing can be more beautiful than the sinuous windings of these sutures in many species, at their union with the exterior shell, adorning it with a succession of most graceful forms, resembling festoons of foliage and elegant embroidery. When these thin septa are converted into iron pyrites, their edges appear like golden filagree work, meandering amid the pellucid spar that fills the chambers of the shelf.”



NAUTILUS. (FOSSIL.)
HORIZONTAL.

We pass by some other fossils

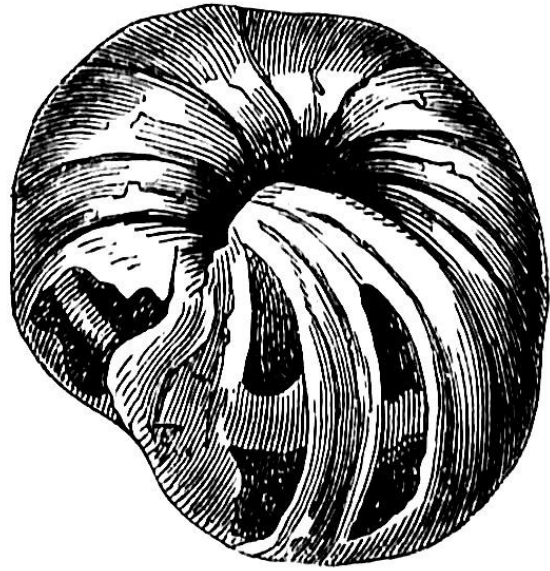


AM. HETEROPHYLLUS. LIAS.

found in the lias, such as the pentacrinite, of which we shall speak when we come to other members of the crinoideal family, gryphites,^[80] of which the *gryphea incurva* is the most common type, broken portions of which, and sometimes good specimens, may be found in most gravel heaps, their peculiar form having obtained for them the name among the rustics of “*devil’s toenails*,” and belemnites,^[81] often met with in vast numbers, and known under the name of *ladies’ fingers*, and *thunderbolts*; and fossil fish, a

few specimens of which are found in the lias.

Passing by these, we next notice the huge Saurians,^[82] by far the most wondrous vertebrated animals with which either the ancient or modern vestiges of creation have made us acquainted. These saurians, sometimes called Enaliosaurians, (*enalios*, the sea, and *sauros*, a lizard,) on account of their peculiar habitat, may all be included in Milton's description of the leviathan, though it is hard to tell what precise creature our great poet had in his "mind's eye" at the time of writing this description—one line of which, from having a syllable too much, reads most unrhythmically—for the crocodile does not go so far out to sea as he represents, and if he did, would be hardly likely to go on to the "Norway foam;" nor can he mean the whale, for the whale has no "scaly rind:"—

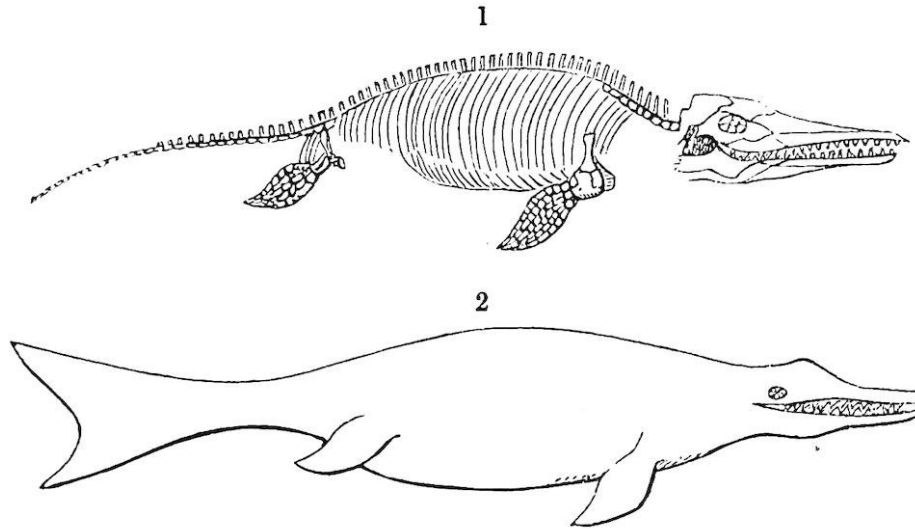


NAUTILUS (FOSSIL), SHOWING
THE SIPHUNCLE.

"That sea beast,
Leviathan, which God of all his works
Created hugest that swim the ocean stream."

Par. Lost, Book I.

But before we enter on a description of these extinct and anomalous creatures, we shall give a sketch of both. No. 1 is the skeleton of the Ichthyosaurus,^[83] restored from the fragments found chiefly at Lyme Regis: it may be seen, with several of its congeners and contemporaries, in the British Museum. No. 2 is a restored outline of this saurian; from both of which sketches it will be seen that the most justly dreaded monsters of our tropical climates sink into insignificance beside the ancient tenants of the mighty deep, in those remote periods of geological antiquity we are now contemplating. The combination of the forms of the fish and the lizard seems more like a troubled dream of Fuseli, than the calm and philosophic deductions of the most eminent anatomists and philosophers.

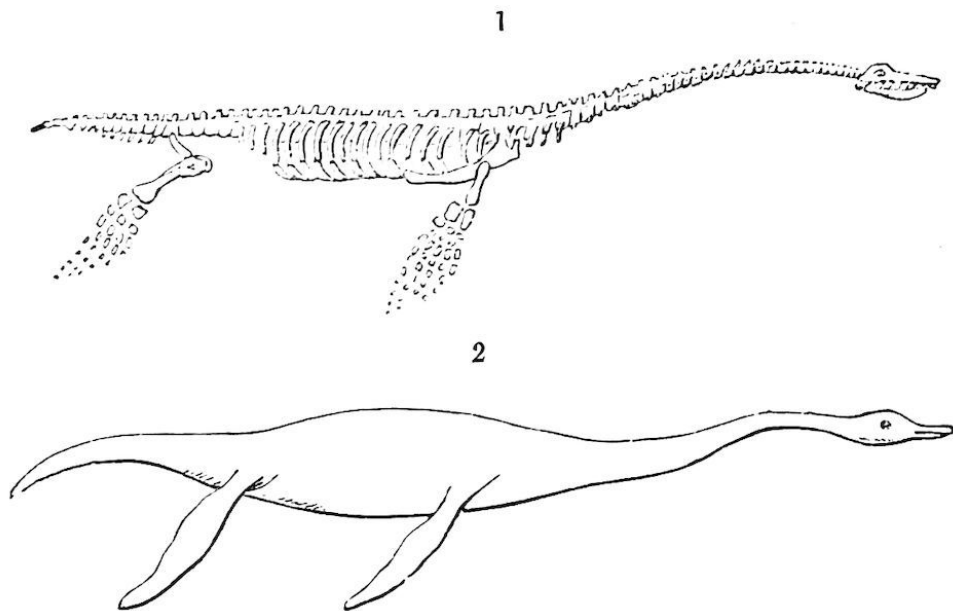


The fossil remains of this creature show that it was intended chiefly, if not entirely, for a marine life. Like the seal, it may occasionally have come to bask on the shore, although, like the seal, it possessed no developed legs or feet, but only paddles. The size of these animals was enormous; they sometimes attained a length of upwards of thirty feet; and to realize such a creature, we must imagine our meeting with a monster thus long in some tropical swamp, having a smooth slimy skin like a whale, a long heavy head like a porpoise, teeth like a crocodile, vertebræ hollow, and therefore light, like the vertebræ of a fish, enabling it to dive swiftly to the bottom, and equally as swift to rise again; and paddles like a whale. Again, look at the head,—that is, go into room No. 4 of the British Museum, and look with wonder, as we often have, on Wall-case A(1), and B(2), and C(3), and you will agree with us that the half of the wonders of this heteroclite^[84] creature have not been told. It had a gape, that is, it could open its jaws seven feet, so that a grenadier guard might walk into his mouth without stooping; it had teeth, not placed in sockets, but arranged in a long continuous trough; it had an eye more marvellous than the eye of the Ancient Mariner, that kept the wedding guest sitting on a stone, who could do nought but hear, for the eye of the Ichthyosaur was often eighteen inches in diameter, so that a man might put his head, hat and all, into its socket,—and this eye was possessed of more wondrous properties than even the eye of the celebrated Irishman that could see round a corner, for the eye of the Ichthyosaur enabled its owner to see all round the country at one time; and as it was a very predatory animal, having doubtless as many enemies as victims, it required

this eye both day and night, and accordingly the eye was placed *close to the nose*, so that the animal could not come to the surface of the water to breathe without being immediately forewarned of danger, or advised of a prize.^[85]

Next follows the *Plesiosaurus*,^[86] which may be seen, in its skeleton parts, restored, and in casts, in room No. 3 of the British Museum, in Wall-cases D(4), E(5), F(6). “The beautiful state of preservation of many of the Plesiosaurs, the entire skeleton, from the point of the muzzle to the extremity of the tail, lying in relief, as if it had sunk down quietly on the soft clay, and become petrified on the spot, manifests how different were the conditions in which the strata of the lias and the wealden were deposited; while the exquisite manner in which the investing stone has been removed, attests the consummate skill and indefatigable zeal of the gentleman (Mr. Hawkins) by whom these superb fossils were developed.”^[87]

Below, No. 1, is a skeleton of the Plesiosaurus, and No. 2 is the restored outline of the animal, whose largest specimen never exceeded seventeen feet.



Cuvier thus describes the Plesiosaur (we borrow the quotation from Buckland):—

“The Plesiosaur is the most heteroclite, and in character the most monstrous, of all the animals that have yet been found amid the ruins of a former world. To the head of a lizard it united the teeth of a crocodile, a neck of enormous length resembling the body of a serpent, a trunk and tail like that of an ordinary quadruped, the ribs of a chameleon, and the paddles of a whale. Such are the strange combinations of form and structure of the Plesiosaurus, a genus the remains of which, after interment for thousands of years amidst the wreck of millions of extinct inhabitants of the ancient earth, are at length recalled to light by the researches of the geologist, and submitted to our examination in nearly as perfect a state as the bones of species that are now existing on the earth.”

We add a word upon the uses of this portion of the crust of the earth, and we do so in the striking words of Hugh Miller. “We have seen how this central district of England has its storehouses of coal, iron, salt, lime,—liberal donations to the wants of the human animal, from the carboniferous, saliferous, and silurian systems; and to this we must now add its inexhaustible deposits of medicine, contributions to the general stock by the oolitic system. Along the course of the lias medicinal springs abound: there is no other part of England where they rise so thickly, or of a quality that exerts a more powerful influence on the human frame. The mineral waters of Cheltenham, for instance, so celebrated for their virtues, are of the number; and the way in which they are elaborated in such vast quantities seems to be as follows:—They all rise in the lias, a formation abounding in sulphate of iron, lime, magnesia, lignite, and various bituminous matters; but they all have their origin in the saliferous marls of the upper new red sandstone which the lias overlies. In the inferior formation they are simply brine springs, but brine is a powerful solvent. Passing through the lias, it acts upon the sulphur and the iron—becomes, by means of the acid thus set free and incorporated with it, a more powerful solvent still—operates upon the lime, upon the magnesia, upon the various lignites and bitumens—and at length rises to the surface, a brine-digested extract of liassic minerals. The several springs yield various analyses, according to the various rocks of the upper formation through which they pass; some containing more, some less lime, sulphur, iron, magnesia, but in all the dissolving menstruum is the same. And such, it would appear, is the mode in which Nature prepares her simples in this rich district, and keeps her medicine-chest ever full.”

Thus wondrous is the machinery of God's universe; every day utters some fresh speech, and every night shows forth some new knowledge.

"The Lord of all, himself through all diffused,
Sustains and is the life of all that lives.
Nature is but a name for an effect,
Whose cause is God."

NOTE.—We do not like to close this chapter without mentioning the name of *Mary Anning*, of Lyme Regis. It is mainly to her practical talent and perseverance that we owe these relics of past ages found in the lias: the history of the British Museum will have to record this humble name, as well as that of Sir Hans Sloane, its founder. The following we borrow from Miss Zornlin's "Recreations in Geology," p. 197—

"Mary Anning died in 1847. Her father, by trade a carpenter, was in his own neighbourhood one of the first collectors of *coorosities*, as they are locally termed, such as *petrified ladies' fingers* and *turbots* (as the fish were termed), *verterbarries* (vertebræ), *cornemonius* (ammonites), and *crocodiles' jaws* (ichthyosauri), &c. He died when his daughter Mary was about ten or eleven years old; and the circumstances of the family being straitened, she went down one day to the beach to search for 'coorosities.' She found a fine specimen of an ammonite; and as she was coming home, a lady who met her in the street offered her half-a-crown for the fossil in her hand. Mary Anning's future destiny was sealed. She prosecuted her searches 'on beach,' and in the following year (1811) observed among the ledges of the rocks a projecting bone of some animal. This enterprising girl (then only eleven years old) traced the fossil in the cliff, and hired some men to dig it out. It proved to be the skeleton of an ichthyosaurus, and has for many years formed an object of interest in the British Museum. Mary Anning afterwards sold this specimen for about 23*l*."

To this we add, that in the "Memoirs of Ichthyosauri," by Thomas Hawkins, Esq. reference is made to Miss Anning, as one "who devoted herself to science, and explored the frowning and precipitous cliffs, when the furious spring-tide conspired with the howling tempest to overthrow them, and rescued from the devouring ocean, sometimes at the peril of her life, the few specimens which originated all the facts and ingenious theories of those eminent persons, whose names must ever be remembered with sentiments of the liveliest gratitude."

CHAPTER IX.

SECONDARY ROCKS.

3. *The Oolite proper.*

“O Lord, how manifold are thy works! in wisdom hast thou made them all.”—DAVID.

Proceeding with our rapid sketch of the crust of the earth and its uses, we now leave the Lias, to enter upon a survey of the Oolite proper. Overlying the Lias, and underlying the Wealden, we find this deposit, which, though it occupies a comparatively narrow track in our own country, is remarkable for the peculiarity and beauty of its fossils, and for the commercial importance of the rocks of which it is composed. Its name is derived from *ōŏn*, an egg, and *lithos*, a stone,^[88] from the remarkable resemblance many of the beds bear to the roe, or eggs of a fish. A good specimen of oolite and the hard roe of a red herring are unlike one another, mostly in the circumstance that one can be cooked, and the other cannot. These egg-like grains are mere agglomerations of calcareous matter, although sometimes a piece of coral, or a broken shell, or a grain of sand, is found to be the nucleus around which these deposits have arranged themselves. It may be as well here to say a word upon the general character of *calcareous* rocks, which are so largely to engage our attention in this and a subsequent chapter. “This division comprehends those rocks which, like chalk, are composed chiefly of lime and carbonic acid. Shells and corals are also formed of the same elements, with the addition of animal matter. To obtain pure lime it is necessary to calcine these calcareous substances,—that is to say, to expose them to heat of sufficient intensity to drive off the carbonic acid and other volatile matter, without vitrifying or melting the lime itself. White chalk is often pure carbonate of lime; and this rock, although usually in a soft and earthy state, is sometimes sufficiently solid to be used for

building, and even passes into a compact stone, or a stone of which the separate parts are so minute as not to be distinguishable from each other by the naked eye.

“Many limestones are made up entirely of minute fragments of shells and corals, or of calcareous sand cemented together. These last might be called ‘calcareous sandstones;’ but that term is more properly applied to a rock in which the grains are partly calcareous and partly siliceous, or to quartz-ore sandstones having a cement of carbonate of lime.

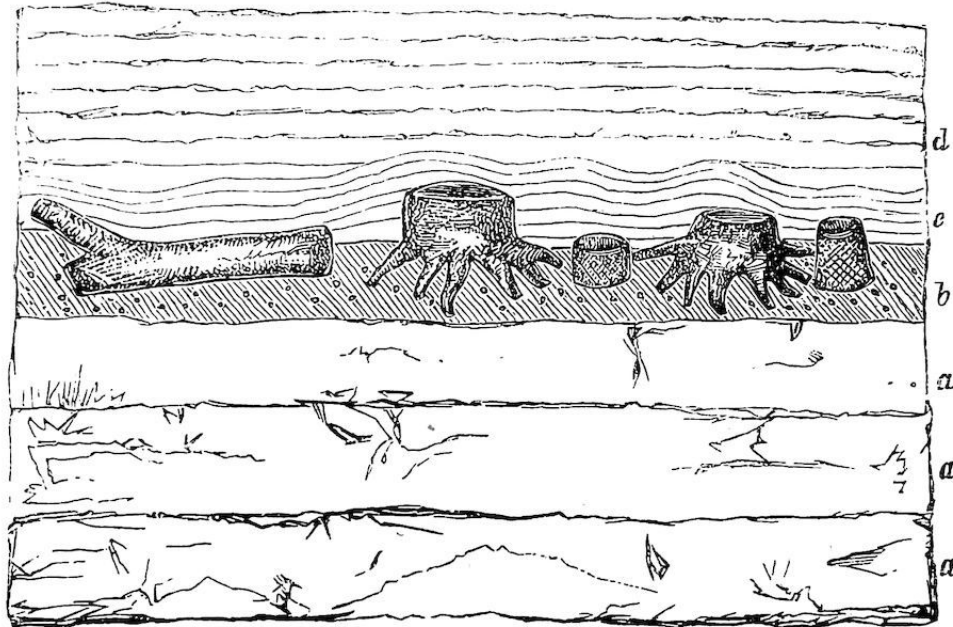
“The variety of limestones called ‘*oolite*’ is composed of numerous small egg-like grains, resembling the roe of a fish, each of which has usually a small fragment of sand as a nucleus, around which concentric layers of calcareous matter have accumulated.

“Any limestone which is sufficiently hard to take a fine polish is called *marble*. Many of these are fossiliferous; but statuary marble, which is also called saccharine limestone, as having a texture resembling that of loaf-sugar, is devoid of fossils, and is in many cases a member of the metamorphic series.”^[89] The geographical distribution of this group of rocks may be traced thus:—Commencing with the *Bill* of Portland, (for there is no *isle* of Portland,) it runs up through part of Dorsetshire, Somersetshire, Gloucestershire, Oxfordshire, Northamptonshire, part of Lincolnshire, terminating in Yorkshire, where the lias and oolite may be seen lying conformably or in sequence; the zone we have thus indicated being about thirty miles in width. The following tabular arrangement will supplement this by pointing out the divisions and subdivisions of the oolite; on which, however, we do not intend to dwell, as our only object, in this most preliminary treatise,—and we shall be pardoned again intruding this thought upon our readers,—is to assist in the investigation of our standard text-books on this science:—

OOLITE PROPER.

1. UPPER.	1. Portland stone, with underlying dirt-beds.
	2. Kimmeridge clay.
2. MIDDLE.	1. Coral rag.
	2. Oxford clay.
3. LOWER.	1. Cornbrash and Forest marble.
	2. Great Oolite and Stonesfield slate.
	3. Fuller’s earth.
	4. Inferior Oolite.— <i>Lyell</i> .

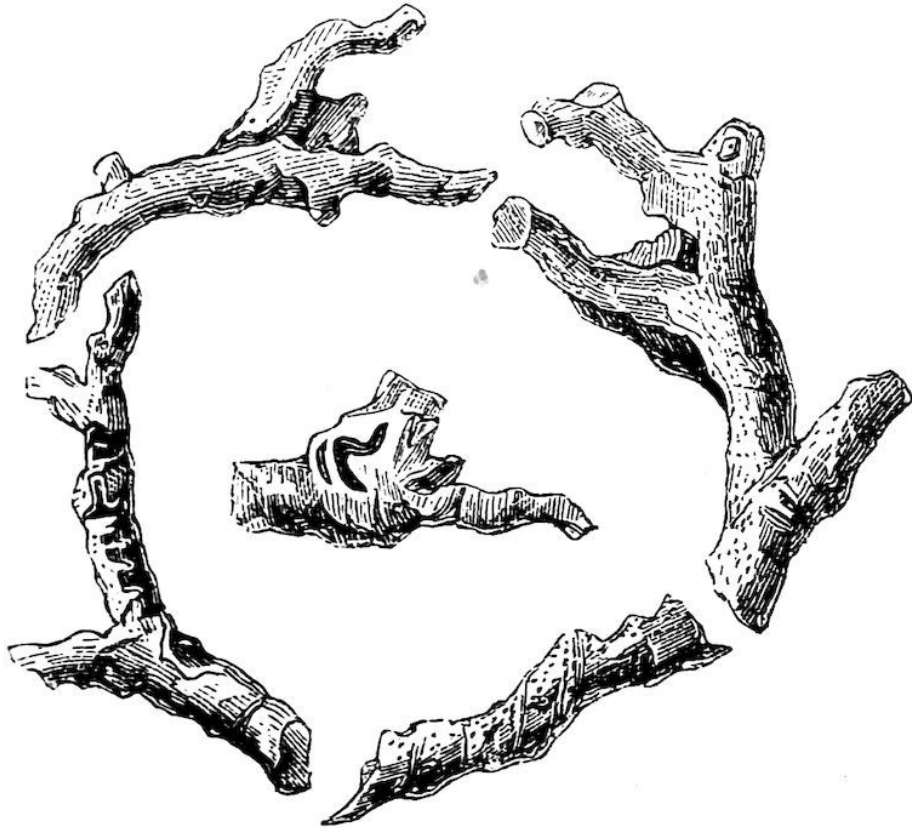
In the upper oolites, it will be seen, is found the famous Portland stone, in which are some of the most remarkable specimens of the extinct fauna of this remote period. “They are found most plentifully in what is locally designated the ‘dirt-bed’ of Portland—a stratum of dark argillaceous mud, which must at one time have been the soil in which they and other vegetables flourished, but which, by a submergence of the land, was converted into the bottom of an estuary, over which other strata of clay, limestone, and sand were deposited. ‘At the distance of two feet,’ says Bakewell, ‘we find an entire change from marine strata to strata once supporting terrestrial plants; and should any doubt arise respecting the original place and position of these plants, there is, over the lower dirt-bed, a stratum of fresh-water limestone; and upon this a thicker dirt-bed, containing not only the cycadeæ, but stumps of trees from three to seven feet in height, in an erect position, with their roots extending beneath them. Stems of trees are found prostrate upon the same stratum; some of them are from twenty to twenty-five feet in length, and from one to two feet in diameter. The following section of a cliff in Dorset exhibits very clearly proofs of the alternation from marine strata to dry land covered with a forest, and of a subsequent submergence of the dry land under a river or lake which deposited fresh-water limestone.’”



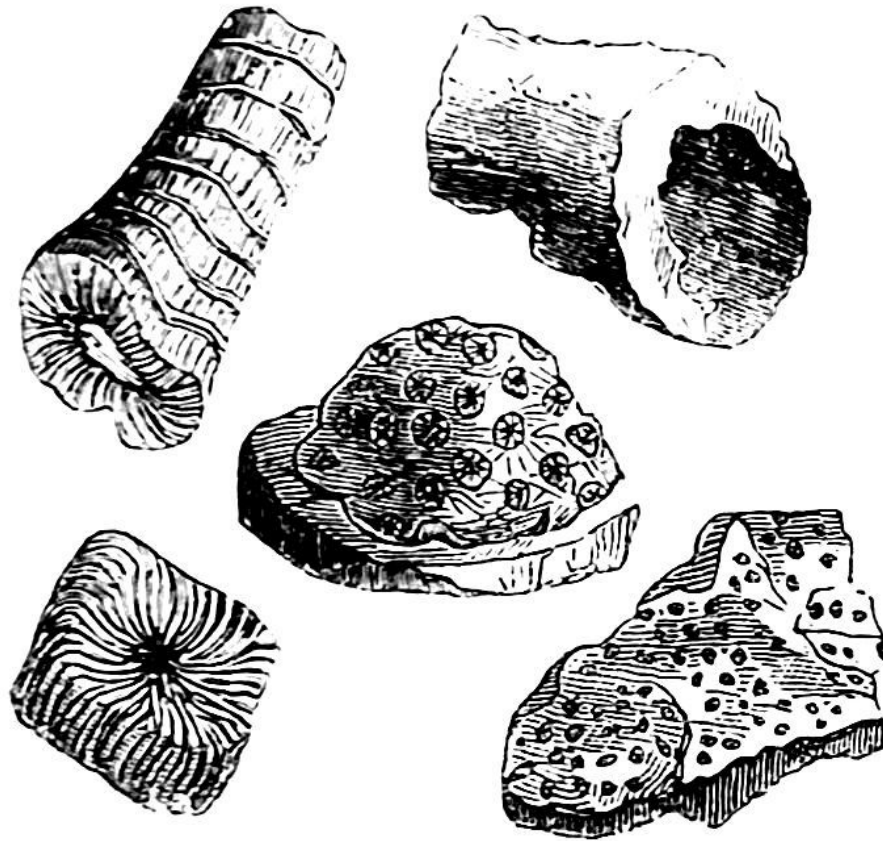
a a a, Portland stone (marine formation); *b*, Dirt-bed, consisting of black mould and pebbles (temporary dry land); *c*, Burrstone, and *d*, Calcareous slate (both of fresh-water formation).—*Chambers' Geology*, p. 128.

From the well-known quarries in Portland, a description of which would tempt us too far astray, have been procured the materials for St. Paul's Cathedral, the Reform Club, and other public buildings. A visit to Portland, and examination of the quarries, such as we have twice paid, is well worth the attention of any summer tourist, and will richly repay, in its romantic scenery, and in the unique simplicity of its people's manners, a week's quiet stay at the King's Arms, the once favoured and favourite inn of George the Third.

Then, if we take the middle oolite, we shall find in it the well-known "coral rag," so called because of the continuous beds of petrified coral found in great abundance, and in many places, apparently, in the same position in which they once grew at the bottom of the sea. We give below a few specimens recently obtained by us from the north of Wiltshire, from which it will be seen how closely they resemble those of existing species.



OOLITE CORAL. (Nat. Size.)



CORALS AND SPONGE FROM THE OOLITE.

But what period was that, and what sunny clime was this, when the ocean poured its waves over what are now our oolitic building-stones, and when the coral insect built its continuous reefs in this our England, just as it is now distributing its labours over so vast an expanse of sea in the tropics and the southern hemisphere?^[90] We cannot but recal the poetic and vivid language of Hugh Miller: “Oh, that hoarse voice of Ocean, never silent since time first began!—where has it not been uttered? There is stillness amid the calm of the arid and rainless deserts, where no spring rises and no streamlet flows, and the long caravan plies its weary march amid the blinding glare of the sand; and the red unshaded rays of the fierce sun. But, once and again, and yet again, has the roar of Ocean been there. It is his sands that the winds heap up, and it is the skeleton remains of his vassals,—shells, and fish, and the stony coral,—that the rocks underneath enclose. There is silence on the tall mountain peak, with its glittering mantle of snow, where the panting lungs labour to inhale the thin bleak air, where no

insect murmurs and no bird flies, and where the eye wanders over multitudinous hill-tops that lie far beneath, and vast dark forests that sweep on to the distant horizon, and along long hollow valleys, where the great rivers begin. And yet, once and again, and yet again, has the roar of Ocean been there. The effigies of his more ancient denizens we find sculptured on the crags, where they jut from beneath the ice into the mist wreath, and his later beaches, stage beyond stage, terrace the descending slopes. Where has the great destroyer not been,—the devourer of continents—the blue foaming dragon, whose vocation it is to eat up the whole land? His ice-floes have alike furrowed the flat steppes of Siberia, and the rocky flanks of Schehallion; and his nummulites and fish lie embedded in the great stones of the Pyramids, hewn in the times of the old Pharaohs, and in the rocky folds of Lebanon still untouched by the tool. So long as Ocean exists, there must be disintegration, dilapidation, change; and should the time ever arrive when the elevatory agencies, motionless and chill, shall sleep within their profound depths, to awaken no more, and should the sea still continue to impel its currents and to roll its waves, every continent and island would at length disappear, and again, as of old, ‘when the fountains of the great deep were broken up,’

‘A shoreless ocean tumble round the globe.’”^[91]

From this digression we return to our third division of the Oolite proper, the *Lower Oolite*, as developed in Somersetshire (around Bath), and in Wiltshire (Tisbury, Braford, &c.), which has its own points of interest and of use. Here occurs the fuller’s earth, mostly found at a village near Bath, called Old Down, which possesses the peculiar property of absorbing the grease or oil remaining in cloth, and thus *fulling* or thickening it. Here also is found that peculiar rubbly limestone, called “*Cornbrash*,”^[92] in Wiltshire, which, on exposure to atmospheric agencies, soon decomposes, and by mixture with the ordinary constituents of the soil, makes an admirable material in agricultural operations. Here also we have the Bath Oolite, the quarries of which are very extensive, abundant in fossils, and the character of whose stone gives to the city of Bath that clean and aristocratic appearance which is so striking to the stranger approaching Bath from the Great Western Railway.

“In this lower division,” says Lieutenant-Colonel Portlock, “also occurs the Bath oolite, which is an excellent stone for the delicate mouldings of

gothic architecture, and is represented in France by the Caen stone, which was imported for the purpose by our early architects, as may be seen in the beautiful Temple Church.”^[93] Of two varieties of Oolite, called “Barnack rag” and “Ketton stone,” obtained from the quarries in Rutlandshire and Northamptonshire, almost all the churches in Cambridgeshire, of the twelfth and thirteenth centuries, are built. As an instance, and illustrative of previous remarks, we may specify King’s College Chapel, Cambridge. Whoever has been to the top of that gem of gothic architecture, and walked outside where the stone is weathered, and has been supplemented in various indifferent ways, may there see the ripple marks of the old tides and winds, marking the ebb and flow of pre-Adamite phenomena; and little shells, some of them most tiny ones, but in beautiful preservation, once left in the soft calcareous sand, now embedded in the hard building stone. And here, before we speak of the fossils of the Oolite, we should like to be thoroughly understood about this same ripple-mark, of which we have more than once spoken, and which it seems desirable to explain. “Another structure, often conspicuous in fine-grained sandstones, is that commonly called “ripple-mark.” Either in quarries or natural cliffs, wherever the upper surface of a bed is exposed, it is often found to be not smooth or flat, but waved in small undulations, exactly like those so often seen on a sandy shore. Now, a good deal of misconstruction has, I think, arisen as to the origin of these small undulations or ripples in the sand, leading sometimes to a possibility of grave error in geological reasoning. People standing on the beach, and observing the gentle rippling motion of the waves, and a very similar form in the sand beneath them, have not unnaturally jumped to the conclusion that the one was the cause of the other, that the ripple on the surface of the water had somehow imprinted its form on the sand at the bottom. Now, really, one is not the *cause* of the other, but they are both caused by the same action, and each is as much a ripple as the other. The wave-like form in the sand is not a ripple *mark*, but a ripple; if it is the mark of anything, it is a ‘current mark,’ and as such I have always preferred to speak of it. Just as a current in the air produces a ripple in the surface of the water below it, so a current in the water produces a ripple in the sand below it. It makes no difference, indeed, whether the sand be acted upon by air or water. Wherever the circumstances are favourable, wind will cause a ripple or current mark on the surface of blown sand, as I observed frequently under very favourable circumstances at Sandy Cape, in Australia, and as has been

observed by Sir Charles Lyell, near Calais. In each case the moving fluid propels the grains of sand forward, piling them up into ridges, which are perpetually advancing by the rolling of particles over the crest of each ridge into the hollow beyond, where they are for a time sheltered from the current, but soon buried under the advancing ridge, to be again turned up and rolled onward, perhaps, as their site becomes exposed to the force of the stream.”^[94]

Other fossil remains found in the Oolite demand our notice before we leave this period of the earth’s history, or this portion of the earth’s crust. In the autumn of last year we were *en route* to Cornwall, but turned aside to visit, among other matters, the Oolite in the neighbourhood of Bath, Chippenham, Bradford, and Trowbridge. Leaving the trunk line of the Great Western, we transferred ourselves at Chippenham to the branch that would take us to Trowbridge, that little Halifax or Preston of the West of England. Many a long day had passed since we made our first acquaintance with the poet Crabbe, and, though fossil hunting, we felt we must pay a pilgrimage to his honoured shrine. We looked at the old church in which for eighteen years he was the representative and expounder of a large and human creed first uttered by the Great Teacher from Galilee; and standing by Chantrey’s monument of Crabbe, we thought how dear to God the man must be, who by dint of purpose, purity, and hope, had risen from an unknown boy to command the ear of the public in a time of poetic dearth, and who, amidst a very chequered life, had realized the fine idea of Goldsmith:

“As some tall cliff, that lifts its awful form,
Swells from the vale, and midway leaves the storm,
Though round its breast the rolling clouds are spread,
Eternal sunshine settles on its head.”

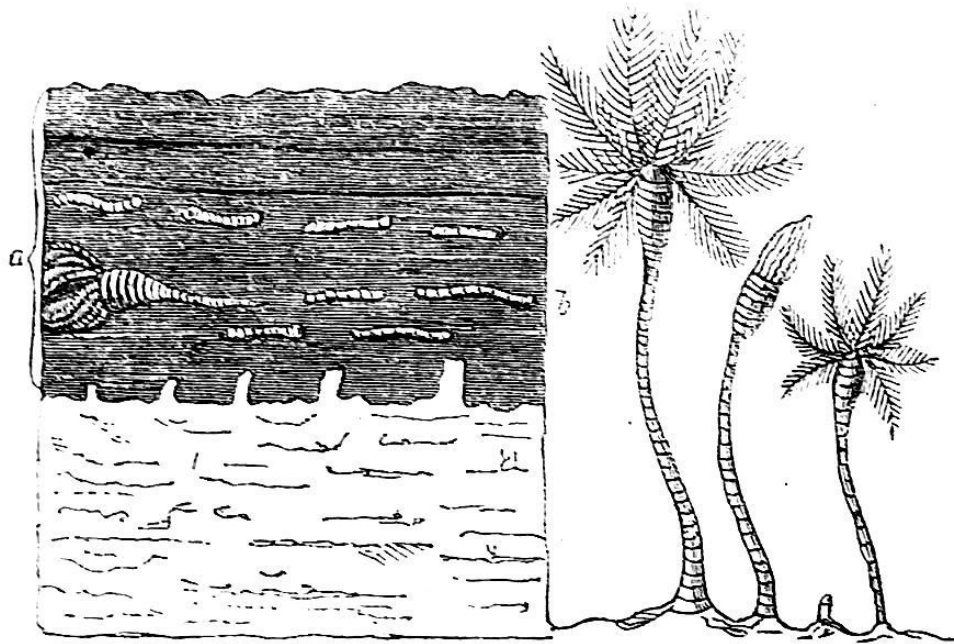
We have heard it objected to Crabbe’s poetry that it wants fire, that it has no romance, that it moves among the homely scenes of rustic life, and seldom soars above parochial records and village history. But Crabbe’s life was a singularly unromantic one; everything with him had been real, hard, up-hill work, in which he had broken down had he not believed in work rather than in genius. At Bungay School he was flogged unmercifully, according to the wise methods of teaching the classics then in vogue, but now happily exploding, except in our antediluvian public schools; at Bury St. Edmunds he was apprenticed to a surgeon, who quickly turned him into an errand boy and servant of all work; at twenty years of age, conscious of

innate power, and aspiring to literary honour, he found himself in London in two equal conditions of misery, without a *friend* and without a *pound*; and then, in this very condition of impecuniosity, the tender place of his heart was touched by an unpropertied girl, Sarah Elwy, and he, the friendless young man, loves her with a pure, deep, passionate, and unchanging love. For eight long years of struggle, her image gave buoyancy to his spirit, and oneness to his purpose, amidst the ups and downs of no common hardships, —now turned from the rich man's door by lacqueys in silver lace, and then politely bowed out by those who might have been his patrons in an hour of need, until, fortune favouring the bold, he became not only the recognised poet of the poor, but better still, the husband of a woman of surpassing worth.

Never aspiring to be a great poet, but only a true and real man, he lived, and laboured, and published much. As a parish clergyman, his memory at Trowbridge is a familiar household word, and having honourably served his generation according to the will of God, he placidly and Christianly fell asleep. As we stood by Chantey's monumental record of this good man, we felt that it did us good to look on such a memorial of a working man; a working man, in the truest sense of the word, for we must take some care, or else this good name will become an empty conventional sham. We must not confine the term to any one class, whether to men who work with sweat of *brow*, or to other men who work as hard, and harder too, with sweat of *brain*; we give the name of working men to all who toil with head or hand, and all such may be bettered by the contemplation of a character like Crabbe's.

But we return; our digression makes geology lag behind. We were fortunate in procuring several fine specimens of the Crinoidean^[95] or Stone-lily family, about which the reader will pardon a few details. At the stations of the Great Northern Railway, the marble mantelpieces and uprights are all made of the Derbyshire encrinital marble, and the mind is filled with singular surprise in the contemplation of these interesting relics of past creations. How multitudinous must have been this one form of life in those ancient days, and how quietly these encrinites, when their brief term of life was over, must have sunk down into the soft calcareous and argillaceous beds lying at the bottom of the old ocean waiting to receive them, only that, mummy like, it might embalm them safely, until in process of time, by upheaval, evaporation, and sun-hardening, their delicate forms should be

brought out to our daylight in all the symmetry and beauty of their pre-Adamite life! But the family of Crinoideans or Stone-lilies, called the “*Pear encrinite*,” or “*apiocrinites rotundus*,” differs very remarkably from the “*Lily encrinite*,” or “*encrinites molliformis*,” and the following descriptions grouped from Lyell and Buckland, aided by the accompanying figures, will introduce these singular animals to our readers. “The Crinoideans or Stone-lilies are almost all confined to the limestone, but an exception occurs at Bath, where they are enveloped in clay. In this case, however, it appears that the solid upper surface of the ‘Great Oolite’ had supported for a time a thick submarine forest of these beautiful zoophytes, (*plant-animals*,) until the clear and still water was invaded by a current charged with mud,” (see diagram below, *a*,) “which threw down the stone-lilies, and broke most of their stems off near the point of attachment. The stumps still remain” (*b*) “in their original position, but the numerous articulations once composing the stem, arms, and body of the zoophyte, were scattered at random through the argillaceous deposit in which some now lie prostrate. Vast strata of entrochal^[96] marble, extending over large tracts of country in Northern Europe and North America, are made up of the petrified bones of encrinites, just as a cornrick is composed of straws. Man applies it to construct his palace or to adorn his sepulchre, but there are few who know, and fewer still who duly appreciate the surprising fact, that much of this marble is composed of the skeletons of millions of organized beings, once endowed with life, and susceptible of enjoyment, which after performing the part that was for a while assigned to them in living nature, have contributed their remains towards the composition of the mountain masses of the earth.”

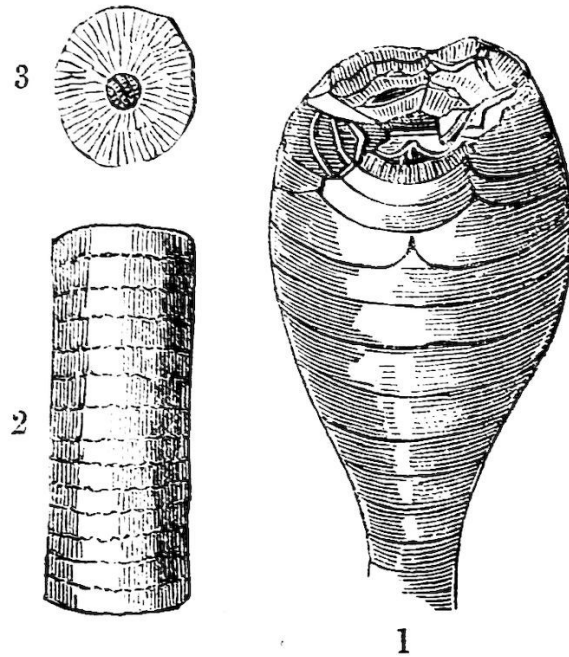


In situ. Restored.

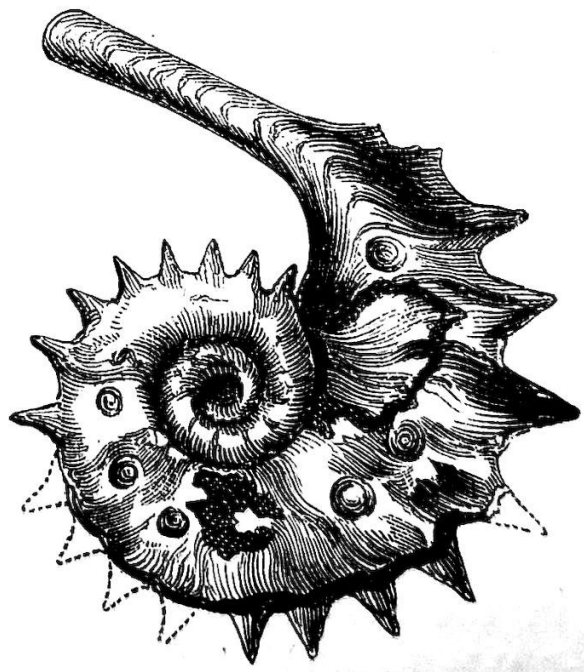
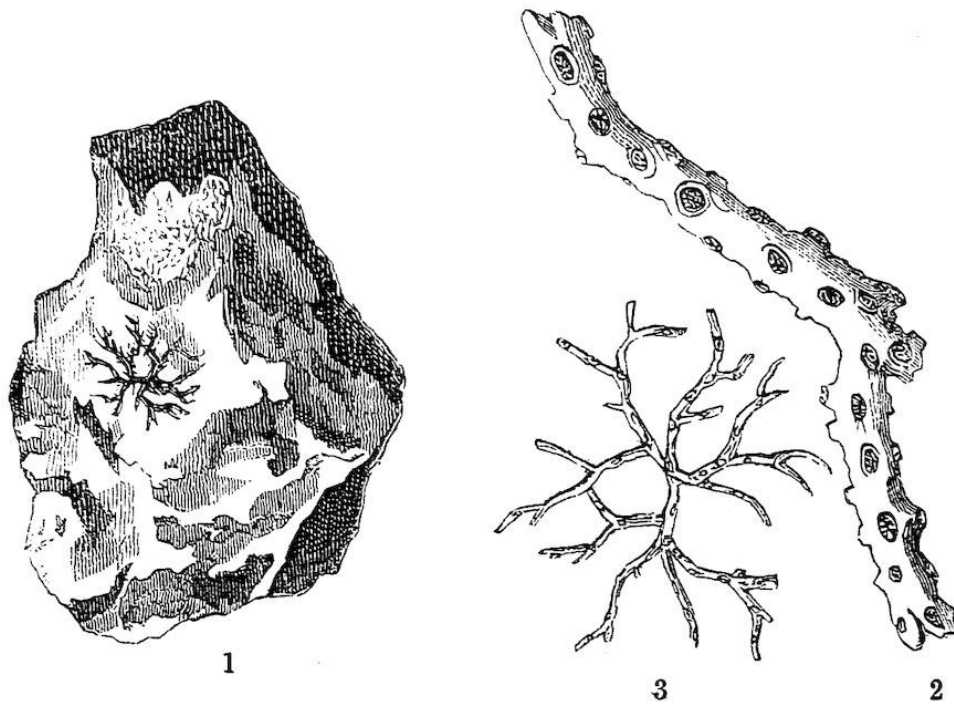
PEAR ENCRINITES, OR APIOCRINITES ROTUNDUS. BRADFORD,
WILTS.

That a better idea still may be formed of this zoophyte, let the following diagrams be added: 1 is the body or bulbous head of the pear-encrinite, curiously laminated, and geometrically divided; 2 is a stem the natural size, and 3 a section, or one of the articulations, in which its entrochal and radiated character will appear.

Below is a root of the pear-encrinite (1); this will give some idea how rudely the graceful stems were broken off by the mud sea that came upon them, and how firm a hold the roots of these beautiful zoophytes took of the ocean bottom; many of these roots are covered with serpulæ and coral, that only reveal their beauty under the lens. In the root before us, a little coral insect has begun to reticulate his tiny links of network; these may be seen in 3, magnified, and by comparison with 2, which is a piece of Jamaica coral, found plenteously on the seashore of the north side of the island, the similarity in structure and symmetry will be immediately detected.



We have already spoken of the Ammonite, but we give here a specimen of a rare kind, preserved in the fine clay near Chippenham. The form of this ammonite is more remarkable than that of any other; its delicate nautilus shell has been sadly compressed, and some of its external protuberances have been broken off, and are here restored by dotted lines; but, more striking still, the delicate horn-like structure that terminated its mouth or aperture has been perfectly embalmed by Nature's kindly hand.



AM. JASON, OR AM. ELIZABETHÆ. *Oolite*.

We have no ammonites afloat now: its twin brother, the nautilus, survives still in sunny climes on the tropic seas; and, perhaps, when a “new heaven

and a new earth” shall usher in another act in the great drama of creation, the nautilus may be superseded by other forms of molluscous life, to show forth the exhaustless resources of His skill and wisdom, who is “wonderful in counsel and excellent in working.” Thus sings a poet and a geologist:—

“The nautilus and the ammonite
Were launch’d in storm and strife;
Each sent to float, in its tiny boat,
On the wide, wild sea of life.

“And each could swim on the ocean’s brim,
And anon, its sails could furl;
And sink to sleep in the great sea deep,
In a palace all of pearl.

“Thus hand in hand, from strand to strand,
They sail’d in mirth and glee,
Those fairy shells, with their crystal cells,
Twin creatures of the sea.

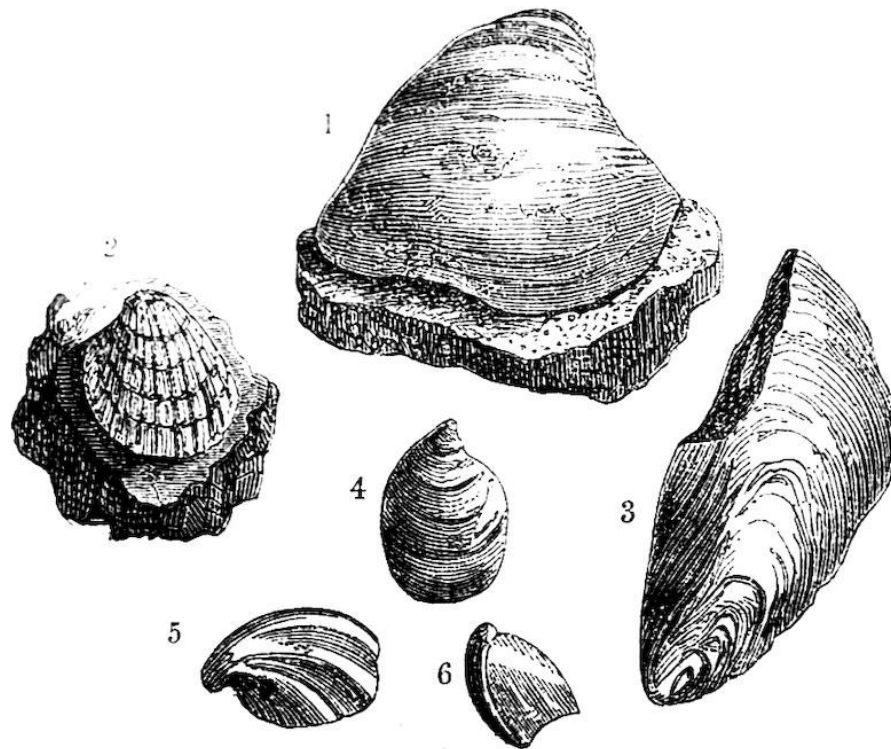
“But they came at last to a sea long past,
And as they reach’d its shore,
The Almighty’s breath spake out in death,
And the ammonite lived no more.

“And the nautilus now, in its shelly prow,
As o’er the deep it strays,
Still seems to seek, in bay and creek,
Its companion of other days.

“And thus do we, on life’s stormy sea,
As we roam from shore to shore,
While tempest-tost, seek the loved—the lost,
But find them on earth no more!”—RICHARDSON.

Below we add a small group of oolitic shells, not on account of any particular beauty of form attaching to them, but as characteristic of this formation, and as lying conveniently near us for being figured. Their hard names—hard to the mere English reader—may possibly alarm the young, who may at present skip the names: let them get to love the science—let them get into the habit of making nature have a meaning in its realities, and into the settled purpose of determining to know what kind of a world this is in which God has cast their lot, and soon these hard names will be only as finger-posts, directing to certain roads on which they may journey to the end of a happy pilgrimage. Aye, better than any old pilgrimage to a fabulous Holy Sepulchre, will be your pilgrimage to sepulchres wrought by the hand

of Infinite Benevolence, for the creatures whom his infinite wisdom had formed, and sustained until he pronounced the decree, “Return!”



1. CUCULLEA CARINATA.
2. CUCULLEA UMBONATA.
3. MODIOLA BIPARTITA.
4. TEREBRATULA GLOBOSA.
5. TEREBRATULA DIGONA.
6. TEREBRATULA MAXILLATA.

Before we quit the Oolite, we have one more stranger to introduce—and in very truth he is the strangest stranger with whom we have yet made acquaintance. We refer to the Pterodactyle.^[97] The geologist, we may suppose, has just lighted on one of these extraordinary remnants of antiquity, say in the Stonefield Quarry, Oxfordshire, and stands aghast as his pick lays open the fragments of this nondescript creature.



But when the researches of science had laid bare the whole of the fossil remains of this heterogeneous creature, when the head and gape of the crocodile, the wing-hands of the bat, and the web-feet of the duck were all revealed; when it was ascertained to be one of those flying reptiles that have no existing type, and that it was “the most extraordinary of all the beings of whose former existence the study of fossils has made us aware,” and “the most unlike anything that exists in the known world;” the blank wonder of the geologist gave way to calm and admiring study, as these undreamt-of relics of the past were made subservient to the wisdom and delight of the present.



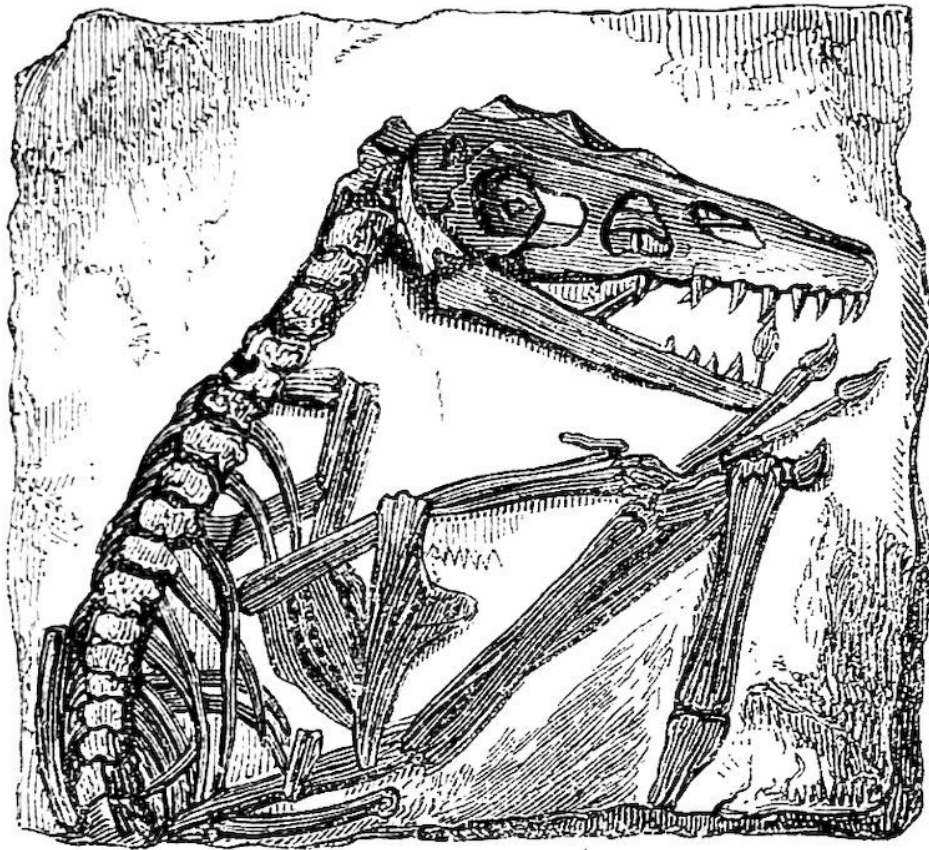
The following is Cuvier's description of this strange creature, borrowed from Ansted, vol. i, p. 418:—

“You see before you,” he says, “an animal which in all points of bony structure, from the teeth to the extremity of the nails, presents the well-known saurian characteristics, and of which one cannot doubt that its integuments and soft parts, its scaly armour, and its organs of circulation and reproduction, were likewise analogous. But it was at the same time an animal provided with the means of flying; and when stationary, its wings were probably folded back like those of a bird, although, perhaps, by the claws attached to its fingers, it might suspend itself from the branches of trees. Its usual position, when not in motion, would be upon its hind feet, resting like a bird, and with its neck set up and curved backwards, to prevent the weight of the enormous head from destroying its equilibrium. Any attempt, however, to picture this strange animal in a living state, would appear to one who has not followed the whole argument to be rather the production of a diseased imagination, than the necessary completion of a sketch of which the main outlines are known to be true. The animal was undoubtedly of the most extraordinary kind, and would appear, if living, the strangest of all creatures. Something approaching to it in form we may

perhaps recognise in the fantastic pictures of the Chinese; but art has, in this respect, not been able to rival nature; and the fabled centaur, or dragon, do not present anomalies more strange than those of the species we have been considering.”

This description of Cuvier will recal to the reader’s mind the well-known words of Milton (Par. Lost, Book II. line 247), in which, in his description of an imaginary fiend, he almost realizes to the life the animal whose extinct and fossil remains have been so recently disinterred.

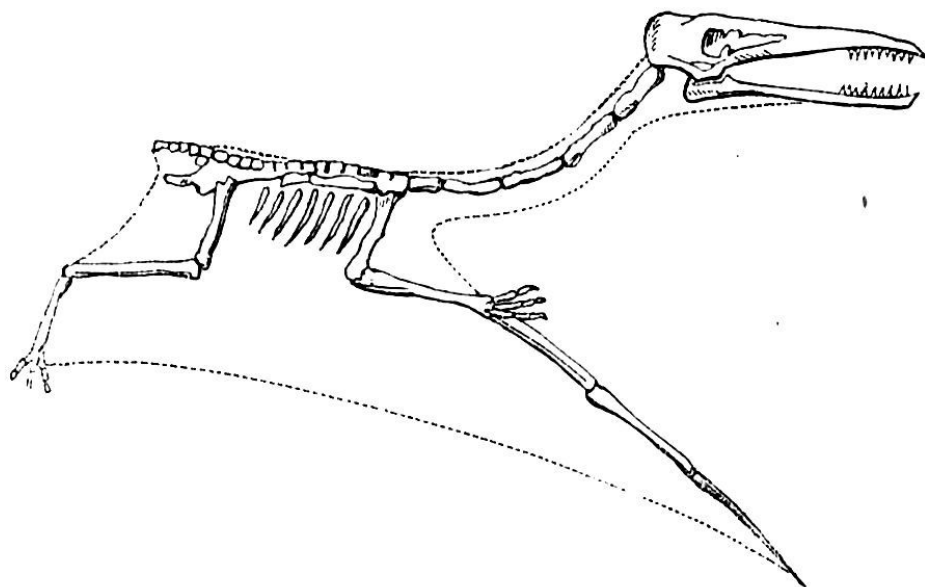
“The fiend
O’er bog, or steep, through strait, rough, dense, or rare,
With head, hands, wings, or feet, pursues his way,
And sinks or swims, or wades or walks, or creeps or flies.”



1. PTERODACTYLE, FROM SOLENHOFEN, (one-third natural size.)

We conclude this chapter by giving drawings of the Pterodactyle. The *first* is the Pterodactyle as found at Solenhofen; the *second* is the skeleton

restored; and the *third* the animal itself, according to the best judges of what a portrait of the Pterodactyle would be.



2. SKELETON RESTORED.



3. THE PTERODACTYLE.

CHAPTER X.

SECONDARY ROCKS.

No. 4. The Wealden.

“The works of the Lord are great, sought out of all them that have pleasure therein.”—DAVID.

Lying immediately between the oolite and the chalk, is a small formation of fresh water, and not of marine deposit, to which the term Wealden has been given. This name has been given to it because it has been found developed chiefly in the Wealds, or Wolds,^[98] of Sussex, Surrey, and Kent; and to Dr. Mantell belongs the honour of investigating this singularly interesting formation, and of giving us its history, after months of patient research and laborious toil conducted on the spot, just as Hugh Miller has become the historian and explorer of the Old Red Sandstone of Cromarty. But although the Wealden is small when compared with the vast extent of some other formations, previously or subsequently added to the material of this “great globe’s” crust, it possesses unusual interest on account of the strange organic remains that are found in it, and of the evidence which these remains supply of vast changes in the conditions and characters of the living beings, found at that period roaming at large in the once tropical swamps of the Wealden.

The Wealden is almost if not wholly of fresh water origin; and a word or two on the formation of the deltas of great rivers, now going on in various parts of the world, will help us rightly to appreciate the character of this formation. It is seldom, some one remarks, that we “can catch a mountain in process of making,” and hence we have much difficulty in arriving at definite ideas concerning the times that the sedimentary rocks occupied in their deposit. But we *can catch* deltas in the process of manufacture,—the

deltas of the Nile, the Mississippi, the Amazon, and other huge rivers that pour down, not their “*golden streams*,” but their muddy accumulations, gathered from the mountain sides whose slopes they wash into the great ocean basin waiting to receive them. “All the rivers run into the sea, yet the sea is not full.”

These deltas,—for we wish to explain as we go along, and would rather fall into the error of explaining where there is no necessity, than of leaving one word that might prove a stumbling-block unexplained,—these deltas are all the mouths, (*les embouchères*,) of *oceanic* rivers, that is, of rivers that pursue their impetuous course towards their native ocean bed; and the character of these deltas necessarily varies with the character of the coast through which they pursue the *uneven* “tenor of their way.” As the level of the sea is approached, the rapidity of the mountain stream is necessarily checked. No longer dashing down steep mountain precipices, as it did in the heyday of its youth, it finds itself grown into a majestic breadth and depth; and as it nears its maternal home, leaving the high land of its origin far behind, it traverses with slow and measured pace the slightly descending planes by which it falls into the ocean. This gradual process enables it to deposit on each side the alluvial soil, and decayed vegetable matter, and so on, that have been held by it in suspension during its rapid progress; and as the current slackens still more as it approaches its final destiny, portions of land gradually rise into view, the results of these deposits, which from their triangular shape have received the name of deltas, from their likeness to the Greek letter Δ.

Thus the mouths of the Ganges form a vast breadth of waters, with intervening islands and strips of land, 200 miles in width. Into this delta the river Ganges runs laden with the rich spoils of clay, sand, vegetable matter, &c., gathered from the Himalaya Mountains, and thus those dismal-looking islands have been formed in the mouth of this river, that have become the home of tigers, crocodiles, &c.

In North America, that “father of waters,” the Mississippi, sometimes called the Missouri, runs through a glorious valley of unexhausted and inexhaustible wealth of 3,000 miles, and runs out into the sea at least fifty miles, while its current miles further. “Like all other great rivers, the Mississippi does not empty itself into the sea in one continuous channel, but in a great variety of arms or mouths, which intersect in sluggish streams the

great alluvial delta, which is formed by the perpetual deposit of the immense volume of water which rolls into the ocean. Between these mouths of the river a vast surface, half land, half water, from 50 to 100 miles in width, and 300 in length, fringes the whole coast; and there the enormous mass of vegetable matter constantly brought down by the Mississippi is periodically deposited. A few feet are sufficient to bring it above the level of the water, except in great floods; and as soon as that is done, vegetation springs up with the utmost rapidity in that prolific slime. No spectacle can be conceived so dreary and yet so interesting, as the prospect of these immense alluvial swamps in the course of formation. As far as the eye can reach, over hundreds of square leagues, nothing is to be seen but marshes bristling with roots, trunks, and branches of trees. In winter and spring, when the floods come down, they bring with them an incalculable quantity of these broken fragments, technically called logs, which not only cover the whole of this immense semi-marine territory, but, floating over it, strew the sea for several miles off to such an extent, that ships have often no small difficulty in making their way through them. Thus the whole ground is formed of a vast network of masses of wood, closely packed and rammed together to the depth of several fathoms, which are gradually cemented by fresh deposits, till the whole acquires by degrees a firm consistency. Aquatic birds, innumerable cranes and storks, water serpents, and huge alligators, people this dreary solitude. In a short time a kind of rank cane or reed springs up, which, by retarding the flow of the river, collects the mud of the next season, and so lends its share in the formation of the delta. Fresh logs, fresh mud, and new crops of cane or reed, go on for a series of years, in the course of which the alligators, in enormous multitudes, fix in their new domain, and extensive animal remains come to mingle with the vegetable deposits. Gradually, as the soil accumulates and hardens, a dwarfish shrub begins to appear above the surface; larger and larger trees succeed with the decay of their more stunted predecessors; and at length, on the scene of former desolation, the magnificent riches of the Louisianian forest are reared.”^[99]

To this glowing description we might add what we ourselves have witnessed on the coast of South America, where the great Amazon, the Orinoco, and the Essequibo, roll their mud-laden waters into the Atlantic, and form vast accumulations of alluvial deposit all along its northern coast; indeed, the coast line of all the Guianas of South America, averaging some

five to ten miles in width, and at least 1,000 miles in length,—Venezuela, British Guiana, Surinam, and Cayenne,—are nothing more than the deposit formed at the base of the mountains by these and other rivers of smaller note. Buried in this mud are pebbles, trunks of trees, animal remains, and so on; and if these in process of time should be elevated and afterwards hardened, or submerged and pressed into clayey slate or sandy stone, the fauna and the flora of these deltas would then be brought to light, just as the leaves of plants are exhibited between the pages of a botanist's collecting book.

In some such method as this was once formed a delta, now constituting the Wealden formation in our own country. What is now the south-east of England, including the Weald clay, the Hastings sand, and the Purbeck beds of limestone and marls, in short, the greater part of the counties already named, with part of Dorsetshire, was once the delta of a mighty river, that so long and so uninterruptedly flowed through this then uninhabited part of the planet, that its accumulated deposits average 1,000 feet in thickness. We have but to pause over those words—*one thousand feet thick*;—once mud, or rather fluviatile deposits of mud and sand, a thousand feet thick in order to receive fresh evidence of the high antiquity of the globe, and of the recent creation of man. We have but for a moment to remember that these deposits consist of innumerable layers of mollusks and crustaceans, a prodigious accumulation of the bones of reptiles and fishes, and of the trunks, branches, and foliage of a long extinct vegetable world, all quietly brought down by “the rivers of waters” of that era, and carefully and without injury deposited in what were then the bottoms of bays or the rising land of deltas, in order to appreciate the evidence which this one deposit alone affords of the immense period of time occupied in its accumulation.

Our next section, the cretaceous division of the secondary rocks, presents us with an exclusively marine deposit. The Wealden, as we have said, is a fresh-water deposit. “Many a long and weary journey,” says Dr. Mantell, “have I undertaken to examine the materials thrown up from a newly-made well, or the section exposed by recent cuttings on the roadside, in the hope of obtaining the data by which this problem is now completely settled.” The data referred to are such as these—the absence of all ammonites, encrinites, corals, terebratulæ, and other marine organisms, which form so large a portion of the cretaceous and other sea deposits.

Here, again, we may anticipate future remarks. It may be thought—by some it has been roundly asserted—that these *fossils were placed in their present situations by the deluge*. Without entering into the theological question, as to the universality or partiality of the Noachian flood, although, as we shall hereafter see, there is but one opinion on that subject held by our best geologists, namely, that the deluge recorded in the Bible was simply the subsidence or submerging of so much of the earth, *and no more*, as was then inhabited by man, and that so partial and so limited was its character, and so brief its duration, compared with those vast geological epochs we have been considering, that there are no traces in nature of that event at all, *i.e.* that we have no one single fossil that can be referred to that event,—without, we repeat, entering upon the investigation of this subject, which will be done in a subsequent part of this volume, we would merely remark, in reply to this absurd notion, that the Wealden is evidently an *alluvial*, and not a *diluvial* formation; and as these terms are so frequently used as if they were synonymous, we shall venture upon our old habit of explanation. An *alluvial* deposit is formed by the *ordinary*, but a *diluvial* deposit by the *extraordinary* action of water. Thus all the straths and carse of Scotland, and our English dales or dells, (may not the remote etymology be *delta*?) are all of alluvial formation; while, as owing their origin to diluvial, or the violent action of water, we are to ascribe the heaps of rubbish, gravel, sand and boulders, that are found in firm compact together.

Thus, if we take a basin of water holding a quantity of earthy matter in solution, and place it on a quiet table, and allow the earthy matter gradually to subside, that which is found at the bottom of the basin is an *alluvial* formation, deposited by the quiet and ordinary action of water.

But when, as sometimes here in Royston, a sudden and heavy rain occurs, when for a while “the windows of heaven” seem opened, and the surrounding hills pour down their rushing streams through the town, filling, as unfortunately they do on such occasions, the cellars of our neighbours in the bottom of the town, with sticks, stones, flint, gravel, bits of chalk, paper, bones, cloth, and a variety of intermingled sundries too numerous to mention, and all huddled together in wild confusion, then is formed in such ill-fated cellars a *diluvial* deposit, occasioned by the extraordinary action of water.

Now, the Scripture flood (and we may be allowed to say, we are not trying to explain away the fact of the deluge,^[100] nor to weaken the strength of the Mosaic narrative, but the very contrary) was a most extraordinary event. Not only were the “windows of heaven opened,” but the “fountains of the great deep” were broken up, and if it had left any traces of its action, they would have been of a heterogeneous and diluvial character; whereas in the Wealden, as elsewhere, the fossils occur in the most orderly and quiet manner, preserving in many cases those exquisite forms of beauty which distinguished them during life.

Without entering more fully into details of the fossil remains of this period, we shall conclude this chapter by a reference to one of the vast saurians whose remains have been disinterred from the Wealden. Many years ago Dr. Mantell discovered, in a quarry at Cuckfield in Sussex, a tooth, which he took up to the Geological Society of London; it was altogether unlike any tooth he had hitherto found, but yet it was so common, that the quarrymen had broken many of them up to mend the roads. The most eminent geologists of the day were puzzled extremely with this tooth. One thought it belonged to a fish; another, that it belonged to an unknown herbivorous mammal; and a third, who was right, that it belonged to a herbivorous reptile. Sir Charles Lyell was at that time about to visit Paris, and the tooth, and that alone, was shown to Cuvier, who at once pronounced it to be the tooth of a rhinoceros. In the process of time other fossil remains were found, portions of jaws, cervical, dorsal, and caudal vertebræ; and on Cuvier seeing these, with the magnanimity of a truly great mind, he frankly avowed his error, and said, “I am entirely convinced of my error in pronouncing it to be the tooth of a rhinoceros.” Shortly after, Dr. Mantell was fortunate enough to procure the skeleton of an iguana,^[101] and he found that the fossil teeth found in Tilgate forest bore a close resemblance to the teeth of the iguana. The teeth of the iguana were found to be small, closely set, and serrated like the edge of a saw, not so much for crushing victims, as for champing and grinding its vegetable food; this corresponded precisely with the fossil teeth found, and after long and careful deliberation, the name given to this crocodile lizard, and the name which it retains, was *Iguanodon*, or the reptile with teeth like the iguana.

The probable size of the *Iguanodon* was thirty feet in length, though probably some exceeded these enormous proportions. Seventy species have been discovered in the quarries opened at Tilgate forest; and this gives us

some idea of the conspicuous part these mighty creatures once played in the eras happily before man was an inhabitant of this planet. Why these remains should generally be found in the same locality we cannot certainly tell, though the previous extract from Alison may shed some light on this curious fact; but it appears beyond doubt that particular spots were selected by these beasts as hospitals and dying places, where, undisturbed by their enemies, they retired to die unseen.

The drawing on page 195, representing the restored fauna of this period, that is, inclusive of the Lias, Oolite, and Wealden formations, will not only illustrate our previous remarks, but enable our readers thoroughly to appreciate the following vivid description of the life of that remote era, which we do not like to abridge. Should any feel sceptical as to whether there were ever such “goings on” in this globe of ours, especially in our eastern counties, now the resort of tourists and invalids having no dread of such creatures before their eyes, we ask them not to reject all this as fable until they have gone to the British Museum, and in Room 3, and in Wall-case C, they will there see some of the magnificent specimens, obtained by Dr. Mantell, of these extinct deinosaurians.^[102] On reading this description, it seems quite justifiable to congratulate ourselves upon the era in which we live. Strange forms and monsters vast have been in the old time before us, and their disinterred remains teach us again and again the good and wholesome lesson of the king of Israel, prefixed as a motto to this chapter.

“Two of these saurians have more especially attracted attention, in consequence of their great abundance in a fossil state in our own country, but they are by no means the only ones known. Of these two, one was more exclusively a tenant of the deep, while the other was probably more frequently met with on the mud banks or on the shore. Both were truly marine in their habits, and both seem to have served as the representatives of the great Cetacean tribe—the whales, the porpoises, and other similar animals now existing.

“It is not easy to imagine the conditions of existence of such animals. We know their form, their proportions, their strange contrivances of structure, their very skin, and the nature of the food which they devoured; and yet, knowing with absolute certainty these points, we hardly dare draw the conclusions which are suggested. I will, however, venture to offer a sketch

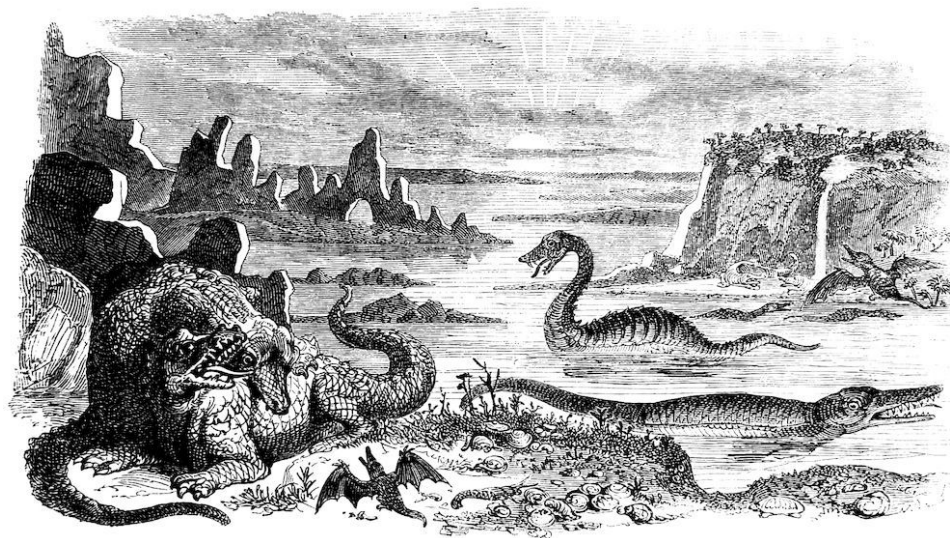
of the appearance of the sea and its inhabitants during this portion of the Reptilian epoch.

“There were then, perhaps, existing on or near the land, some of those reptiles which I shall describe in the next chapter; and with them were associated true crocodilians, not much unlike the fresh water gavial inhabiting the Ganges. These, also, might occasionally swim out to sea, and be found in the neighbouring shoals.

“But these shoals were alive with myriads of invertebrated animals; and crowds of sharks hovered about, feeding upon the larger tribes. There were also numerous other animals, belonging to those remarkable groups which I have attempted to describe in some detail. Imagine one of these monstrous animals, a plesiosaurus, some sixteen or twenty feet long, with a small wedge-shaped crocodilian head, a long, arched, serpent-like neck, a short compact body, provided with four large and powerful paddles, almost developed into hands; an animal not covered with brilliant scales, but with a black slimy skin. Imagine for a moment this creature slowly emerging from the muddy banks, and, half walking, half creeping along, making its way towards the nearest water. Arrived at the water, we can understand from its structure that it was liable to exhibit greater energy. Unlike the crocodile tribe, however, in all its proportions, it must have been equally dissimilar in habit. Perhaps, instead of concealing itself in mud or among rushes, it would swim at once boldly and directly to the attack. Its enormous neck stretched out to its full length, and its tail acting as a rudder, the powerful and frequent strokes of its four large paddles would at once give it an impulse, sending it through the water at a very rapid rate. When within reach of its prey, we may almost fancy that we see it drawing back its long neck as it depressed its body in the water, until the strength of the muscular apparatus with which this neck was provided, and the great additional impetus given by the rapid advance of the animal, would combine to produce a stroke from the pointed head which few living animals could resist. The fishes, including perhaps even the sharks, the larger cuttle-fish, and innumerable inhabitants of the sea, would fall an easy prey to this monster.

“But now let us see what goes on in the deeper abysses of the ocean, where a free space is given for the operations of that fiercely carnivorous marine reptile, the ichthyosaurus. Prowling about at a great depth, where

the reptilian structure of its lungs, and the bony apparatus of the ribs would allow it to remain for a long time without coming to the air to breathe, we may fancy we see this strange animal, with its enormous eyes directed upwards, and glaring like globes of fire. Its length is some thirty or forty feet, its head being six or eight feet long; and it has paddles and a tail like a shark; its whole energies are fixed on what is going on above, where the plesiosaurus or some giant shark is seen devouring its prey. Suddenly striking with its short but compact paddles, and obtaining a powerful impetus by flapping its large tail, the monster darts through the water at a rate which the eye can scarcely follow towards the surface. The vast jaws, lined with formidable rows of teeth, soon open wide to their full extent; the object of attack is approached—is overtaken. With a motion quicker than thought the jaws are snapped together, and the work is done. The monster, becoming gorged, floats languidly near the surface, with a portion of the top of its head and its nostrils visible, like an island covered with black mud, above the water.



FAUNA OF THE OOLITIC PERIOD, RESTORED.

“But a description of such scenes of carnage, enacted at former periods of the earth’s history, may perhaps induce some of my readers to question the wisdom that permitted them, and conclude rashly that they are opposed to the ideas which we are encouraged to form of the goodness of that Being, the necessary action of whose laws, enforced on all living beings, gives rise

to them. By no means, however, is this the case. These very results are perfectly compatible with the greatest wisdom and goodness and even according to our limited views of the course of nature, they may be shown not to involve any needless suffering. To us men, constituted as we are, and looking upon death as a punishment which must be endured, premature and violent destruction seems to involve unnecessary pain. But such is not the law of nature as it relates to animal life in general. The very exuberance and abundance of life is at once obtained and kept within proper bounds by this rapacity of some great tribes. A lingering death—a natural decay of those powers which alone enable the animal to enjoy life—would, on the contrary, be a most miserable arrangement for beings not endowed with reason, and not assisting each other. It would be cruelty, because it would involve great and hopeless suffering. Death by violence is to all unreasoning animals the easiest death, for it is the most instantaneous; and therefore, no doubt, it has been ordained that throughout large classes there should be an almost indefinite rate of increase, accompanied by destruction rapid and complete in a corresponding degree, since in this way the greatest amount of happiness is ensured, and the pain of misery and slow decay of the vital powers prevented. All nature, both living and extinct, abounds with facts proving the truth of this view; and it would be as unreasonable to doubt the wisdom and goodness of this arrangement, as it would be to call in question the mutual adaptation of each part in the great scheme of creation. No one who examines nature for himself, however superficially, can doubt the latter; and no one certainly, who duly considers the laws ordained for the general government of the world, can believe it possible for these laws to have acted without a system of compensation, according to which the vital energies of one tribe serve to prepare food for the development of higher powers in another.”^[103]

CHAPTER XI.
SECONDARY ROCKS.
No. 5. The Chalk.
THE CRETACEOUS SYSTEM.

“His hands formed the dry land.”—DAVID.

At the conclusion of the Wealden period, the crust of our globe underwent another great change. The fresh-water rocks we have just been studying were probably submerged by some violent subterranean or subaquean agency, and upon a great part of them a new deposit was gradually formed. Foot by foot, and inch by inch, slowly accumulated those cretaceous particles (Lat. *creta*, chalk) which now constitute the chalk hills and cliffs of our own country, to say nothing of the extent of territory in Europe occupied by this well-known mineral. To those who like ourselves live in a chalk district, this formation possesses special interest, simply because the “bounds of our habitation” are fixed there; and though we miss here all traces of the Ichthyosaurus and Iguanodon, although no Deinotherium nor Mastodon as yet make their appearance on the stage of being, yet with a very little knowledge of geology, coupled with a desire not to walk through the world with “eyes and no eyes,” it is in our power to invest many an otherwise motiveless walk with pleasurable interest. A cutting in the hills may reveal some unthought-of geological curiosity; a heap of flints may disclose some “rich and rare” fossils; and even a walk over the Downs, those breezy “downs” or “heaths” of southern England, covered with their short herbage, and dotted with their browsing sheep—those downs that undulate in such well-known forms of beauty, may be

fruitful of suggestions concerning their origin, character and antiquity, that may be of healthful power in the development of an intellectual life.

We live—that is, we aforesaid—not on a fresh-water deposit, like the people of Purbeck, Hastings, and so on, but upon an *old marine deposit*. The huge cliffs of chalk that gave the name of *Albion*, the “white isle,” to our country, with all these downs, which rest upon chalk, having a depth or thickness of often more than two thousand feet, were once formed at the bottom of pre-Adamite oceans; and in process of time, as the secondary period drew near to its destined close, and other and higher types of life were to make their appearance, were upheaved by Him, “whose hands thus formed the dry land,” to serve the future purposes and contribute to the comforts of the coming lord of all.

Of this Albion we may well be proud, as the home of religion, the birth-place of true freedom, and the sanctuary of the oppressed. With Coleridge we exclaim:—

“Not yet enslaved, nor wholly vile,
O Albion, O my mother isle!
Thy valleys fair, as Eden’s bowers,
Glitter green with sunny showers;
Thy grassy upland’s gentle swells
Echo to the bleat of flocks;
Those grassy hills, those glittering dells,
Proudly ramparted with rocks;
And Ocean, ’mid his uproar wild,
Speaks safety to his island-child.
Hence for many a fearless age
Has social quiet loved thy shore;
Nor ever proud invader’s rage
Or sack’d thy towers or stain’d thy fields with gore.”

If we were to tell the story of the Chalk formation, it would run something after this fashion:—Once upon a time—but what time we don’t know, for man’s records only go back a few thousand years; but long, long ages before there was a man to till the earth—all this chalk lay out of sight at the deep bottom of the old, old sea. The tall cliffs at Dover, which Shakspeare would hardly recognise now with their tunnels and railroads, the wavy downs of Sussex and Surrey, the chalk hills of Hampshire, all that piece of Wiltshire called Salisbury Plain, and this line of country stretching away north to bluff Flamborough Head, and running down south to Weymouth and the isle of Purbeck, taking up in its course a good part of

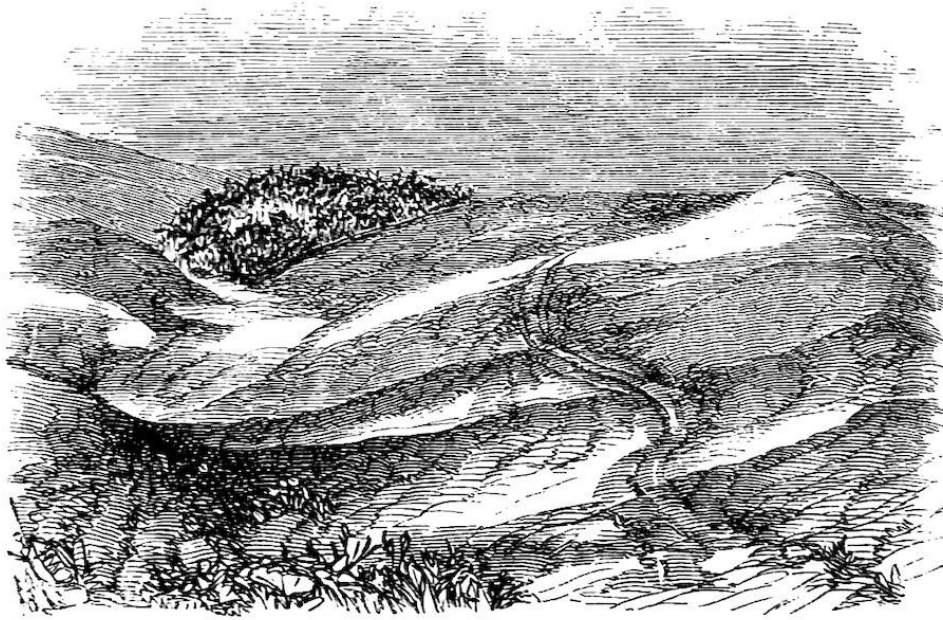
Buckinghamshire, Bedfordshire, Cambridgeshire, and Norfolk,—all this immense district, averaging probably a thousand feet in thickness, lay quietly at the bottom of the sea. Yes! it wasn't created just pat in a moment, and set up in these hills and valleys, any more than sand was created sand, but gradually, grain by grain, it accumulated, ages rolling by while thus the earth was a-preparing. These undulating downs knew then no boys pursuing on them their healthful and gymnastic games; no fair riders then crossed these pleasant turfy swards, inhaling life and vigour from the air of heaven; no evening promenades for men of business, recreating the overworked physical nature,—then diversified the scene: but instead thereof crumbling particles of older rocks covered the floor of the ocean with their dust, so making green sand, while the waste and débris of coral reefs, ground up and pulverized, poured in their lavish contributions, so making chalk; (?) while above this deposit thus forming, the ocean waves rolled to and fro, and in their rise and fall, their daily ebb and flow, sounded their eternal bass in the ear of old father Time, who certainly then had no human children. Had we been there we should have seen as tenants of the “vasty deep,” pectens, plagiostamas, hamites, belemnites, and other odd-looking crustaceans, roaming about at their own sweet will; some ganoid fishes that have now such hard names, that to write them would interfere with the strain of this description; while here and there we might have spied a veteran remnant of the oolite come down by strange chance into the chalk, a solitary pterodactyle, snapped at and pounced upon by an ugly mososaurus, a sea crocodile standing four feet high. These, and such as these, were the tenants of that ocean, which then rolled over our heaths and downs; and when at length their period of life and purpose of creation were answered, these too all perished, and the secondary formation ended its wondrous career. The old ocean still rolls on, unchanged, unaltered still; but what changes has it seen in these long distant epochs of which geology tries to tell the story!

“Time writes no wrinkle on thine azure brow;
Such as creation's dawn beheld, thou rollest now!”

No one can write the history of the Chalk formation, without remembering his indebtedness to the patient and laborious toil of Dr. Mantell in this department, from whose “Geological Excursions round the Isle of Wight,”^[104] we give the following extract. “The features of a chalk district are so well known, that a brief notice will render local details

unnecessary. The rounded summits of the hills, covered with short verdant turf—the smooth undulated outline of the downs unbroken, save by the sepulchral mounds of the early inhabitants of the country—the coombes and furrows ramifying and extending into the deep valleys, which abruptly terminate at the base of the hills, and appear like dried-up channels of rivulets and streams, though free from all traces of alluvial débris, thus bearing the impress of physical operations, of which the agents that produced them have long since passed away—are phenomena familiar to every one who has travelled over the downs of the south-east of England, and are displayed in the chalk districts of the ‘beautiful isle.’ These features are restricted to the hilly districts of the *white chalk*, and have resulted from the peculiar nature of the sedimentary detritus,^[105] of which the strata comprised in the upper division of the Cretaceous system are composed: for in the lower groups, clays, marls, sands, and sandstones prevail; and where these deposits approach the surface, and form the subsoil, the country is broken and diversified, and the landscape presents a striking contrast with the down scenery, as may be observed in the picturesque district which flanks the escarpment of the chalk hills. It may, perhaps, be necessary to remind the unscientific reader, that these strata are but an insulated portion of an ancient sea-bottom, or in other words, a mass of consolidated sediments formed in the profound depths of the ocean, in a very remote period of the earth’s physical history. This detritus is made up of inorganic and organic materials. The former consist of the débris of the cliffs and shores which encompassed the ancient sea, of the spoils of islands and continents brought down into the ocean by fresh-water currents, and of chemical deposits thrown down from mineral solutions. The organic substances are the durable remains of animals and plants which lived and died in the ocean, and of fluviatile and terrestrial species that were transported from the land by rivers and their tributaries; the whole forming such an assemblage of sedimentary deposits as would probably be presented to observation, if a *mass of the bed of the Atlantic ocean*, 2,000 feet in thickness, were elevated above the waters, and became dry land.”

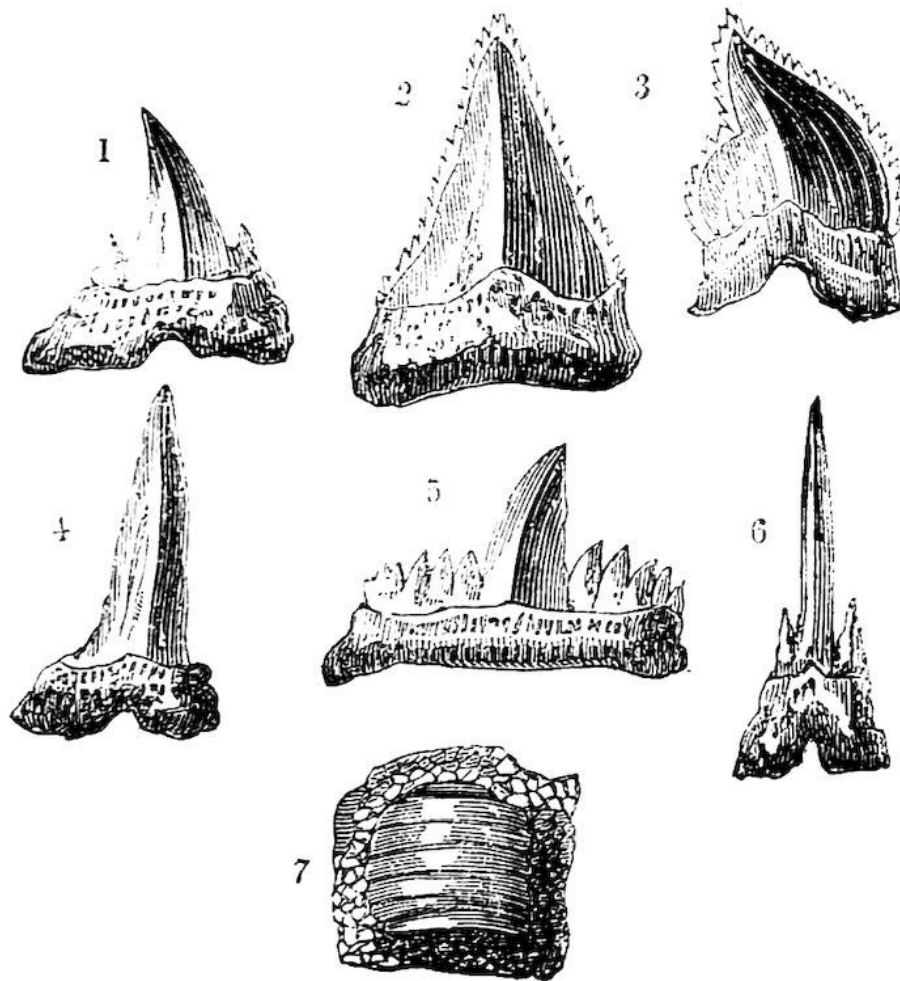
We have alluded to the undulating character of the downs, so “well known,” as Mantell says, that “local details are unnecessary.” How correct this is may be seen in the following drawing, which represents a portion of Royston Heath.



All over this heath is found the “Royston crow,” during the winter months. This fine bird migrates hither from Norway, to avoid its severe winters, and is scientifically known as the “Hooded-crow, *corvus cornix*.” On its first arrival, when it is in its best plumage, it is comparatively tame, allowing the sportsman to approach very near; but as the season advances, acquaintance with the gun makes it very knowing and shy. It associates freely with the other crows, but its nest has never yet been found in England. About March the hooded-crow wholly disappears. The head, throat, and wings are black; the back and breast a “clear smoke-grey.” Norman, the bird-stuffer of this town, has always several fine specimens on hand.

As, in the case of the Carboniferous system, we ventured to say to the reader that it was not all *coal*, so in the Cretaceous system, we would remind him that it is not all *chalk*; but without going minutely into the subdivisions which the chalk formation has received, because this unpretending elementary treatise does not profess to *teach* geology, but simply aims, as we have ventured again and again to repeat, to infuse into the mind a desire of acquaintance with the marvels and truths of this science, we will just indicate the leading divisions and nomenclature of this deposit. First, there is the green sand; that is, *first*, beginning at the bottom or lower part of the formation: this may be well seen and studied in the

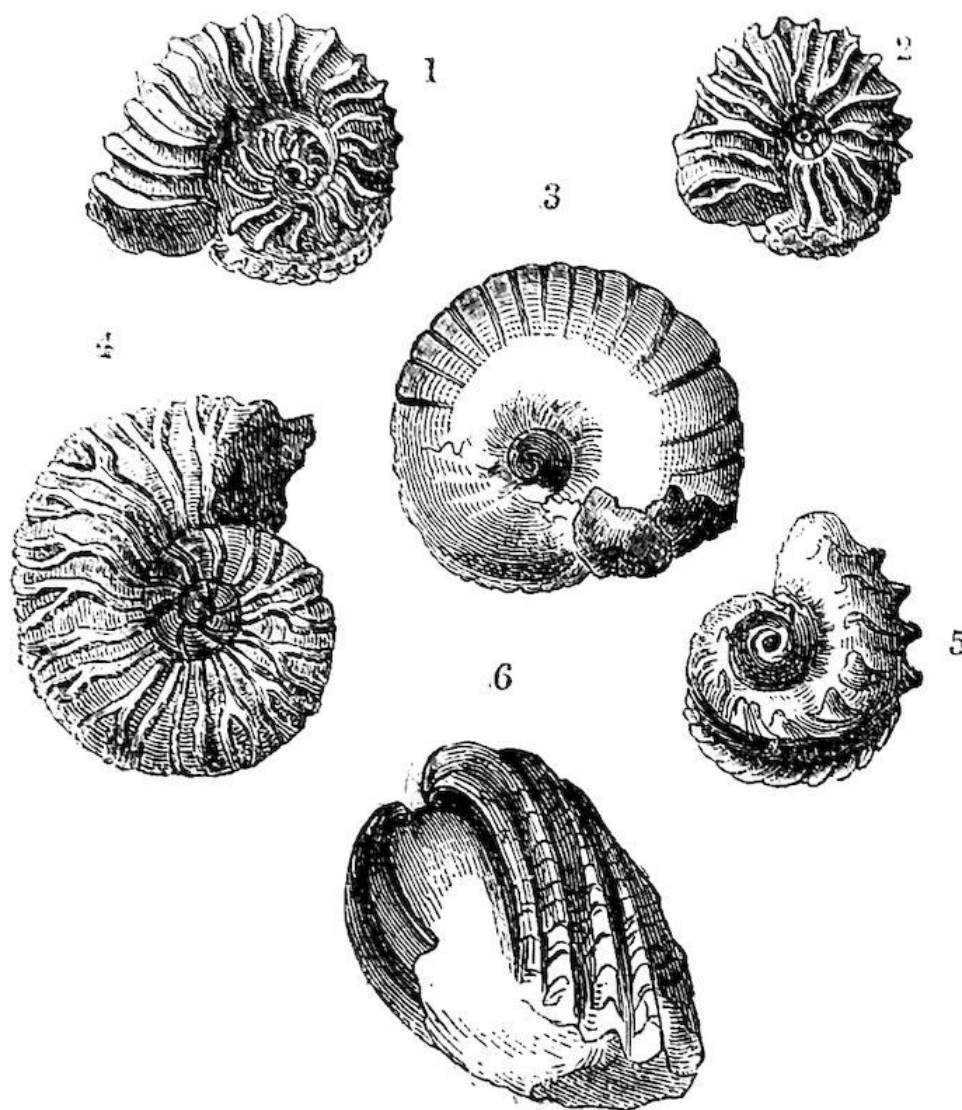
neighbourhood of Cambridge, where we have procured many of its characteristic fossils, including several vertebræ and teeth of the otodus, a fish allied to the shark family, such as are figured in the opposite diagram.



FOSSIL TEETH OF FISHES:

FROM UPPER GREEN SAND,
CAMBRIDGE.

1. OTODUS.
2. CARCHARIAS.
3. CORAX.
4. OXYRHINA.
5. NOTIDANUS.
6. LAMNA.
7. PTYCHODUS.

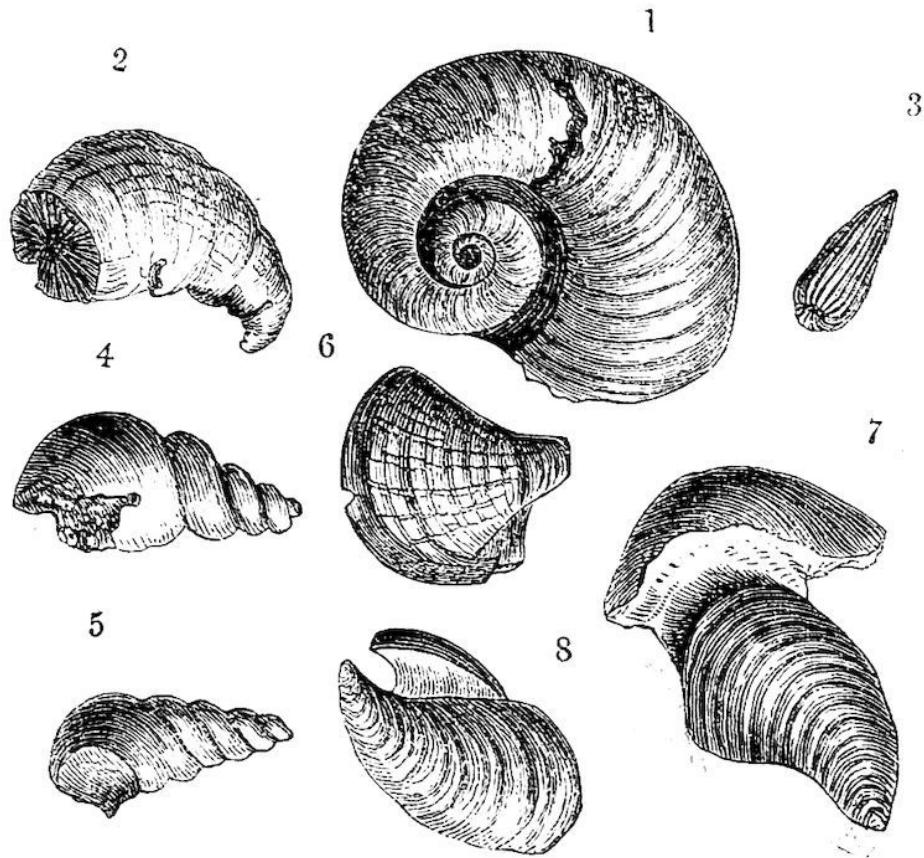


FOSSILS FROM THE GAULT,
FOLKSTONE.

1. AMMONITE DENTATUS.
2. AM. LAUTUS.
3. AM. SPLENDENS.
4. AM. CRISTATUS.
5. AM. DENARIUS.
6. CATILLUS SULCATUS.

At Potton and Gamlingay in Bedfordshire, and in the neighbourhood, this green sand is highly ferruginous, and the roads and fields present that peculiarly dark-red colour which is first singular and then wearisome to the

eye. In the case of the Potton beds, the red colour is caused by oxidization or rust of iron; in the neighbourhood of Cambridge, &c., where there is the *green sand*, this is owing to the influence of “chloritous silicate of iron.” Then we have the *galt* or *gault*, a local term of which we cannot trace the etymology. The *gault*, however, is not of great thickness, but is likely to be the most interesting department of the Chalk to the beginner, on account of the abundance and peculiar appearance of its fossils. A ramble under the cliffs at Folkstone,^[106] where the *gault* may be seen in perfection, will amply repay any one for toil, dirt, and a few slips and bruises. He will there find evidences of a prolific and prodigal bestowment of life in the innumerable fragments of organic remains every where observable; and if he be patient,—if he won’t go running on from spot to spot, saying, as some do, “Oh, there’s nothing here;” if he will just persevere in a minute examination of every spot where organic remains may be detected, he will not come away without his reward in ammonites, hamites, and other cephalopodous mollusks, and most of them with that peculiar nacreous or mother-of-pearl lustre upon them which renders the fossils of this period so beautiful and attractive. Only we caution the explorer not to buy of the so-called guides. At Dover and Folkstone the rogues have a knack of getting a lump of *gault*, and sticking into it one or two common pyrites, which are very abundant in the cliff, bits of shell, ammonites, &c.; they then offer this conglomerate for sale, all rounded and smooth, assuring you upon their “sacred honour,” the honour of men who always draw upon their imagination for their facts, that they would not ask so much for it, only on account of its excessive rarity. As good economists always avoid cheap houses, and go to the best shops, so let the young geologist always go to the best shop: let him go to the cliff with his hammer, and work for himself. We picture a few fossils from the *gault*, only regretting that it is out of the power of our artist to convey their lustrous colours, as well as their curious forms.



1. NATICA CANICULATA.

2.} VENTICRULITES.

3.}

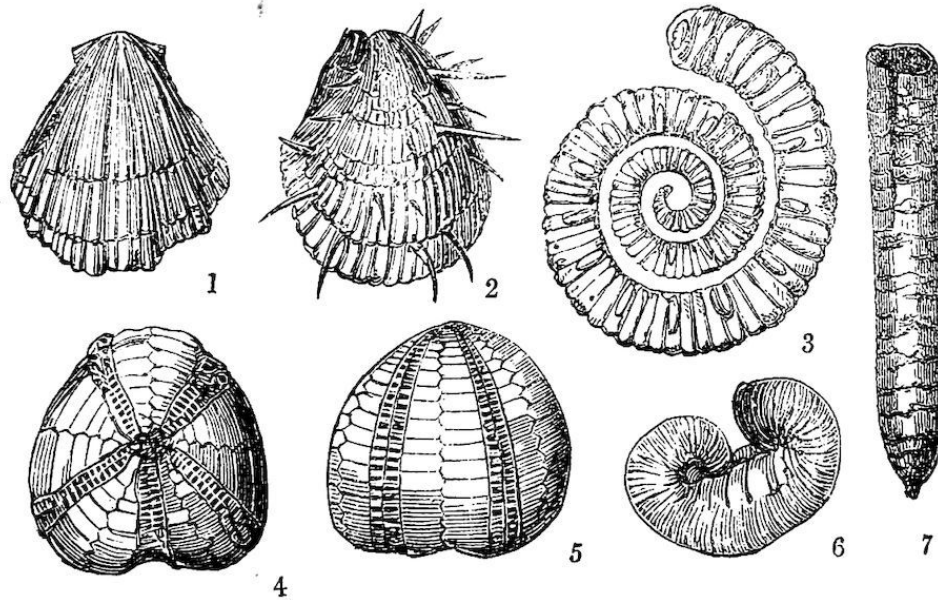
4.} ROSTELLARIA MARGINATA.

5.}

6.}

7.} CATILLUS CRISPI.

8.}

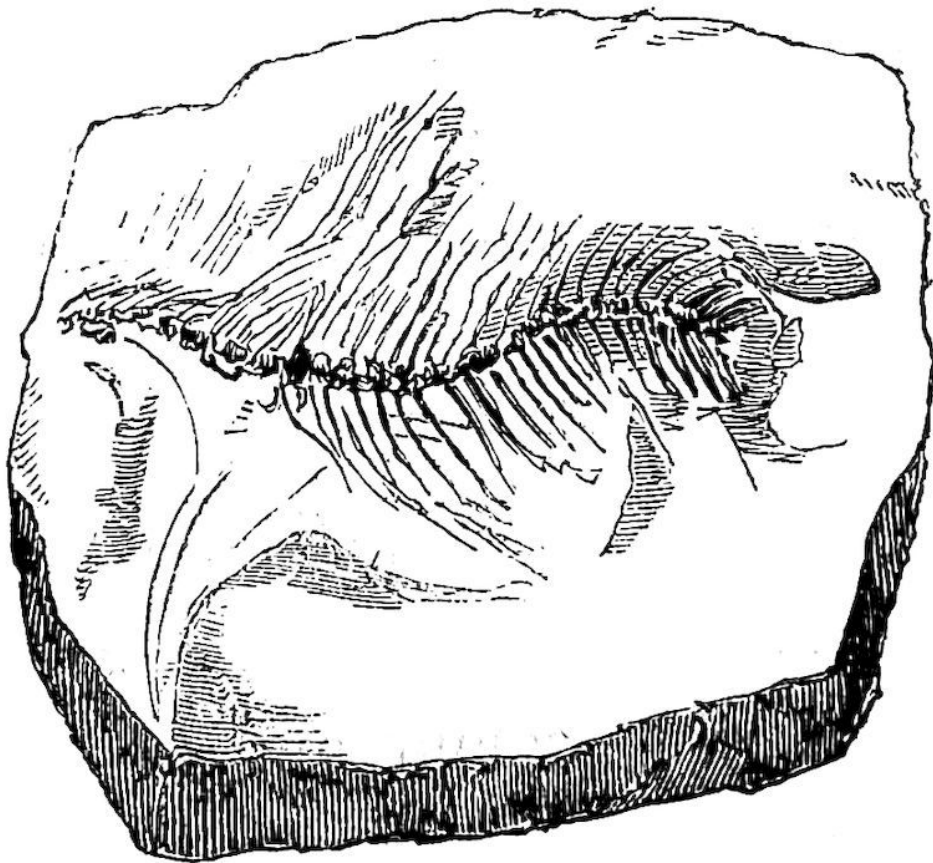


Then comes, lastly, the *Chalk*: that is, the *white* chalk, divided into lower and upper; the *lower* being harder and mostly without flints, and the upper characterised by layers and bands of flint, sometimes nodular, as in Cambridgeshire, and sometimes flat almost as a pancake, as in the neighbourhood of Woolwich.

Above are some of the most characteristic fossils of the Chalk. No. 1 is a *pecten*, or oyster, called the “five-ribbed,” or *quinque costatus*; No. 2 is the *plagiostoma spinosa*, so called on account of its spines, a shell found frequently in our chalk or lime-pits; No. 3 is the intermediate hamite (Lat. *hamus*, a hook), “*hamites intermedius*,” No. 4 is the *spatangus coranguinum*, a very common fossil *echinus* in the chalk; No. 5 is the *ananchytes ovata*, found frequently in the Brighton and Ramsgate cliffs; No. 6 is a scaphite (Gr. *skaphē*, a skiff or boat); and the last is our old friend the belemnite, who has survived so many of this earth’s changes, and now finds himself a contemporary of the cretaceous inhabitants of the globe.

In many respects, the Chalk presents us with remarkable anomalies: we have *sand*, the green sand, but unlike in colour and in texture the sand of the old and new red sandstone, where we find it compressed and hardened into solid and compact masses of stone; we have *clay*, argillaceous beds such as the gault, but it is not clay hard and pressed into slaty rocks, but soft and compressible; and we have *carbonate of lime*, the chalk constituting the

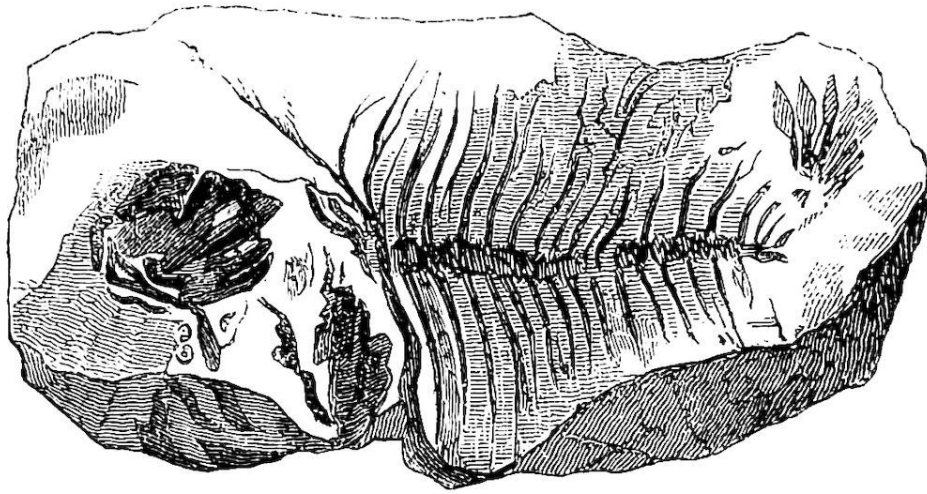
calcareous beds of this formation; but where we have met with it before it has been hard and solid limestone, and marble, not pliable and soft as in the Cretaceous system; and yet apparently it is all the same material as we have found in the earlier stages of the earth's crust—the washings, degradations, and deludations of older and harder rocks, along with the secretions and remains of organized animals that once peopled this ancient earth; thus affording us, on a large scale, another illustration of the economy observable in all the works of God.



FOSSIL FISH FROM LEWES.

Having spoken of the fossil fishes of the Chalk, we here give drawings of two procured from the neighbourhood of Lewes, the famous fossil fishing-ground of the late Dr. Mantell; and it is due to the name and memory of the Chalk historian and geologist, to inform the reader that Dr. M. was the first who succeeded, by skilful removal of the surrounding chalk, in procuring a perfect ichthyolite from the cretaceous formation of England. The British

Museum is now enriched by Dr. Mantell's collection of fossil fishes, that once so much excited the admiration of Agassiz, when he saw them at Brighton.



FOSSIL FISH (ICHTHYOLITE) FROM LEWES.

Here let us again advert to the Deluge theory, not because our own minds are not satisfied on the point, but because theology and science alike demand a true statement of the facts of the case. We believe, as we said in a previous chapter, in the plenary inspiration of nature, just as we believe that the Scriptures were given by inspiration of God; and we are quite sure that both books, if they are not misinterpreted, will declare the glory of God in one common speech, and elevate the mind of man, to whom they speak, up to a more adoring trust and a profounder reverence. With Dr. Hitchcock we say, "It seems to me that the child can easily understand the geological interpretation of the Bible and its reasons. Why, then, should it not be taught to children, that they may not be liable to distrust the whole Bible, when they come to the study of geology? I rejoice, however, that the fears and prejudices of the pious and the learned are so fast yielding to evidence; and I anticipate the period, when on this subject the child will learn *the same thing in the Sabbath school and in the literary institution*. Nay, I anticipate the time as not distant when the high antiquity of the globe will be regarded as no more opposed to the Bible, than the earth's revolution round the sun, and on its axis. Soon shall the horizon, where geology and revelation meet, be cleared of every cloud, and present only an unbroken and magnificent

circle of truth.”^[107] But to return; this Deluge theory refers all existing fossils to “Noah’s flood,”—to that violent diluvial action, the graphic account of which is in the book of Genesis. Now, it is impossible to believe this if we look at a fossil: look, for instance, at this terebratula, and observe how perfectly uninjured it is, frail as is its shelly covering; or at this plagiostoma spinosa, and mark how susceptible it is of injury, and yet that its brittle spines are all unbroken; or at this inoceramus or catillus, and observe its delicate flutings, still in exquisite preservation, without fracture or distortion; or these specimens of echinites, the ananchytes ovata, or the spatanguscor-anguinum, and see the markings on the shell, the apertures of the mouth and stomach still perfect; who can see all this and not come to the conclusion, that these creatures, and thousands such as these, endured not only no violence in death, such as a deluge would suppose, but that at death they subsided quietly to the bottom of the sea, there to find a fitting sepulchre of soft cretaceous matter prepared for them, which in process of time was lifted up, to exhibit in a hard chalky bed their forms of pristine beauty?

In the upper chalk every one has seen the layers of flint, and marked their singular distribution, in layers; and here we would add, that the existence of flint in chalk is one of those hard nuts which geology has not yet cracked. The geologist, the chemist, and the zoologist have all puzzled themselves in vain to find a truly satisfactory origin for these nodules of siliceous matter. We have heard it suggested that they may be *coprolites*; but no one who examines the texture of a flint, can hold that theory, to say nothing of the idea that the coprolites have been preserved, while the animal remains have perished. We may sum up all we have to say about flints in the following words, from that useful little book, Chambers’s “Rudiments of Geology:”

“*The formation of flint*, within a mass so different in composition as chalk, is still in some respects an unsolved problem in geology. It occurs in nodular masses of very irregular forms and variable magnitude; some of these not exceeding an inch, others more than a yard in circumference. Although thickly distributed in horizontal layers, they are never in contact with each other, each nodule being completely enveloped by the chalk. Externally, they are composed of a white cherty crust; internally, they are of a grey or black silex, and often contain cavities lined with chalcedony and crystallized quartz. When taken from the quarry they are brittle and full of moisture, but soon dry, and assume their well-known hard and refractory

qualities. Flints, almost without exception, enclose remains of sponges, alcyonia, echinida, and other marine organisms, the structures of which are often preserved in the most delicate and beautiful manner. In some specimens the organism has undergone decomposition, and the space it occupied either left hollow, or partially filled with some sparry incrustation. From these facts, it would seem that flints are as much an aggregation of silex around some organized nucleus, as septaria are aggregations of clay and carbonate of iron. This is now the generally received opinion; and when it is remembered that the organisms must have been deposited when the chalk was in a pulpy state, there can be little difficulty in conceiving how the silex dissolved through the mass would, by chemical affinity, attach itself to the decaying organism. Chalk is composed of carbonate of lime, with traces of clay, silex, and oxide of iron; flint, on the other hand, consists of 98 per cent. of pure silex, with a trace of alumine, oxide of iron, and lime. Silex is quite capable of solution: it occurs in the hot-springs of Iceland and most thermal waters; has been found in a pulpy state within basalt; forms the *tabasheer* found in the cavities of the bamboo, and the thin pellicle or outer covering of canes, reeds, grasses, &c.; and siliceous concretions are common in the fruits and trees of the tropics. All these facts point to a very general diffusion of silex in a state of solution; and whatever may have caused its abundance in the waters during the deposition of the upper chalk, there can be little doubt respecting the mode in which it has been collected around the organic remains of these early seas."

At Scratchell's Bay in the Isle of Wight, it will be seen that the flints are in a vertical position; and to the most casual observer the perpendicular arrangement of these flints will supply the strongest evidence of disturbance by upheaval from below. The bay in front, called Scratchell's Bay, is a small but romantic indentation in the coast of the south side of the island, in which are the famous Needles. In the face of the cliff is a noble archway between 200 and 300 feet high, which has been created by the constant action of water eating and wearing away the lower beds; while the Needles themselves are only isolated masses of chalk, separated or eroded from the main land by the same erosive action. "To the late Sir Henry Inglefield belongs the merit of having first observed and directed attention to the highly interesting phenomena of vertical chalk strata, occasioned by the disruption and elevation of the eocene and cretaceous formations, which are so remarkably displayed in the Isle of Wight, where the vertical position of

the strata, and the shattered condition of the flint nodules, thought still embedded in the solid chalk, may be conveniently studied in the cliffs in the neighbourhood of Scratchell's Bay."^[108]

With the study of the Chalk formation, we close what has been appropriately termed the "secondary period, or middle epoch of the ancient world;" of which it has been well said, "In reviewing the characters of the Cretaceous group, we have evidence that these varied strata are the mineralized bed of an extensive ocean, which abounded in the usual forms of marine organic life, as algæ, sponges, corals, shells, crustacea, fishes, and reptiles. These forms are specifically distinct from those which are discovered in the tertiary strata; in many instances, the genus, in all the species, became extinct with the close of the Cretaceous period. It affords a striking illustration of creative power, that of the hundreds of species which composed the Fauna and the Flora of the Cretaceous group, not one species passed into the succeeding epoch."^[109]

Of that old ocean with its countless tenants we have already spoken, and conclude by applying to it the well-known lines of Montgomery, in his celebration of the coral insect in his "Pelican Island:"—

Millions of millions here, from age to age,
With simplest skill, and toil unwearable,
No moment and no movement unimproved,
Laid line on line, on terrace, terrace spread,
To swell the heightening, brightening gradual mound,
By marvellous structure climbing towards the day.
Omnipotence wrought in them, with them, by them;
Hence what Omnipotence alone could do,
Worms did.



WALTONIAN AND MANTELLIAN FISHERMEN.

CHAPTER XII.

THE TERTIARY SYSTEM.

“And God made the beast of the earth after his kind.”

MOSES.

In our rapid sketch of the materials constituting the crust of the earth, we first of all, in that imaginary section which we supposed to have been laid bare to us, studied the characters of the hypogene rocks,^[110] that make up the Azoic period, in which, with the exception of a few zoophytes, all nature was void of animal, life and possessed only by the genius of dread silence. Rising higher, we surveyed the Palæozoic or primary rocks, where the fishes of the Old Red Sandstone convinced us of progress in the forms of life, and taught us our first lesson in the ascending scale of those types of life with which Palæontology has now made us familiar. Leaving this period at the Carboniferous era, we entered upon the Mesozoic,^[111] or Secondary period, ushered in amidst strange convulsions that must again and again have rendered the earth “without form and void;” and here we found ourselves in company with the strange and gigantic remains of a higher order of vertebrated animals, the saurians, the crocodile-kings of a bygone period; and as we pondered these hieroglyphics of past generations, our souls “were seized the prisoners of amaze;” and now, in our upward ascent, leaving behind us the scenes

“Where eldest Night
And Chaos, ancestors of nature, held
Eternal anarchy,”

we come to the Cainozoic,^[112] or Tertiary Rocks, where other and higher types of life are found. Huge mammals, beasts of prodigious size, are now found inhabitants of the earth, the precursors of man—reasoning,

intelligent, responsible man, who is presently to make his appearance on this great theatre of life, “made a little lower than the angels,” to have dominion over the works of Jehovah’s power.

Sir H. de la Beche proposes, for tertiary, the term “*supercretaceous*;” it is, however, of little consequence which term is adopted, the meaning in each case being the same, that all the rocks or strata lying above the chalk are to be considered as belonging to the tertiary system or series. Confessedly, it is a dark period in the history of those successive creations which have been engaging our attention, for we can trace no near connexion between the secondary or older, and tertiary or newer formations. That is to say—and the bare statement appears so sufficient and final a refutation of what has been termed the “development hypothesis,” now recognised as contradictory to fact and to Scripture—that there are not known to exist in any of these newer strata the same beings, or the descendants of the same beings, that were found upon the earth at the termination of the chalk deposit.

Nor is this all; not only are none of the old fossils found in any one of the three divisions of this system, but we are introduced at once to so many new ones, that their species and genera are almost endless; and he is not only a geologist of mark, but a most singularly accomplished geologist, who thoroughly understands their minute subdivisions, and can appropriately classify the fossils of this most fossiliferous era. To make the matter as simple as possible, let us add that “the broad distinction between tertiary and secondary rocks is a *palæontological* one. None of the secondary rocks contain any fossil animals or plants of the *same species* as any of those living at the present day. Every one of the tertiary groups *do* contain some fossil animals or plants of the *same species* as those now living.”^[113]

Having alluded to the threefold division of this series of rocks, we shall proceed to notice them, dwelling a while upon each, and showing the principle on which each is based, as originated and enunciated by Lyell. Of the three divisions, the first is called *Eocene* (*ēōs*, the dawn, and *kainōs*, recent), by which term is represented the oldest or lowest of this tripartite series. Then we have the *Miocene* (*meiōn*, less, and *kainōs*, recent)—a name, we think, not the most appropriate, and likely to mislead the beginner, because really it represents a series of beds, *more* and not *less* recent than the *Eocene*; but the idea of the name is this (and it must be

carefully borne in mind), that although it is *more* recent than the series of beds *below*, it is *less* recent than those *above*; it is nearer the “dawn” of our present era than the *Eocene*, but not so near the dawn as the *Pliocene*. This last term, the *Pliocene* (from *pleiōn*, more, and *kainōs*, recent), is applied to the newest of the beds of the tertiary series in which there are found many *more recent* than extinct shells. The Tertiary system or series, then, is divided into these three sections: viz. 1. The older Tertiary or Eocene; 2. The middle Tertiary or Miocene; and 3. The newer Tertiary or Pliocene.

We used a term just now, in quoting from Mr. Juke’s most useful manual, which we will explain; we said, the “distinction between the secondary and tertiary rocks is wholly a *palæontological* one;” that is, it is a distinction founded not on the character of the rocks, but on the character of the *organic remains* found in the rocks. “This character,” says Sir Charles Lyell, “must be used as a criterion of the age of a formation, or of the contemporaneous origin of two deposits in different places, under very much the same restrictions as the test of mineral composition.

“First, the same fossils may be traced over wide regions, if we examine strata in the direction of their planes, although by no means for indefinite distances.

“Secondly, while the same fossils prevail in a particular set of strata for hundreds of miles in a horizontal direction, we seldom meet with the same remains for many fathoms, and very rarely for several hundred yards, in a vertical line, or a line transverse to the strata. This fact has now been verified in almost all parts of the globe, and has led to a conviction, that at successive periods of the past, the same area of land and water has been inhabited by species of animals and plants even more distinct than those which now people the antipodes, or which now coexist in the Arctic, temperate, and tropical zones. It appears that, from the remotest periods, there has been ever a *coming in of new organic forms*, and an *extinction of those which pre-existed on the earth*, some species having endured for a longer, others for a shorter time; while none have ever reappeared after once dying out. The law which has governed the creation and extinction of species seems to be expressed in the verse of the poet—

‘Nature made him, and then broke the die;’

and this circumstance it is which confers on fossils their highest value as chronological tests, giving to each of them, in the eyes of the geologist, that authority which belongs to contemporary medals in history. The same cannot be said of each peculiar variety of rock; for some of these, as red-marl and red sandstone for example, may occur at once upon the top, bottom, and middle of the entire sedimentary series; exhibiting in each position so perfect an identity of mineral aspect, as to be undistinguishable. Such exact repetitions, however, of the same mixtures of sediment, have not often been produced, at distant periods, in precisely the same parts of the globe; and even where this has happened, we are seldom in any danger of confounding together the monuments of remote eras, when we have studied their imbedded fossils and relative position.”^[114]

Let us now briefly explain the very simple but satisfactory basis on which this threefold division of the Tertiary rocks rests, and then proceed to a brief explanation of each. The reader will already have noted a statement on a preceding page, to which we shall be pardoned, if, a second time, we ask attention to it. In imparting elementary instruction on geology we have always found our classes more or less puzzled by the tertiary system, on account of its nomenclature and minute subdivisions, and we have learnt from experience the importance of presenting this statement over and over again. We have remarked that in the *secondary rocks*—that is, up to the end of the chalk system—there are *no* organic remains found precisely similar to any species existing at the present day; but when we come to the tertiary rocks, although we find many strangers, we find also a good many organic remains of the same kind and character as the shells, that are now found on our shores. In one part of the Tertiary the number of fossils that belongs to existing genera is many, in another more, in another more still; and upon this simple idea of positive, comparative, and superlative, the present division is based. Taking the *percentage* principle as a guide, Sir Charles Lyell and a distinguished French geologist, M. Deshayes, have ascertained that in the lowest beds of this system there were only 3½ per cent. of fossil shells similar to existing species, and this, for “the sake of clearness and brevity,” says Sir Charles, “was called the *Eocene* period, or the period of the dawn, the dawn of our modern era so far as its testaceous fauna are concerned.” Rising higher in the examination of these rocks, certain strata were found containing 18 per cent. of fossil shells, similar to shells found now; and to this was given the name of *Miocene*, the puzzling name already

spoken of, because it means *less* recent, whereas it is in reality *more* recent, and is to be understood in relation to the series *below*, and not to the series *above*. A step higher up in this system revealed deposits of a coralline and craggy character, in which 41 per cent. of fossil shells like those of the present era were found; and to this the name of *Pliocene*, or more recent still, was given; and latterly, in Sicily chiefly, a series of strata has been discovered, referable to the Tertiary, in which 95 per cent. of recent species of shells have been found, and to this series the name of Post-Pliocene, or *Pleistocene*, has been given. Before our description of each of these divisions, let us add, that “the organic *remains of the system* constitute its most important and interesting feature. The fossils of earlier periods presented little analogy, often no resemblance, to existing plants and animals; here, however, the similitude is frequently so complete, that the naturalist can scarcely point out a distinction between them and living races. Geology thus unfolds a beautiful gradation of being, from the corals, molluscs, and simple crustacea of the grauwacke—the enamelled fishes, crinoidea, and cryptogamic plants of the lower secondary—the chambered shells, sauroid reptiles, and marsupial mammalia of the upper secondary—up to the true dicotyledonous trees, birds, and gigantic quadrupeds of the tertiary epoch. The student must not, however, suppose that the fossils of this era bring him up to the present point of organic nature, for thousands of species which then lived and flourished became in their turn extinct, and were succeeded by others long before man was placed on the earth as the head of animated existence. Of *Plants*, few marine species have been detected; but the fresh-water beds have yielded cycadeæ, coniferae, palms, willows, elms, and other species, exhibiting the true dicotyledonous structure. Nuts allied to those of the cocoa and other palms have been discovered in the London clay; and seeds of the fresh-water *characeæ*, or stoneworts, known by the name of *gyrtonites* (Gr., *gyros*, curved, and *gonos*, seed), are common in the same deposit. Of the *Radiata*, *Articulata*, and *Mollusca*, so many belong to existing genera, that this circumstance has suggested the classification of tertiary rocks according to the number of recent species which they contain.”—*Chambers’ Outlines*, p. 147.

Let us now begin *the Eocene period*. The most remarkable formations of this period are the London and Hampshire basins. Of the London basin we have already spoken in a previous chapter; a few additional remarks will be sufficient. The diagrams 4 and 5, p. 25, will explain these tertiary deposits

better than any verbal explanation; and when it is remembered that this bed of clay is probably a thousand feet in thickness, we get a passing illustration of the folly of those puerile reports which a few years since were industriously circulated about a coming earthquake in London. Poor, uneducated people took up the alarm rather anxiously, never dreaming of what any tyro in science would have told them—that supposing there was a subterranean chimney on fire down below, there was a wet blanket under their feet composed of a thousand feet of sodden and solid clay, a blanket of the material they may see in the deep cuttings of the Great Northern Railway in and about London, that would most effectually have put out any fire, or checked the progress of any earthquake, just as a cannon-ball is stopped dead by a woolsack.

A run down the river Thames will take any one who has a day to spare to the isle of Sheppey, where he will be amply rewarded by seeing, on the north side of the island, an exposure of this formation in the cliff laid bare to the height of 200 feet, and which pleasure trip will be amply rewarded by the discovery *in situ* of the fossil tropical plants, &c., that once flourished in the neighbourhood of our cold and foggy London. “At the entrance of the Thames, the London clay extends on both sides of the river, and is admirably exhibited in the isle of Sheppey, which consists entirely of this stratum. The cliffs on the north side of the island are upwards of 200 feet high, and are cut down vertically by the action of the sea; they have long been celebrated for the remarkable abundance and variety of the organic remains obtained from them, amongst which, perhaps, the most interesting are the fruits, berries, and woody seed-vessels of several hundred species of plants. From the same locality there have also been obtained the remains of upwards of fifty species of fish, and a considerable number of crustaceans, and many other invertebrata; besides some remarkable bones which have been described by Professor Owen, and which indicate the former existence in this island of large serpents, and of such birds as prey upon small reptiles and mammalia. Many of these fossils, especially those of plants, are very difficult to preserve, owing to the great tendency of the iron pyrites, which enter largely into their composition, to effloresce and be destroyed by exposure to the atmosphere.”^[115]

Passing from the London basin to the Hampshire basin or Barton beds, we shall first give a group of the shells found here; and we wish our readers could look at them as they lie before us in their condition of most exquisite

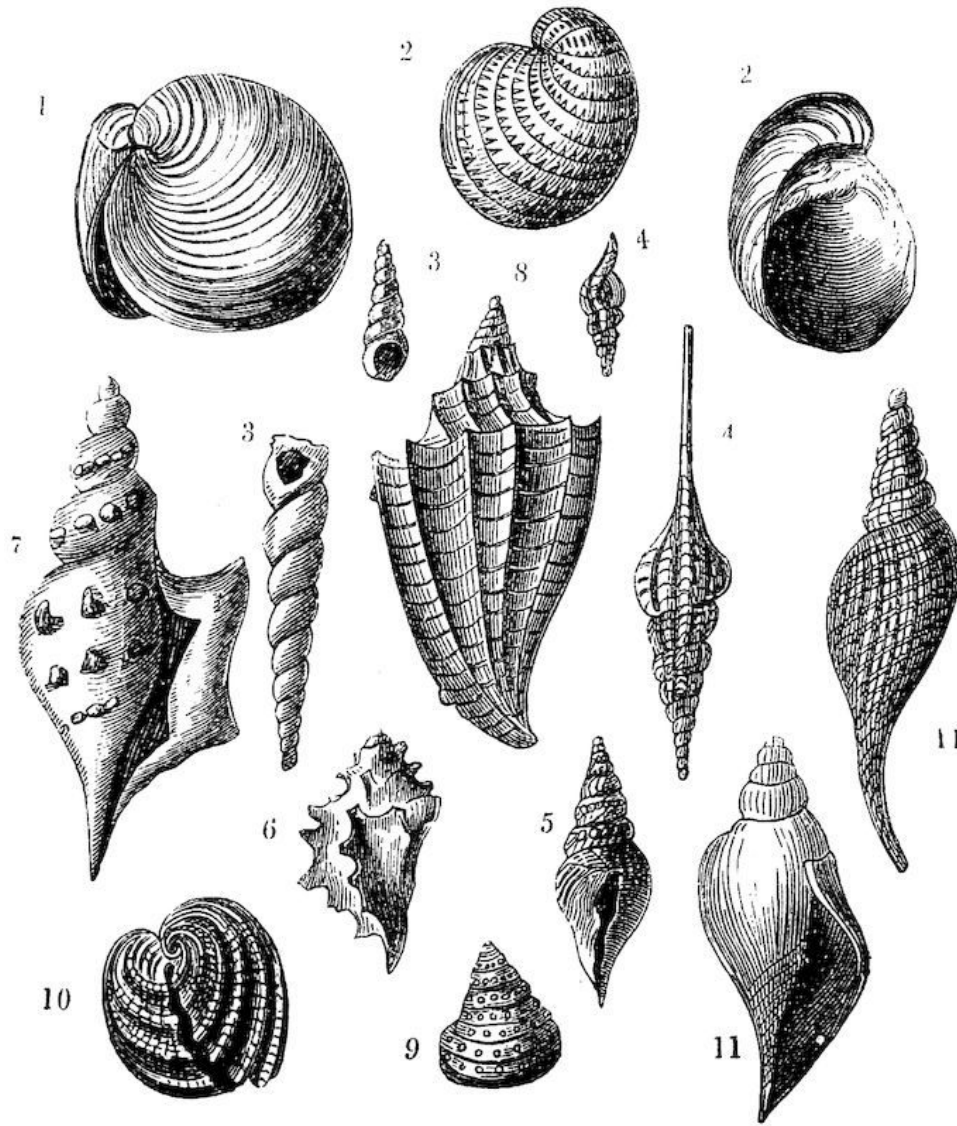
preservation, so exquisite, that those who have seen them have involuntarily and frequently exclaimed, “*But these can’t be fossils!*” I know of no picture-painting of past history so touching, and yet so true, as these lovely specimens of the shells of the pre-Adamite condition of England in all their native simplicity. To those who see in them shells, and only shells, why, in the name of the prophet, give them *figs*, while we again remember Wordsworth’s hero,—

“A primrose on the river brim,
A yellow primrose was to him,
And it was nothing more.”

To us they speak a wondrous story, replete with the knowledge that maketh glad the heart of man, because it is purifying, elevating knowledge; and though it does not teach the peculiar truths of theology, and we heartily wish that geology had been allowed to tell only its own tale of Creation—for here, as elsewhere,

“Nature, when unadorned,
Is then adorned the most”—

instead of being put to the rack, and made to suggest the special truths of Revelation,^{[\[116\]](#)} with which it has nothing to do;—although, we say, it does not teach the peculiar and special truths for which a Revelation was needed, it everywhere throws light on the boundless treasures of wisdom and care and beneficent Providence of the God of revelation. But again we catch ourselves sermonizing: to the diagram.

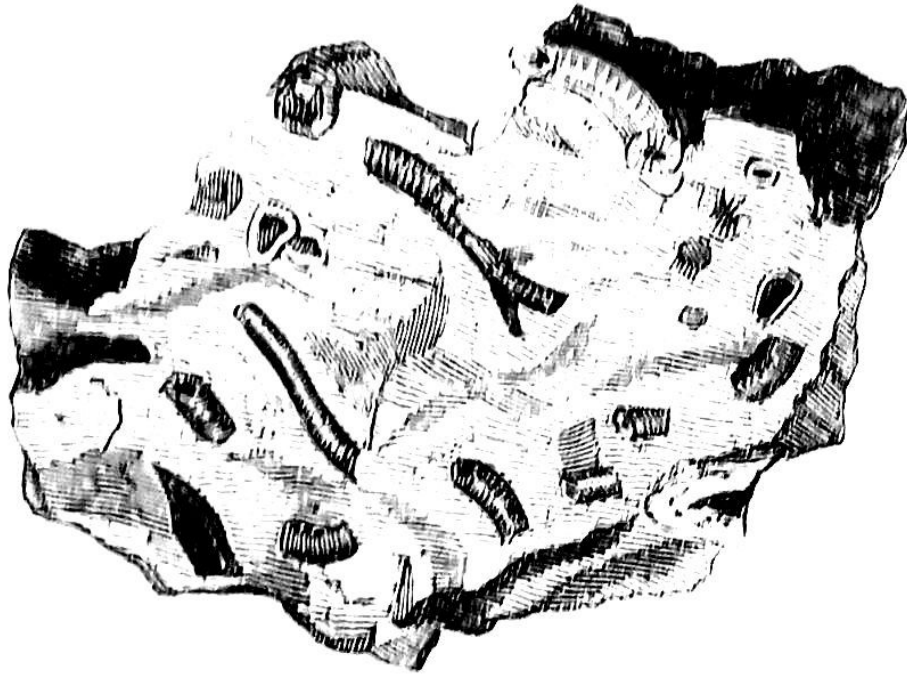


FOSSILS FROM THE LONDON CLAY.

1. *TELLINA CRASSA*.
2. *CHAMA SQUAMOSA*.
3. *TURRITELLA IMBRICATARIA*.
4. *FUSUS ASPER*.
5. *PLEUROTOMA COLON*.
6. *MUREX TUBIFER*.
7. *APORHAIS PES-PELICANI*.
8. *VOLUTA LUCTATOR* (or *luctatrix*).
9. *TROCHUS MONOLIFER*: the necklace trochus.
10. *VENERICARDIA COR-AVIUM*.
11. *FUSUS BULBIFORMIS*: the bulb fusus.

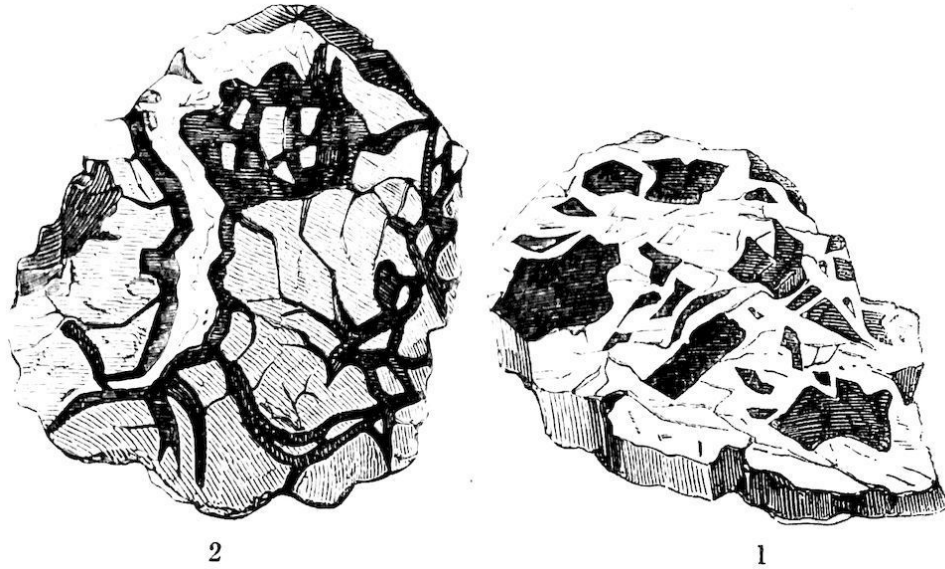
These fossils we obtained from the neighbourhood of Christchurch; and as these sheets were being written, we received from Dr. Mantell's "Geological Excursions in the Isle of Wight," the following appropriate description of them: "The numerous marine fossil shells which are obtained from this part of the coast of Hampshire, are generally known as Hordwell fossils; but it is scarcely necessary to remark, that they almost entirely belong to the London clay strata, and are procured from Barton cliffs. These fossils are most conveniently obtained from the low cliff near Beacon Bunny, and occur in greatest abundance in the upper part of the dark green sandy clay. There are generally blocks of the indurated portions of the strata on the beach, from which fossils may be extracted. A collection of Hordwell fossils, consisting of the teeth of several species of sharks and rays, bones of turtles, and a great variety of shells, may be purchased at a reasonable price of Jane Webber, dealer in fossils, Barton cliff, near Christchurch."—(P. 124.)

Before leaving the Eocene, or rather the London clay of the Eocene, we will give a drawing of a fossil in our possession. The drawing opposite represents a piece of fossil wood, pierced through and through by *Teredinæ*, a boring mollusk allied to the *Teredo*, which still proves so destructive to our vessels. Although the wood is converted into a stony mass, and in some parts covered by calcareous matter, the same as is found in the septaria, so common in these beds, to which we shall presently direct attention, still the grain and woody texture are most distinct. This wood was once probably floating down what we now call the Thames, when these piercing, boring mollusks seized hold upon it, penetrated its soft texture, and lived, moved, and had their being down at the bottom of the river in their self-constructed chambers. Time rolled on, and the log of wood is floated upon the shore, and there it lies to harden and to dry; again the log is drifted away, and, buried in some soft bed of clay, is preserved from rotting. In process of time it again sees the light; but now saturated by argillaceous material, and when hardened by the sun, becomes the petrification such as we see it.



WOOD PERFORATED BY *TEREDINA PERSONATA*, LONDON CLAY.

Here let us refer to the septaria, of which we have just spoken; two specimens lie before us, which we will briefly describe. In one (1) the clay is in distinct lozenge-shaped masses of a blue colour, while veins of calcareous spar or crystallized carbonate of lime surround these, which are capable of a beautiful marble-like polish; in the other (2) the clay is of the same colour, only in larger proportions, and the spar is of a deep brown colour, while here and there portions of iron pyrites may be seen; they become beautiful ornaments in a room when cut and polished. It should be added, that the septaria are not without their economic uses, being extensively used as cement after being stamped and burnt.



SEPTARIA

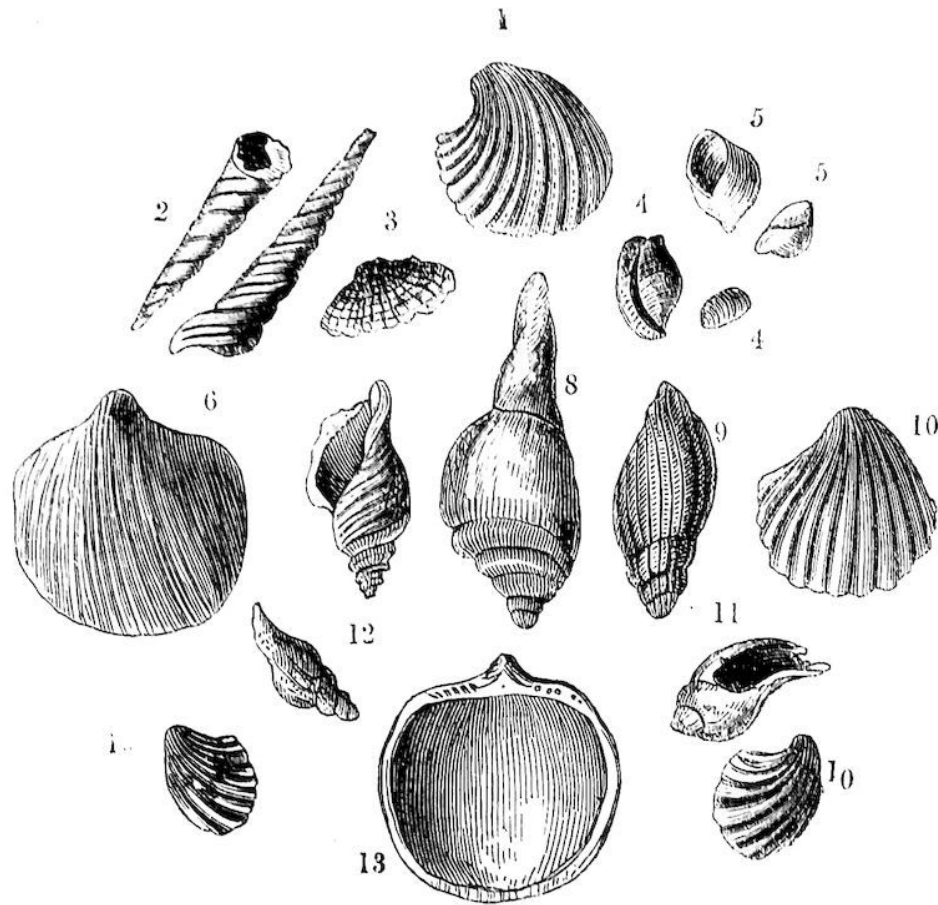
Here we may leave this brief sketch of the Eocene, or lowest beds of the Tertiary. A new creation has been introduced to our view; and although we still wait for the coming of man—the lord and interpreter of all—the contemplation of these successive acts and centres of creation fills our minds with renewed admiration and reverence of Him for whom, and by whom, and to whom are all things. Thus “even Geology, while it has exhumed and revived long buried worlds, peopled with strange forms in which we can feel little more than a speculative interest, and compared with which the most savage dweller in the wilderness of the modern period—jackal, hyæna, or obscene vulture—is as a cherished pet and bosom friend, has made for us new bonds of connexion between remote regions of the earth as it is, on account of which we owe it a proportionate share of gratitude.”^[117]

No. II.—*The Miocene.*

We shall briefly pass over this period. At Bordeaux, Piedmont, and in Lisbon, this formation is seen; as well as in various other parts of the Continent of Europe. The supposition of Geology is, that during this period “whole regions of volcanoes burst forth, whose lofty but now tranquil cones can be seen in Catalonia, in Spain, in France, Switzerland, Asia, and in America. The Alps, the Carpathian Mountains, and other lofty ranges were at this period partially upheaved. The researches of Sir Robert Murchison have established this fact, by his finding deep beds of limestone, characteristic of the Tertiary period, on the summit of one of the loftiest of the Alps, fully ten thousand feet above the level of the sea.”

No. III.—*The Pliocene Period.*

This term has already been explained. We shall only detain the reader by a few words respecting the organic remains that characterize this formation. In England it is confined to the eastern part of the county of Suffolk, where it is called “Crag.” This is a mere provincial name, given particularly to those masses of shelly sand which are used to fertilize lands deficient in calcareous matter. The geological name given to this strata is the “Red or Coralline Crag;” and it is so called on account of the deep ferruginous colour its fossils have through extensive oxidization of iron. We give drawings of the fossils of the Red Crag, obtained from the neighbourhood of Ipswich.



FOSSILS FROM THE RED CRAG,
NEAR IPSWICH.

1. VENERICARDIA SENILIS.
2. TURRITELLA.
3. PATELLA ÆQUALIS.
4. CYPREA.
5. PALUDINA LENTA.
6. PECTUNCULUS VARIABILIS.
7. MUREX.
8. FUSUS CONTRARIUS.
9. BUCCINUM ELONGATA.
10. VENERICARDIA SCALARIS.
11. VOLUTA LAMBERTI.
12. FUSUS ASPER.
13. PECTUNCULUS PILOSUS.

But these are not the only fossils of this period; it is here we meet, and that for the first time, with the highest form of animal life with which the researches of geology have made us acquainted. We have traced life in

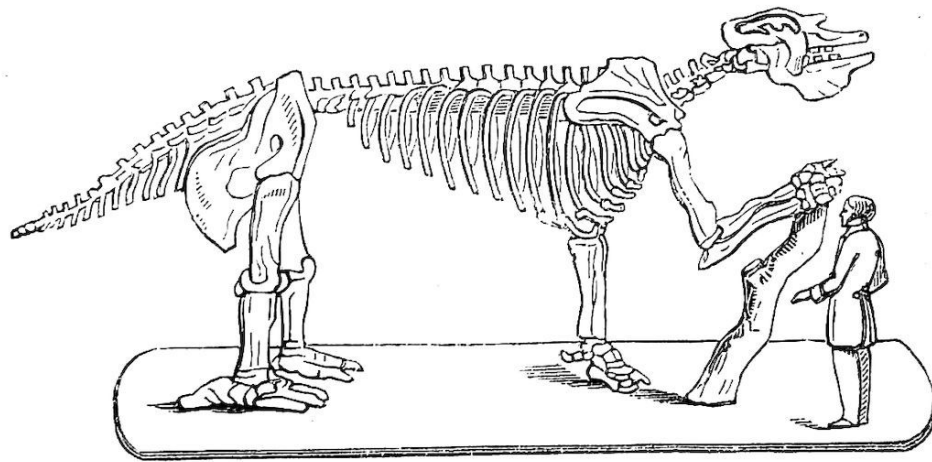
various forms in the different rocks that have passed under our rapid survey, and in all we have seen a wondrous and most orderly gradation. We began with the coral zoophytes, and from them proceeded to the mollusks and crustacea of the hypogene rocks; ascending, we discovered “fish with glittering scales,” associated with the crinoids and cryptogamous plants of the secondary series of rocks; and then we arrive where we are now, among the true dicotyledonous and exogenous plants and trees, with the strange birds and gigantic quadrupeds of the tertiary period. But the student must not imagine that even the fossils of this epoch bring him up to the modern era, or the *reign of man*; for even in the tertiary system numberless species lived and flourished, which in their turn became extinct, to be succeeded by others long before man, the chief of animals and something more, made his appearance, to hold dominion over these manifold productions of creative skill and power. But amidst these creations,

“God was everywhere, the God who framed
Mankind to be one mighty human family,
Himself their Father, and the world their home.”

It would be altogether beside the purpose of this preliminary treatise to enter into any details respecting the animals that have been found in such abundance in the Norwich Crag, that it has been called the “Mammaliferous Crag.” Those who desire full and deeply interesting information on this question should consult Owen’s noble work, entitled “British Fossil Mammals and Birds,” where, under the respective divisions of Eocene, Miocene, and Pliocene, he will see a complete chart of our riches in the possessions of a past creation. For the discovery of the Siberian Mammoth so often quoted, we shall refer to the same work, page 217, &c.; from which we shall only quote one brief extract, illustrative of the abundance of these remains in our own coasts in the ages past.

“Mr. Woodward, in his ‘Geology of Norfolk,’ supposes that upwards of two thousand grinders of the mammoth have been dredged up by the fishermen of the little village of Happisburgh in the space of thirteen years. The oyster-bed was discovered here in 1820; and during the first twelve months hundreds of the molar teeth of mammoths were landed in strange association with the edible mollusca. Great quantities of the bones and tusks of the mammoth are, doubtless, annually destroyed by the action of the waves of the sea. Remains of the mammoth are hardly less numerous in

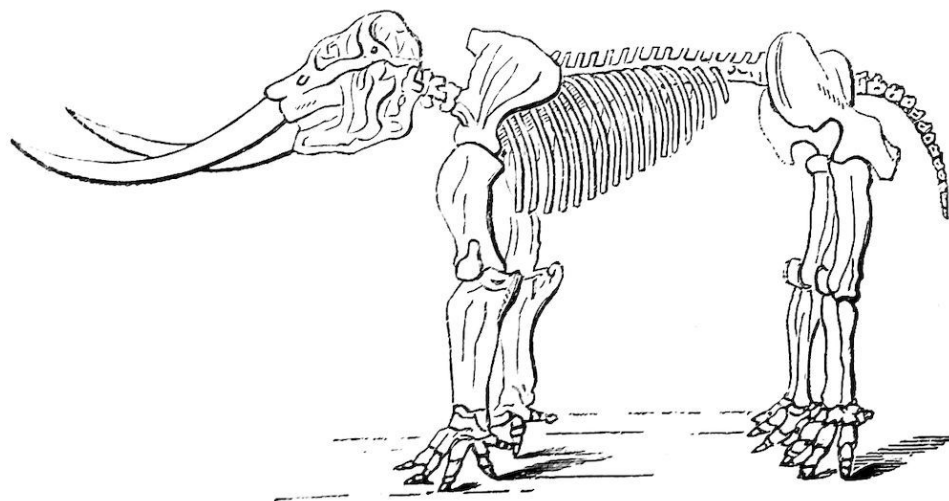
Suffolk, especially in the pleistocene beds along the coast, and at Stutton;—they become more rare in the fluvio-marine crag at Southwold and Thorp. The village of Walton, near Harwich, is famous for the abundance of these fossils, which lie along the base of the sea-cliffs, mixed with bones of species of horse, ox, and deer.”^[118]



SKELETON OF THE MEGATHERIUM CUVIERI (AMERICANUM).

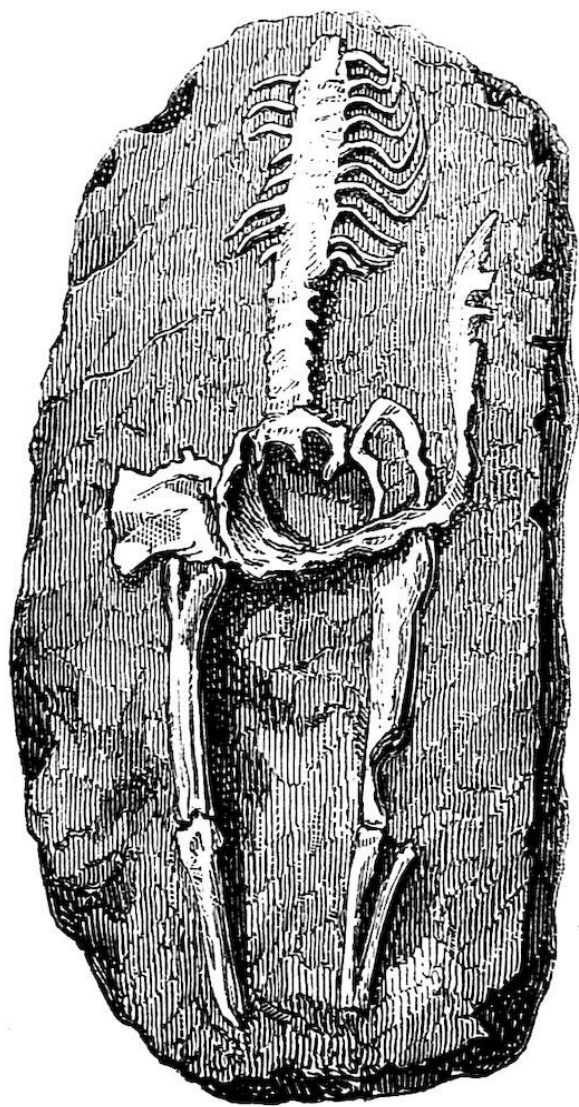
All the animals of this period are called theroid animals: from *therion*, a wild beast; and looking at the skeletons as they have been arranged from the few existing fossils, or from nearly complete materials—a matter not of guess-work, but of the most rigid application of the principles of comparative anatomy—we stand astounded at the prodigious sizes of these mammoths of the tertiary era. There is the *deinotherium*, or fierce wild beast; the *palæotherium*, or ancient wild beast; the *anoplotherium*, or unarmed wild beast, and others. We give above a drawing of the well-known *megatherium*, or great wild beast, to be seen in the British Museum, and add the following from Mantell’s Guide to the Fossils of the British Museum:—“This stupendous extinct animal of the sloth tribe was first made known to European naturalists by a skeleton, almost entire, dug up in 1789, on the banks of a river in South America, named the Luxon, about three miles south-east of Buenos Ayres. The specimen was sent to Madrid, and fixed up in the Museum, in the form represented in numerous works on natural history. A second skeleton was exhumed at Lima, in 1795; and of late years Sir Woodbine Parish, Mr. Darwin, and other naturalists have sent bones of the megatherium, and other allied genera, to England. The model

of the megatherium has been constructed with great care from the original bones, in the Wall-cases 9, 10, and in the Hunterian Museum. The attitude given to the skeleton, with the right arm clasp[ing] a tree, is, of course, hypothetical; and the position of the hinder toes and feet does not appear to be natural. Altogether, however, the construction is highly satisfactory; and a better idea of the colossal proportions of the original is conveyed by this model, than could otherwise be obtained.”^[119]

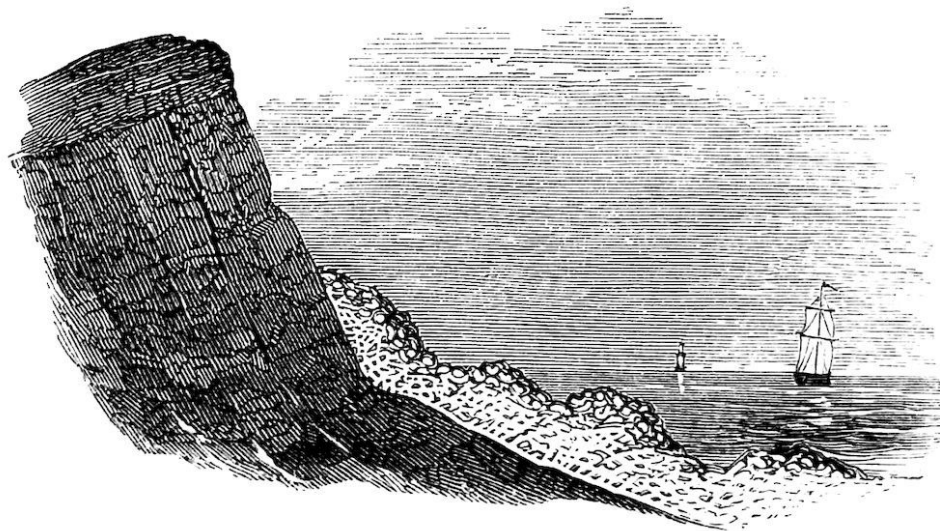


SKELETON OF THE MASTODON OHIOTICUS, FROM NORTH AMERICA.
(Height, 9½ feet; length, 20 feet.)

We give below a drawing of the “*Mastodon Ohioticus*,” for the following account of which we are indebted to the same source. It will be found in Room 6, figure 1.—“This fine skeleton was purchased by the trustees of the British Museum of Albert Koch, a well-known collector of fossil remains; who had exhibited, in the Egyptian Hall, Piccadilly, under the name of the Missouri, or Leviathan of the Missouri, an enormous osteological monster, constructed of the bones of this skeleton, together with many belonging to other individuals—the tusks being fixed in their sockets so as to curve outwards on each side of the head. From this heterogeneous assemblage of bones, those belonging to the same animal were selected, and are articulated in their natural juxtaposition.”^[120]



FOSSIL HUMAN SKELETON, FROM GUADALOUPE. (The original 4 feet 2 inches long, by 2 feet wide.)



PLAN OF THE CLIFF AT GAUDALOUPE.

In Wall-case D of the British Museum, may be seen a fossil skeleton of a human being, brought from the island of Guadaloupe, the consideration of which must for ever remove any idea that may exist about man being contemporaneous with the theroidal mammals of which we have been speaking.^[121] Professor Whewell has remarked that the “gradation in form between man and other animals is but a slight and unimportant feature in contemplating the great subject of the origin of the human race. Even if we had not revelation to guide us, it would be most unphilosophical to attempt to trace back the history of man, without taking into account the most remarkable facts in his nature: the facts of civilization, arts, government, speech—his traditions—his internal wants—his intellectual, moral, and religious constitution. If we will attempt a retrospect, we must look at all these things as evidence of the origin and end of man’s being; and we do thus comprehend, in one view, the whole of the argument—it is impossible for us to arrive at an origin homogeneous with the present order of things. On this subject, the geologist may, therefore, be well content to close the volume of the earth’s physical history, and open that divine record which has for its subject the moral and religious nature of man.”

“Mysterious framework of bone, locked up in the solid marble,—unwonted prisoner of the rock! an irresistible voice shall yet call thee out of thy stony matrix. The other organisms, thy partners in the show, are incarcerated in the lime for ever,—thou but for a term. How strangely has

the destiny of the race to which thou belongest re-stamped with new meanings the old phenomena of creation!... When thou wert living, prisoner of the marble, haply as an Indian wife and mother, ages ere the keel of Columbus had disturbed the waves of the Atlantic, the high standing of thy species had imparted new meanings to death and the rainbow. The prismatic arch had become the bow of the covenant, and death a great sign of the unbending justice and purity of the Creator, and of the aberration and fall of the living soul, formed in the Creator's own image,—reasoning, responsible man.”^[122]

FINIS:—THE GEOLOGIST'S DREAM.



ALARM, BUT NO DANGER.

CHAPTER XIII.

SCRIPTURE AND GEOLOGY; OR, APPARENT CONTRADICTIONS RECONCILED.

“By Him were all things created, that are in heaven, and that are in earth, visible and invisible, whether they be thrones, or dominions, or principalities, or powers; all things were created by Him, and for Him; and He is before all things, and by Him all things consist.”—PAUL.

We have in the course of the previous volume alluded to certain discrepancies, supposed to exist between the statements of Scripture, and the teachings of Geology. We have more than once intimated our intention of discussing at some length these questions, that have so long been tabooed by truly religious people, and often needlessly exaggerated by those who have possessed a “microscopic eye,” in the discovery of the weak points of Christian faith or opinion. Dropping the convenient and euphonious form of egotism, which allows sovereigns and authors to adopt the plural, we shall crave to stand before our readers in our personal and singular, rather than in our impersonal and plural, form of speech.

To the reader let me first say, that while I do not wish to appear before him as an advocate, as if I held a brief or had a retaining fee on behalf of Moses, I nevertheless feel rather keenly that the “reverend” put before my name may give something like this aspect to all my remarks. It may be thought, and that honestly enough, that because mine is the clerical profession, I am bound, *per fas aut nefas*, to contend for the authority of Scripture. It may be thought—in fact, it is daily alleged against us—that the particular “stand-point” we occupy is an unfair one, inasmuch as a preacher is bound to “*stick to the Bible*;” and indeed that he always comes to it with

certain *à priori* conclusions, that to a great extent invalidate his reasonings, and destroy the morality of his arguments.

Possibly there may be more truth in this than any of us dream of: *fas est ab hoste doceri*. I therefore make no professions of honesty, and appeal to no one's feelings; let us go and look at the Bible, and at the earth's crust, and be guided by our independent researches. Should this happen to be read by any one whose mind is out of joint with Scripture; who no longer reposes with satisfaction on the old book of his childhood and his youth; who has begun to fear,—perhaps to think that it is only a collection of “cunningly devised” fables; and who is on the verge of giving up Christianity and all “that sort of thing;” to such an one I shall speak, supposing him to be as honest in his doubts as I am in my convictions. I cannot deal with man as if he had no right to doubt; I have never yet “pooh-poohed” any one's unbelief; but I have always striven to regard all doubts expressed in courteous phrase, as the result of investigation, even though it may be partial, as the fruit of study, although it may have been misguided, and as the painful conclusions of a thinking mind, and not cherished for the sake of “having a fling” at moral truth or a righteous life, or at the mothers and sisters whose life “remote from public haunt” has saved them from ever doubting the truth of revelation.

Doubting and scorning are very opposite phases of mind: we here address the *doubter*; with the *scorner* we have nothing to do; if *ridicule* is his substitute for argument, by all means let him enjoy it; and if *calling names* is his substitute for patient investigation, let him enjoy that pastime also—hard words break no bones; but for the *doubter*, for the man who has his honest difficulties, and finds large stumbling-blocks in the path of unresisting acquiescence in household faiths, for such an one I have much to say in this chapter, if he will read it,—to him I stretch forth my hand in cordial greeting, and invite him to examine evidence, and consider facts; and then, whatever may be the result, whether I shake *his doubts* or he shake *my faith*, we shall at least have acted a manly and a straightforward part. At any rate, we ought ever to meet as friends, and to be candid and forbearing, as men liable to err through manifold besetments and biasses.

Having thus thrown myself upon my reader's candour, by a clear avowal of the spirit in which such controversies ought to be conducted, let us together proceed to the purpose of this chapter. Between Geology and

Scripture interpretation there *are* apparent and great contradictions—that all admit: on the very threshold of our future remarks, let us allow most readily that between the usually recognised interpretations of Scripture and the well-ascertained facts of Geological science, there are most appalling contradictions; and the questions arising thence are very important, both in a scientific and in a theological point of view. Is there any method of reconciliation, by which the harmony of the facts of science with the statements of the Bible can be shown? Where is the real solution to be found? Are we mistaken in our interpretations, or are we mistaken in our discoveries? Have we to begin religion again *de novo*, or may the Bible and the Book of Nature remain just as we have been accustomed to regard them; both as equally inspired books of God, waiting only the service and worship of man, their priest and interpreter?

These are questions surely of no common importance. Neither the Christian nor the doubter act a consistent part in ignoring them. Should the Christian say, “I want no teachings of science: I want no learned phrases and learned researches to assist me in understanding my Bible: for aught I care, all the ‘ologies’ in the world may perish as carnal literature: I know the Book is true, and decline any controversy with the mere intellectual disputant;” and if the Christian should go on to add, as probably he would in such a state of mind, and as, alas! too many have done to the lasting disgust and alienation of the thoughtful and intelligent: “These are the doubts of a ‘philosophy falsely so called:’ science has nothing to do with Revelation: they have separate paths to pursue; let them each go their own way: and should there come a collision between the two, we are prepared to give up all science once and for ever, whatever it may teach, rather than have our views upon Revelation disturbed:”—now, if the Christian talks like that, he is acting a most unwise part. He is doing in his limited sphere of influence what the Prussian Government intended to have done when Strauss’ “Life of Christ” appeared. It was the heaviest blow that unbelief had ever struck against Christianity, and the Government of Prussia with several theological professors were disposed to prosecute its author, and forbid the sale of the book. But the great Neander deprecated this course, as calculated to give the work a spurious celebrity, and as wearing the aspect of a confession that the book was unanswerable. He advised that it should be met, not by *authority*, but by *argument*, believing that the truth had

nothing to fear in such a conflict. His counsel prevailed, and the event has shown that he was right.

If, on the other hand, the doubter should say, “The intelligence of the day has outgrown our household faiths; men are no longer to be held in trammels of weakness and superstition, or to be dragooned into Religion;—the old story about the Bible, why, you know we can’t receive that, and look upon those compilations that pass by that name as divinely inspired Books; we have long since been compelled to abandon the thought that Christianity has any historic basis, or that its Books have any claim upon the reverence or faith of the nineteenth century, as of supernatural origin.”

To such an one I should say, that this begging of the question, this *petitio principii*, is no argument; these are statements that require every one of them a thorough demonstration before they are admitted; you deny the Christian one single postulate: you deny him the liberty of taking *anything* for granted; and then begin yourself with demanding his assent unquestioned to so large a postulate as your very first utterance involves, “that the intelligence of the age has outgrown our household faiths.” Before you proceed you must prove that; and we must know what is meant by those terms, before we can stand upon common ground, and hold anything like argument upon these debated points.

From such general observations let us come to the precise objects before us: Geology and Scripture are supposed to be at variance specially on three points. *The age of the earth: the introduction of death: and the Noachian Deluge.* These apparent contradictions are the most prominent difficulties, and cause the most startling doubts among those who imagine Science to be antagonistic to Christian revelation. I propose to devote a little attention to each of these questions, while I endeavour honestly to show how, in my opinion, apparent contradictions may be reconciled. The questions are these, to state them in a popular form: 1. *Is the world more than 6,000 years old?* and if it is, how are the statements of Scripture and Geology to be reconciled? 2. *Was death introduced into the world before the fall of man?* and if it was, how are the truths of Scripture on this question to be explained? and, 3. *What was the character of the Noachian Deluge?* was it partial or universal? and what are the apparent discrepancies in this case, between science and the Bible?

Perhaps before I proceed a step further I ought to add that, in my belief, the age of the earth, so far as its material fabric, *i.e.* its crust, is concerned, dates back to a period so remote, and so incalculable, that the epoch of the earth's creation is wholly unascertained and unascertainable by our human arithmetic; whether this is contradicted in the Scripture, is another question.

With regard to the introduction of death, I believe that death upon a most extensive scale prevailed upon the earth, and in the waters that are under the earth, ages, yea countless ages, before the creation of man—before the sin of any human being had been witnessed; that is what Geology teaches most indisputably: whether the Scriptures contradict this statement, is another question.

With regard to the Noachian Deluge, I believe that it was quite partial in its character, and very temporary in its duration; that it destroyed only those animals that were found in those parts of the earth then inhabited by man; and that it has not left one single shell, or fossil, or any drift or other remains that can be traced to its action. Whether the Scriptures teach any other doctrine, is another question.

By this time the ground between us is narrowed, and I may probably anticipate that I shall have objections to answer, or misapprehensions to remove, quite as much on the part of those who *devoutly believe*, as on the part of those who *honestly doubt* the Christian Scriptures.

First then,

I. *How old is the world?* How many years is it since it was called into being, as one of the planets? How many centuries have elapsed since its first particle of matter was created?

The answer comes from a thousand voices, “How old? why, 6,000 years, and no more, or closely thereabouts! Every child knows that;—talk about the age of the world at this time of day, when the Bible clearly reveals it!”

Now I ask, Where does the Bible reveal it? Where is the chapter and the verse in which its age is recorded? I have read my Bible somewhat, and feel a deepening reverence for it, but as yet I have never read that. I see the age of man recorded there; I see the revelation that says the *human* species is not much more than 6,000 years old; and geology says this testimony is true, for no remains of man have been found even in the tertiary system, the latest of all the geological formations. “*The Bible, the writings of Moses,*”

says Dr. Chalmers, “*do not fix the antiquity of the globe; if they fix any thing at all, it is only the antiquity of the species.*”

It may be said that the Bible does not dogmatically teach this doctrine of the antiquity of the globe; and we reply, Very true; but how have we got the idea that the Bible was to teach us all physical science, as well as theology. Turretin went to the Bible for *Astronomy*: Turretin was a distinguished professor of theology in his day, and has left behind him large proofs of scholarship and piety. Well, Turretin went to the Bible, determined to find *his system* of astronomy in it; and of course he found it. “The sun,” he says, “is not fixed in the heavens, but moves through the heavens, and is said in Scripture to rise and to set, and by a miracle to have stood still in the time of Joshua; if *the sun did not move, how could birds, which often fly off through an hour’s circuit, be able to return to their nests, for in the mean time the earth would move 450 miles?*” And if it be said in reply, that Scripture speaks according to common opinion, then says Turretin, “We answer, that the Spirit of God best understands natural things, and is not the author of any error.”

We smile at such “ecclesiastical drum” noise now, and we can well afford to do so: but when people go to the Bible, determined to find there, not a *central truth*, but the truths of physics, in every department of natural science, are we to be surprised that they come away disappointed and angry? As Michaelis says, (quoted by Dr. Harris, in his “Man Primeval,” p. 12,) “Should a stickler for Copernicus and the true system of the world carry his zeal so far as to say, that *the city of Berlin sets at such an hour*, instead of making use of the common expression, that the sun sets at Berlin at such an hour, he speaks the truth, to be sure, but his manner of speaking it is pedantry.”

Now, this is just the way to make thoughtful men unbelievers: and we will not adopt that plan, because it is not honest, neither is it clever; the Gordian knot of every question might easily be solved in this way.

Bearing in mind that the question we are now trying to solve is this, “What is the evidence afforded by Geology, as to the history of creation, and in what way does the geological age of the world affect the supposed statement of Scripture, that the world is only 6,000 years old?” I reply thus, and I prefer to use the words of others rather than my own, lest it should be supposed that I am introducing mere novelties of opinion on this subject:

—“That the first sentence in Genesis is a simple, independent, all-comprehending axiom to this effect; that matter, elementary or combined, aggregated only or organized, and dependent, sentient and intellectual beings have not existed from eternity, either in self-continuity or in succession: but had a beginning; that their beginning took place by the all-powerful will of One Being, the self-existent, independent and infinite in all perfection, and that the date of that beginning is not made known.”

These are the words of Dr. Pye Smith,^[123] of whose name as an authority, both in matters of science and philology, no one need be ashamed.

Dr. Redford says, “We ought to understand Moses as saying, indefinitely, far back, and concealed from us in the mystery of eternal ages, prior to the first moment of mundane time, ‘God created the heavens and the earth.’”

“My firm persuasion is,” says Dr. Harris, “that the first verse of Genesis was designed by the Divine Spirit to announce the absolute origination of the material universe by the Almighty Creator, and that it is so understood in the other parts of Holy Writ; that, passing by an indefinite interval, the second verse describes the state of our planet immediately prior to the Adamic creation; and that the third verse begins the account of the six days’ work.”

Dr. Davidson, in his “Sacred Hermeneutics,” says,—“If I am reminded, in a tone of animadversion, that I am making science, in this instance, the interpreter of Scripture, my reply is, that I am simply making the works of God illustrate His word in a department in which they speak with a distinct and authoritative voice; that it is all the same whether our geological or theological investigations have been prior, if we have not forced the one into accordance with the other. And it may be deserving consideration whether or not the conduct of those is not open to just animadversion who first undertake to pronounce on the meaning of a passage of Scripture, irrespective of all appropriate evidence, and who then, when that evidence is explored and produced, insist on their *à priori* interpretation as the only true one.”

But I quote no more: such are some of the eminent theological contributions to this department of science:—satisfactory in this respect, that a fair interpretation of Scripture does not require us to fix any precise date, much less the inconsiderable one of six thousand years, as the period of the earth’s formation.

Geology teaches the same thing.—Of the various formations that compose the earth's crust, to the ascertained extent of ten miles, suppose we select two,—the Old Red Sandstone and the Chalk formations. Laborious and scientific men have been at the pains to calculate the gradual increase of some of these *now proceeding deposits*,—such as the Deltas, in course of formation at the mouth of the Nile, and at the gorges of the Ganges; and they find that the progress of the depth of increase is exceedingly small,—probably not more than a foot in many years. Mr. Macculloch, a name standing very high for accurate investigation, states, from his own observation, that a particular Scottish lake does not form its deposit at the bottom, and hence raise its level, at the rate of more than half-a-foot in a century; and he observes, that the country surrounding that lake presents a vertical depth of far more than 3,000 feet, in the single series of the Old Red Sandstone formation; and no sound geologist, he hence concludes, will, therefore, accuse the computer of exceeding, if, upon the same ratio as the contiguous lake, he allows 600,000 years for the production of this series of rock alone.

A last instance which may here be adduced, of the apparent length of time required for the construction of a particular rock, offers itself in the Chalk formation. The enormous masses of this rock, presenting their tall white precipices in such simple grandeur to our view, might well excite our astonishment at the periods which would seem needful for their collection and deposition, even if they were mere *inorganic* concretions of calcareous matter. But what shall we say when the investigations of the microscope have lately revealed to us that these mountains of chalk, instead of being formed of mere inert matter, are, on the contrary, mighty congeries of decayed animal life,—the white apparent particles, of which the chalk masses are composed, being each grain a well-defined organized being, in form still so perfect, their shells so entire, and all their characteristics so discoverable, as to cause no doubt to naturalists as to the species in the animal economy to which they belonged. How justly does Sir Charles Lyell, who in his “Elements” records at length this surprising discovery, exclaim,

“The dust we tread upon was once alive!”

“Look at the lofty precipices which lay naked a slight section of the Chalk at the Culvers, or the Needles in the Isle of Wight, or the still loftier

Shakspeare Cliff at Dover, and let the mind form a conception, if it can, of the countless generations of these minutest of living creatures it must have required to build up, from their decayed bodies and their shelly exuviae, layer on layer, those towering masses thus brought to our view. Who shall dare to compute the time for this entire elaboration? The contemplation almost advances us a step towards forming a conception of infinitude.”^[124]

I need not dwell longer on the antiquity of the globe:—Geology and Scripture present no conflicting testimonies on this subject. Our *interpretation* of Scripture has, undoubtedly, been modified; but the living Word itself abideth, in all its grandeur and purity, for ever. And “the time is not far distant when the high antiquity of the globe will be regarded as no more opposed to the Bible than the earth’s revolution round the sun and on its axis. Soon shall the horizon, where geology and revelation meet, be cleared of every cloud, and present only an unbroken and magnificent circle of truth.”^[125]

The reader shall not be detained so long on the second point of inquiry, which is

II. “Was death introduced into the world before the fall of man? and if it was, how are the statements of Scripture, on this question, to be explained?” To this I have replied by anticipation, that, in my opinion, death, upon a most extensive scale, prevailed upon the earth, and in the waters that are under the earth, countless ages before the creation of man. Into the proof of this position allow me to go very briefly, although I am well aware that I run the risk of incurring the charge of heterodoxy, when I state my full conviction, that *death*, as well as the *world*, was *pre-Adamite*. The general impression is the contrary; but general impressions are not always right:—“general impression” is a very unsubstantial ghost to deal with, very like that cant phrase we spoke of at the beginning of this lecture,—“the intelligence of the age.” “General impression” has it, that death was *not* pre-Adamite; that there was *no* death before the fall; and that, to *say* the contrary, is, at least, to tread on very dangerous ground. In vain does Geology—“now happily a true science, founded on facts, and reduced to the dominion of definite laws”—lay bare the Silurian rocks, and discover even *there* extinct forms of life in exquisitely beautiful preservation. In vain does Geology, after showing us the fossil trilobite and coral, unfold the volume of the Old Red Sandstone, and show us *there* the fossil remains of

fish—so perfect that we might imagine them casts rather than fossils. In vain does Geology open its vast Oolitic system, and show us *there* other forms of extinct life in fossil insects, tortoises, mighty saurians, and huge iguanodons. In vain does Geology lay bare the Chalk, with its marine deposits; and the Tertiary formation, with its enormous theroid mammalia, far surpassing in size the largest animals we are acquainted with. In vain are all these fossil remains exhibited imbedded in the earth; and in vain do we search, amidst all these, for one fossil remain of man, or one fossil vestige of man's works. The easy, the cheap, the unreflective answer is, "Oh! these things were created there, or else Noah's flood left them there."

Of course, we can fall back upon a miracle as having done all this; but to have recourse to miracles when no miracle is recorded, is just to shake our faith in that all-inspired testimony, that supernatural Book, the existence of which is the great miracle of time. But there are the fossils! How did they come there if the forms of animal life, once inhabiting those remains, had not previously lived and died? Created! What? Created fossils? Then why not, when the Almighty created man, did he not create, at the same time, some skeletons of *man*, and place them in the earth, as he put skeletons of trilobites, fishes, reptiles, and mammals there? Our common sense and reverence both reject the idea. As to the puerile notion that Noah's flood put them there, did not Noah's flood overwhelm man as well as animals? and as the bones of man are as durable as the bones of animals, how is it that we never meet with a fossil human skull or thigh bone, or house?

We believe that death was a part of the divine plan of God's creation; that death is a law of all organic life—a necessary law and a most benevolent provision; that the living structure of all animals derives its substance from dead organic matter. We believe that, altogether apart from human sin, preceding and successive generations must be the order of being; for if there were no death, animals would soon pass beyond the limit of provision sufficient for nutritive support, or of localities for suitable habitations. We believe that if there had been no death prior to man's sin, it would involve the supposition that all animals were herbivorous; whereas, even the little *ladybird* cannot live without its meal of *aphides*; and, so believing, we find our faith in Scripture deepened when, seeing on every hand the extensive proofs of death, we find man, the moment he lost his lordship and proud eminence, and reduced himself voluntarily to the condition of animalism, immediately brought *penally* within the influence of that law of death,

whose existence he must have recognised in the death of animals from the first day of his creation.

Does any one reply, "This is contrary to Scripture?" I ask them what Scripture teaches that the death of *animals* is the result of man's sin?—rather would not Scripture sanction the thought that death was a part of the divine plan of God's creation, and that the certainty of man's transgression was the reason for giving this constitution to nature? True, Milton sings, in his noble poem, that will live as long as the English language lives—

"Of man's first disobedience, and the fruit
Of that forbidden tree, whose mortal taste
Brought death into the world, and all our woe:"

but we are not obliged to call the *Paradise Lost* our Bible; or to quote Milton as a physiological authority, although the prevalence of the opinion that death was not pre-Adamite, and a good deal of theology besides, is more of Miltonic than of Scripture teaching.

I leave this branch of my subject far before it is exhausted: so far from that, each of the three points enumerated might easily be expanded into a lecture; and I can only hope that my brevity in treating these topics will not be misconstrued into a desire to shirk any of the difficulties with which their investigation is surrounded.

III. I come, lastly, to the question of the *Noachian Deluge*, and shall again repeat my own words: "What was the character of the Noachian Deluge?—was it partial or universal? and what are the apparent discrepancies, in this case, between science and the Bible?" And I have added to this my belief that the Noachian Deluge was quite partial in its character, and very temporary in its duration: that it destroyed only those animals that were found in those parts of the earth habitable by man, and that it has not left a single shell or fossil, or any drift boulders or pebbles, or any other remains that may be traced to its action.

Very briefly we shall try and prove this; and perhaps the most popular way will be the best remembered,—only that the reader will bear in mind that this little book does not pretend to exhaust the subject, but only to realize the idea expressed at the beginning of this chapter. Presuming, that all have in their recollection the Scriptural account of the Noachian Deluge, instead of quoting words with which all are familiar, I will only remark, as the basis of my illustrations, that rain descended, and probably the ocean

overflowed, for forty days; that the waters lay upon the land, and covered them one hundred and fifty days; that at the end of that time they began to subside, and that in twelve months and twenty-seven days they were gone from the face of the earth, and the Noachian family liberated from the ark.

The question is, was this flood universal, and were all kinds of animals preserved in the ark? To which my answer, as involving my belief, is this, that the flood was *local*, and that only the animals peculiar to Armenia were provided for in Noah's ark.

"Oh! but the Bible says it was universal," says everybody. Yes; but that, you know, is just the question between us. The terms "all the earth" *seem* to imply universality, but they do not necessarily involve this. *"All countries came to Egypt to buy corn;"* certainly not all the world literally, but all the surrounding countries. So there were once dwelling at Jerusalem devout Jews *"from out of every nation under heaven;"* but not *literally* out of every nation, for the names of the nations are immediately given, and we find the nations to have been a few between Egypt and the Black Sea, and between Italy and Palestine. There are many other illustrations of a similar character: these will suffice: I only adduce these to show that at the beginning Scripture does not oblige us to consider "all" as meaning "every one;" or to understand literally "all the inhabitants of the earth" as meaning every creature.

Now, looking at the structure and composition of the earth's crust, especially its fossiliferous rocks, I am driven to one of three conclusions, each of them involving difficulty, I acknowledge, but the one that involves the least is, of course, the most preferable. Either I must admit—

1. That the fossils in these rocks were all deposited in order and in succession, without injury, through a crust of rocks ten miles in thickness, during twelve months' violent diluvial action:

2. Or that they were all deposited there during the 15,000 or 16,000 years that had elapsed since the creation of man prior to the Deluge; that is, supposing the creation of man and the creation of the earth to have been synchronous. Or, lastly, which theory I accept—

3. That the date of the earth's physical being is unknown to us, and that the fossiliferous rocks were deposited in decades of ages before the creation of man.

For, on the other hand, let us suppose the flood to have been universal, in the strict and literal sense of the term; then let me suggest some of the consequences and difficulties of such a theory.

1. One consequence would be that some remains of man or of his works would have been found; but nothing of this kind has occurred. Even Armenia has been geologically examined, and no human remains have been found; and surely man's bones would last as long as the shells of a trilobite or terebratula?

2. And, secondly, the organic remains, the fossils themselves, would have been found confusedly heaped together; whereas, the remains in the crust of the earth are as carefully arranged as the contents of a well-ordered cabinet. We know always to a certainty what fossils will be found in any rock before we examine that rock.

3. Besides which, some, at least, of the organic remains found ought to correspond with existing beings and species: yet the contrary is the case, except only a few fossils found near the surface of the earth, in that portion of the earth's crust occupied by the tertiary system.

Nor is this all. Consider the *vast difficulties* the universal flood theory has to contend with, all of which are removed by the theory we have adopted.

1. *There is the quantity of water required.* If all over the earth the water rose twenty-two feet six inches above the tops of the highest mountains, the quantity of water required would be eight times the whole quantity of water now existing. Where all this could have come from first and gone to afterwards, are prodigious stumbling-blocks. Of course we can resort to miracle; but this is not the way to get rid of difficulty in a manly and honest spirit.

2. *Then consider the number of animals the ark must have contained.* There are 1,000 species of mammalia, 5,000 species of birds, 2,000 species of reptiles, and 120,000 well-ascertained and distinct species of insects. Do we pretend that all these were housed and fed for nearly thirteen months in a vessel that was only 450 feet long, 75 feet broad, and 45 feet high; and that such a vessel contained room for them, and their food, besides that of man, for such a long period. The little toys of Noah's ark are certainly pretty, but very mischievous, and most of the popular notions of the flood have grown up from our nurseries as much from the use of this toy in this

case, as from the reading of *Paradise Lost* in the other: and the result is, the Bible is made responsible for it all.

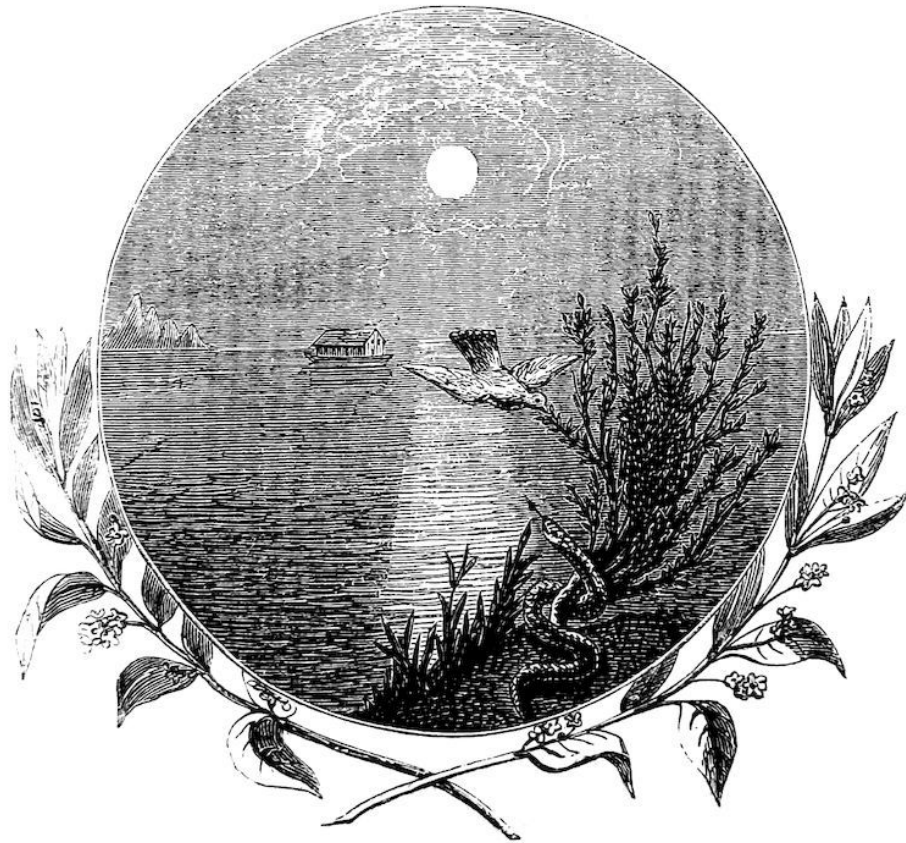
3. Then consider the subsequent distribution of animals: the polar bear and the tropical elephant, the ferocious tiger and a young fawn, going out together in order, and without violence: of course we can suppose another miracle to repress passions and violence. Besides which, in addition to the fauna, the animal kingdom, we must ask what became of the flora or vegetable kingdom during this period, if the flood were universal? We have at least twenty-five botanical provinces, with their peculiar and numberless farms of vegetable life; what became of them? Were they preserved in the ark, or under the water?—for such questions must be answered by those who charge us with inconsistency in attempting to reconcile the facts of science with the words of Scripture. And as a last difficulty, (suggested first, I believe, by Dr. Pye Smith, and which I shall therefore state in *his* words, lest it should seem that I use “plainness of speech,”) let us look at the descent from Ararat out of the ark, into Armenia, with all these animals, birds, insects, plants and trees. “That mountain is 17,000 feet high, and perpetual snow covers about 5,000 feet from its summit. If the water rose, at its liquid temperature, so as to overflow that summit, the snows and icy masses would be melted; and on the retiring of the flood, the exposed mountain would present its pinnacles and ridges, dreadful precipices of naked rock, adown which the four men and the four women, and with hardly any exception the quadrupeds, would have found it utterly impossible to descend. To provide against this difficulty, to prevent them from being dashed to pieces, must we again suppose a miracle? Must we conceive of the human beings and the animals as transported through the air to the more level regions below; or that, by a miracle equally grand, they were enabled to glide unhurt adown the wet and slippery faces of the rock?”

Such are some of the difficulties and some of the consequences that must flow from an acceptance of any other theory than the one I have proposed: that the flood was partial in its character, extending only over the habitable parts of the earth; and that it was so temporary in its character as not to have left a single trace of its influence visible on rock or fossil.

I have thus endeavoured to suggest points of reconciliation between the accepted facts of Geology and the recorded statements of Scripture; and if this slight contribution be accepted as an aid to faith, and a proof of candour

on my part to meet those who linger on the border land of doubt, my purpose will be fully answered.

Let me add, in the words of Chenevix Trench—words uttered in the University of Cambridge not long since: “May we in a troubled time be helped to feel something of the grandeur of the Scriptures, and so of the manifold wisdom of that Eternal Spirit by whom it came; and then petty objections and isolated difficulties, though they were multiplied as the sands of the sea, will not harass us. For what are they all to the fact, that for more than 1,000 years the Bible collectively taken, has gone hand in hand with civilization, science, law—in short, with the moral and intellectual cultivation of the species, always supporting and often leading the way? Its very presence as a believed book, has rendered the nations emphatically a chosen race; and this, too, in exact proportion as it is more or less generally studied. Of those nations which in the highest degree enjoy its influences, it is not too much to affirm that the differences, public and private, physical, moral, and intellectual, are only less than what might be expected from a diversity in species. Good and holy men, and the best and wisest of mankind, the kingly spirits of history enthroned in the hearts of mighty nations, have borne witness to its influence, and have declared it to be beyond compare the most perfect instrument and the only adequate organ of humanity: the organ and instrument of all the gifts, powers, and tendencies, by which the individual is privileged to rise beyond himself, to leave behind and lose his individual phantom self, in order to find his true self in that distinctness where no division can be,—in the Eternal I AM, the ever-living WORD, of whom all the elect, from the archangel before the throne to the poor wrestler with the Spirit until the breaking of day, are but the fainter and still fainter echoes.”



M. CLAY, PRINTER HEAD STREET HILL.

[1.](#) Whewell's Astronomy and Physics, p. 48.

[2.](#) From *παλαιός*, *ancient*, and *ζωόν*, *life*; ancient-life period.

[3.](#) Hughes, Physical Geography. 3d ed. p. 21.

[4.](#) Hughes, Physical Geog. p. 22.

[5.](#) Dr. Pye Smith.

[6.](#) As Chimborazo in South America, 21,414; Ararat, 16,000; Dhawalagiri, in the Himalayas, 28,000 feet above the level of the sea; compared with which what a mole-hill is Vesuvius, only 8,947 feet; or Blue Mountain Peak, 8,600, or even Mont Blanc, that monarch of mountains, which is 15,816 feet above the sea!

[7.](#) Hughes, p. 16.

[8.](#) Chambers' Rudiments of Geology, p. 71.

9. These wells are so frequently spoken of as to need no explanation, further than to remind the reader that they are so called from having been first introduced in the province of Artois, the ancient Artesium in France.

10. The deepest Artesian well is the famous one in the Plaine de Grenelle, Paris. This well yields 516 gallons a minute; its temperature is 81° Fahr.; and its depth is nearly 1,800 feet.

11. How truly hieroglyphics—sacred carvings; (*ieros*, sacred, *glupho*, I carve;) and in this sense there is a holier meaning than Shakspeare could have dreamt of in his well-known lines, when applied by the geologist to his researches:—

“And this our life, exempt from public haunt,
Finds tongues in trees, books in the running brooks,
Sermons in stones, and good in everything.”

12. And I may say, *my friend* also, to whom, during my residence in Jamaica, I was frequently indebted for contributions on natural history to the *Jamaica Friendly Instructor*, of which I was Editor.

13. A Naturalist's Sojourn in Jamaica, by P. H. Gosse, Esq. pp. 496–7.

14. So called because of its *grained* or *granular* appearance.

15. First brought from Syene, in Egypt.

16. Feld-spar, written also felspar, a compound of *feld*, field, and spar.

17. See Ansted's Ancient World, p. 21.

18. Memnon, or Ramesis. This famous head is in the British Museum; the body is of greenstone, the head of syenite, and the bust one continuous mass.

19. From *dis* and *integer*. The separation of the whole parts of a rock, without chemical action, by means of the light, the air, or the rain, is called disintegration.

20. Lieut. Portlock on Geology, p. 93.

21. Ansted's Geology, Descriptive and Practical, vol. ii. pp. 290, 291.

22. Ansted's Geology, p. 291.

23. As the ancients did not know or use the compound metal brass, though bronze was common amongst them, we must in this verse, and all others in which the word “brass” is used, understand it to mean copper.—Hughes' Scripture Geography, Art. Geology of Palestine, p. 133.

24. Murray's Hand-book for Cornwall, p. 199.

25. Ansted, vol. ii. p. 418.

26. Whewell, Anniversary Address to Geol. Society, 1839.

27. In Memoriam.

- [28.](#) Dr. Pye Smith says 140,000 feet.
- [29.](#) See a valuable map of fossils published by the Christian Knowledge Society.
- [30.](#) Trilobite: *treis*, three, and *lobos*, a lobe; having three lobes.
- [31.](#) Bridgewater Treatise, vol. i. p. 396.
- [32.](#) A fossil shell allied to the Argonauta and Carinaria.
- [33.](#) “Man has no tail, *quantum mutatus*; but the notion of a much-ridiculed philosopher of the last century is not altogether without foundation; for the bones of a caudal extremity exist in an undeveloped state in the os coccygis of the human subject.” Poor *man*!—Vestiges of Creation, p. 71.
- [34.](#) Sedgwick, p. 216, “On the Studies of the University of Cambridge.”
- [35.](#) “My School and Schoolmasters,” by Hugh Miller.
- [36.](#) “Old Red Sandstone; or, New Walks in an Old Field;” by Hugh Miller, p. 48.
- [37.](#) “By mind, by hand, and by hammer.”
- [38.](#) “Old Red Sandstone,” p. 66.
- [39.](#) Ichthyolite: *ichthus*, a fish, *lithos*, a stone: fossil fish, or the figure or impression of a fish in the rock.
- [40.](#) “Old Red Sandstone,” pp. 41, 42.
- [41.](#) “Old Red Sandstone,” p. 69.
- [42.](#) From *akanthos*, a thorn, and *pterugion*, the fin.
- [43.](#) From *malakos*, soft, and *pterugion*, the fin.
- [44.](#) 1. Ganoid, from *ganos*, splendour, because the scales are coated with a bright enamel.
- [45.](#) 2. Placoid, from *plax*, a plate; sometimes large, sometimes reduced to a point; *e.g.* shark.
- [46.](#) 3. Ctenoid, from *kteis* (gen. *ktenos*, a comb); scales jagged like a comb.
- [47.](#) 4. Cycloid, from *kuklos*, a circle; scales smooth and simple: *e.g.* salmon, &c.
- [48.](#) From *kephalē*, the head; *aspis*, a buckler.
- [49.](#) Coccosteus, from *kokkos*, a berry, and *osteon*, a bone.
- [50.](#) “Old Red Sandstone,” p. 86.
- [51.](#) Pterichthys: *pteron*, a wing, and *ichthus*, a fish.
- [52.](#) “Old Red Sandstone,” pp. 80, 81.

[53.](#) Osteolepis: *osteon*, a bone, and *lepis*, a scale.

[54.](#) Operculum, the flap which covers the gill.

[55.](#) “Old Red Sandstone,” p. 111.

[56.](#) “Vast quantities:” let any reader go and turn over the non-bituminous shale lying on the waste heaps of every coalpit, and he will see that this is no exaggeration.

[57.](#) Capillus Veneris.

[58.](#) Corruption of *arrière-dos*, a fire-place. See a view and description of one in “A Visit to Penshurst,” in Howitt’s “Visits to Remarkable Places,” Second Series.

[59.](#) Juicy and soft, as peas, beans, plantains, bananas, &c.

[60.](#) “Ancient World,” pp. 76, 77.

[61.](#) This may seem strange at first; but I have journeyed through tropical forests that realized completely this sketch, so far as stillness and silence are concerned. A modern and most accomplished naturalist says of a Jamaica virgin forest, “Animal life is almost unseen; the solitude is scarcely broken by the voices of birds, except that now and then the rain-bird or the hunter (large cat-tailed cuckoos that love the shade) sound their startling rattle, or the mountain partridge utters those mournful cooings which are like the moans of a dying man.”—Gosse’s *Jamaica*, p. 198.

[62.](#) From κάλαμος (*calamus*), a reed.

[63.](#) Ansted’s “Ancient World,” p. 82.

[64.](#) Mesozoic: *i.e.* middle life period; *mesos*, middle, *zoos*, life.

[65.](#) The Religious Tract Society.

[66.](#) Lyell’s “Manual of Elementary Geology.” Postscript, p. 13.

[67.](#) Ansted’s *Geology*, vol i. p. 306.

[68.](#) Ichnites; from *ichnon*, a footstep, and *eidos*, like.

[69.](#) *Ornithos*, a bird, and *ichnon*.

[70.](#) Marsupial, from *marsupium*, a pouch; animals of the fourth order of Cuvier, that have a pouch in which the young are carried.

[71.](#) Batrachian, from *batrachos*, a frog; animals in Cuvier’s fourth class of reptiles.

[72.](#) *Cheir*, the hand, *therion*, a wild beast; a wild beast with a foot like a hand.

[73.](#) From *labyrinthus*, a labyrinth, and *odous*, a tooth; so called from the labyrinthine structure of the tooth.

[74.](#) In some cases we find, corresponding to a set of footmarks, a continuous furrow, presumed to be the impression of a tail dragged along the sand by the animal while walking.

- [75.](#) Ansted's Ancient World, pp. 125–127.
- [76.](#) Knight's Cyclopædia of Arts, &c.
- [77.](#) Quarterly Review, May, 1852. Article on Roger de Coverley.
- [78.](#) This is a corruption, we are inclined to think, of the word “layers;” one of those provincial corruptions of the Queen's English that get stereotyped.
- [79.](#) Buckland's Bridgewater Treatise. pp. 351, 352.
- [80.](#) A fossil bivalve, allied to the oyster, and very abundant in the secondary strata.
- [81.](#) Belemnite, from *belemnos*, a dart, and so called from its arrow-headed shape.
- [82.](#) Saurian, from *sauros*, a lizard, the name by which the great family of lizards is designated.
- [83.](#) From *ichthus*, a fish, and *sauros*, a lizard; so called from its resemblance to both.
- [84.](#) Heteroclite; *heteros*, another, and *klitos*, inclining; a word applied to any thing or person deviating from common forms.
- [85.](#) Very unlike the alligator, whose eyes are placed at a considerable distance behind the nose.
- [86.](#) From *pleiōn*, more, and *sauros*, a lizard; because it is more like a lizard than the Ichthyosaurus.
- [87.](#) Mantel's Fossils of the British Museum, p. 341.
- [88.](#) This formation is sometimes called the Jurassic system.
- [89.](#) Lyell's Manual of Elementary Geology, p. 12, ed. 1852.
- [90.](#) “So vast an expanse!” Mr. Darwin traced coral reefs in the Pacific, 4,000 miles long and 600 broad. Between the coasts of Malabar and Madagascar there is a chain of coral reefs, called the Maldives and Laccadives, 480 miles long and 50 miles wide. On the east coast of Australia there is an unbroken reef of 350 miles long; and between Australia and Guinea, coral reefs extend 700 miles in length. Truly the coral animals, like the “conies,” are a “feeble folk,” but their habitations survive our proudest monuments.
- [91.](#) Hugh Miller's First Impressions, pp. 203, 204.
- [92.](#) Brash is a Wiltshire word for short or brittle; and thus a quick-tempered, irritable person, is said to have a *brashy* temper.
- [93.](#) Geology for Beginners (Weale's Series), p. 147.
- [94.](#) Jukes's Popular Geology, pp. 42–44.
- [95.](#) From *krinos*, a lily, and *eidos*, like; lily-shaped animals of the Radiated division, forming a link between the animal and vegetable world.
- [96.](#) From *trochos*, a wheel; wheel-shaped crinoideans.

[97](#). From *pteron*, a wing, and *dactulos*, a finger; the wing-fingered animal.

[98](#). The term *Weald* or *Wold* is the old Saxon for our present *Wood*; and now, altered by pronunciation, is found in connexion with many words and names of places: *e.g.* *Waltham* (*Weald-ham*), the wood house or home; *Walthamstow*, the wood house store, and so on. Thus it is that words are “fossil poetry.”

[99](#). Alison’s description of South America, in *History of Europe* (Article, *South American Revolution*); vol. viii.

[100](#). “Our disposition is, and has been, not to multiply miracles after the sort in which this has been done by many more *zealous* than *wise* friends of revelation. In all cases we allow the miracle without question, which is distinctly claimed to be such in the Scriptures, and where the circumstances clearly indicate that a miracle was necessary,—we say ‘necessary,’ because we are persuaded that the Almighty has almost invariably chosen to act through natural agencies, and under the laws which he has imposed on nature, whenever they are adequate to produce the required result. We believe it is one of the beautiful peculiarities of the Bible, that it has none of those gratuitous and barren wonders, which form the mass of the pretended miracles which the various systems of false religion produce.... For our own part, we do not wish to hear of small miracles, which leave us doubtful whether there be any miracle at all. If we are to have miracles, let them be decidedly miraculous, and let not our veneration for the Divine character be offended by exhibitions of the Almighty, as laying bare his holy arm to remove the small remaining difficulty which theorists leave him to execute.”—*Dr. Kitto’s Biblical History of Palestine*.

[101](#). We have a fine specimen before us which we brought from Demerara, answering well to Gosse’s description of the iguana found in Jamaica. “In the eastern parts of the island the great iguana (*Cyclura lophoma*), with its *dorsal crest, like the teeth of a saw, running all down its back*, may be seen lying out on the branches of the trees, or playing bo-peep from a hole in the trunk.” It is considered a great delicacy by many, but it never seemed Christian food to us, and we never ventured to provoke our palate with a taste.

[102](#). *Enaliosaurians* are sea lizards, such as those found in the Lias; and *deinosaurians* are terrible lizards, such as those found in the Wealden.

[103](#). Ansted’s *Ancient World*, pp. 164–168.

[104](#). Just published by Bohn, in his valuable “Scientific Library;” a marvel of cheapness and value.

[105](#). Since writing the above we have met with the following, which proves that this origin of chalk is not so fabulous as some think it:—“Lieut. Nelson, Mr. Dance, and others have shown, that the waste and débris derived from coral reefs produces a substance exactly resembling chalk. I can corroborate this assertion from my own observations, both on some very white chalky limestones in Java and the neighbouring islands, which I believe to be nothing else than raised fringing coral reefs, and on the substance brought up by the lead over some hundreds of miles in the Indian Archipelago, and along the north-east coast of Australia, and the coral sea of Flinders,”—*Juke’s Physical Geology*, p. 263.

[106](#). We take the origin of the word *Folkstone* to mean, that that old town was once built of the brick that may be made of the *galt*: it was the *folk’s-stone*.

[107](#). “The Religion of Geology,” &c., by E. Hitchcock, LL.D. &c. p. 70.

[108.](#) Mantell's "Geological Excursions," p. 145.

[109.](#) Richardson, p. 391.

[110.](#) *Under-borne* rocks; *upo*, below, and *ginomai*, to be formed.

[111.](#) *Middle life* period: *mesos*, middle, and *zoos*, life.

[112.](#) Recent-life period: *kainōs*, recent, and *zoos*, life.

[113.](#) Juke's Practical Geology, p. 265.

[114.](#) Lyell's Manual of Geology, pp. 97, 98.

[115.](#) Ansted's Geology, vol. ii. p. 14.

[116.](#) Even Hitchcock's good book is sadly disfigured and damaged, by trying to make geology prove too much. How can geology teach or suggest sin and the resurrection?

[117.](#) British Quarterly, Feb. 1852.

[118.](#) Owen's British Fossil Mammals and Birds, p. 255.

[119.](#) Mantell, pp. 477–479.

[120.](#) Mantell, p. 471.

[121.](#) The skeleton is not more than 150 years old, and is probably one of an Indian who fell in war; and has been covered with carbonate of lime, held in solution in some spring.

[122.](#) Hugh Miller's "First Impressions of England and its People," p. 362.

[123.](#) No sooner did geology give signs of being able to speak from her subterranean abode, and say something new about the history of this old world, than Dr. Smith was among the foremost of the geologists, intent upon the interpretation of these mysterious, and at first incoherent sounds. At times the sounds seemed unscriptural, but his faith never failed; at other times it seemed in confirmation of Scripture, and he was filled with delight. There were sepulchres older than what he had accounted the era of death, and he must solve the mystery. Mineralogy, to which from his youth he had given considerable attention, became to him history more ancient than that of Moses, and poetry more fascinating than that of Homer. His minerals became books of wonderful tales; his fossils, before riddles of nature, the pictures of things in ancient worlds. The earth was a land of monuments, and the rock which before seemed nothing more than the solid masonry of the foundation on which men might build their dwellings, became the enduring chronicle of the millions of years in which extinct ages had risen, flourished and decayed. From that time he suffered no discovery of the geologists to escape his attention; and every valuable book upon the subject in English, German, or French, contributed its supplies to mitigate his insatiate craving after further information.

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[124](#). Gray's Antiquity of the Globe, pp. 57–59.

[125](#). Hitchcock, p. 70.

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